

Genetic response to supplements (999 supplements)

Genetic response predicted to 999 popular dietary supplements, by category
Report generated in October 2 of 2026 · revised the 3 of October 2026 (aligned with the current protocol)

Genetic basis: DNA crude genotyping chip (GRCh38, 684 967 SNPs) + local imputation (Beagle 5.5, public reference panel of genomes complete to 30x; only genotypes with DR2 ≥ 0.8). Genetic data did not leave the local computer.

Evidence base: published gene-nutrient interactions (nutrigenomics), safety genes (liver, G6PD, HLA, TERT), plant pharmacogenetics (CYP3A4/P-gp, CYP2C9, CYP2D6, UGT1A1) with the same profile of the 632 drug report, LiverTox (NIH), NIH ODS and NCCIH chips, upper limits EFSA and CPIC guidelines used by analogy.

List of supplements: 999 products (substances, chemical forms, strains and combined formulas), constructed from NIH ODS and NCCIH chips and completed by curation of the usual commercial categories (vitamins and forms, minerals, amino acids, lipids, polyphenols, Western plants, Ayurvedic, Chinese/Kampo, Americas and Africas, mushrooms, probiotics, enzymes, nootropics, sport, hormones, longevity, joints/skin, metabolism) + current stack. Duplicates removed.

Important: This is a genetic reading of chip data, not a diagnosis or a prescription. It uses only genetics (without analysis). All indications are suggestion (not applied) and should be decided with the doctor. The evidence of nutrient is, in general, weaker than that of pharmacogenetics.

red avoid/pause **orange** adjust dose, shape or time **yellow** monitoring/confirming **green** normal (relevant gene evaluated) **gray** indeterminate **compact list** with no known genetic relevance ★ stack ➕ favourable genetic indication

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1. Executive summary

Class	Number of supplements	%
Avoid / pause	53	5.3%
Adjust dose/form/time	66	6.6%
Monitoring / confirm	111	11.1%
Undetermined	11	1.1%
Normal (gene evaluated)	221	22.1%
No known genetic relevance	537	53.8%
Total analyzed	999	100%

Fast reading: 230 Supplements require attention (avoid, adjust or monitor) and 67 have a favourable genetic indication (⊕). Almost everything is explained by eight axes: (1) genetic liver (PNPLA3 I148M + MBOAT7 + Pi*MZ probable) → avoid hepatotoxic plants (green tea in extract, kava, ashwagandha, red rice yeast, comfrey, He Shou Wu... (Complete abstract click electronic access below) (2) UGT1A1 *1/*28 (Gilbert) + inhibition of CYP3A4/P-gp → curcumin, quercetin, luteolin and apigenin (pause while ALT is high, current protocol); (3) COMT Val/Val + MTHFR compound (homocysteine p92 score) → high dose catechols, SAMe/methionine, L-dopa; prefer 5-MTHF, TMG, glycine and creatine; (4) DHFR CC → folic acid substituted by 5-MTHF; (5) HFE H63D → without iron by routine and vitamin C away from meals; (6) normal to intermediate CYP1A2 + MTNR1B G → caffeine ≤ 200 mg in the morning and melatonin 0.3–1 mg; (7) TERT risk + nevi p88 → avoid telomerase activators; (8) CYP3A5 expresser + sirolimus → nothing that induces or inhibits CYP3A4 in the window of rapamycin (St John's wort, schisandra, berberine, piperine...). Current stack (44 items): 4 to pause, 11 to adjust, normal 22, indeterminate 1 and 6 without specific gene — all consistent with the genetic plan and its coordination note. Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.

Avoid / pause (53)

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
vitamin A (retinyl acetate)	Vitamins and forms	BCO1: slightly ↓ conversion PNPLA3/Pi*MZ (liver): unfavourable	avoid preformed retinol	The conversion of β-carotene is only slightly reduced (BCO1 R267S het; without A379V), so there is no genetic need for retinol. With the isotretinoin of the stack, retinol supplemental sum retinoids (toxicity, transaminases); the genetic liver (PNPLA3/Pi*MZ) reinforces. Obtain vitamin A from carotenoids (suggestion, not applied).
vitamin A (retinol, retinyl palmitate)	Vitamins and forms	BCO1: slightly ↓ conversion PNPLA3/Pi*MZ (liver): unfavourable	avoid preformed retinol	The conversion of β-carotene is only slightly reduced (BCO1 R267S het; without A379V), so there is no genetic need for retinol. With the isotretinoin of the stack, retinol supplemental sum retinoids (toxicity, transaminases); the genetic liver (PNPLA3/Pi*MZ) reinforces. Obtain vitamin A from carotenoids (suggestion, not applied).
vitamin A + D (oil)	Vitamins and forms	BCO1: slightly ↓ conversion PNPLA3/Pi*MZ (liver): unfavourable	avoid preformed retinol	The conversion of β-carotene is only slightly reduced (BCO1 R267S het; without A379V), so there is no genetic need for retinol. With the isotretinoin of the stack, retinol supplemental sum retinoids (toxicity, transaminases); the genetic liver (PNPLA3/Pi*MZ) reinforces. Obtain vitamin A from carotenoids (suggestion, not applied).

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
cod liver oil	Fatty acids	BCO1 : slightly ↓ conversion PNPLA3/Pi*MZ (liver) : unfavourable	avoid preformed retinol	The conversion of β-carotene is only slightly reduced (BCO1 R267S het; without A379V), so there is no genetic need for retinol. With the isotretinoin of the stack, retinol supplemental sum retinoids (toxicity, transaminases); the genetic liver (PNPLA3/Pi*MZ) reinforces. Obtain vitamin A from carotenoids (suggestion, not applied).
EGCG / green tea extract	Polyphenols and other phytochemicals	PNPLA3/Pi*MZ (liver) : unfavourable COMT : Val/Val (high) HLA-B*35:01 : probable negative UGT1A1 : IM (*1/*28)	avoid (liver)	Concentrated green tea extract is a known cause of liver injury (LiverTox, probability B; risk in fasting and > 800 mg EGCG/day). Genetic liver (PNPLA3, MBOAT7, Pi*MZ); EGCG is catechol (COMT Val/Val). In favor: HLA-B*35:01 probable negative. Tea in drink is OK (suggestion, not applied).
★ apigenin	Polyphenols and other phytochemicals	UGT1A1 : IM (*1/*28) CYP3A5:expressor (*1/*3) CYP1A2 : normal to intermediate CYP2C9 : IM (*1/*3)	avoid / pause (test-pause)	Current stack (100 mg) — plan S28: PAUSE (test-pause). Test-pause (current protocol). Test pause for the same reasons of luteolin: inhibits UGT1A1 (*1/*28: up to bilirubin and confuses liver reading) and CYP3A4/P-gp (sirolimus; CYP3A5 expressor). It is not catechol, but also inhibits CYP1A2 (already normal to intermediate: *1F het + rs2472297 C/C), which rises melatonin and caffeine, and CYP2C9 (*1/*3). Pause; pause while ALT is high (current protocol). Coordination note: in the current protocol, the pause lasts while ALT is high; if it returns, 25-50 mg at night, outside the rapamycin window (suggestion, not applied).
curcumin (95% extract)	Polyphenols and other phytochemicals	UGT1A1 : IM (*1/*28) PNPLA3/Pi*MZ (liver) : unfavourable HLA-B*35:01 : probable negative CYP3A5:expressor (*1/*3)	avoid/pause	UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).
curcumin + piperine	Polyphenols and other phytochemicals	UGT1A1 : IM (*1/*28) PNPLA3/Pi*MZ (liver) : unfavourable HLA-B*35:01 : probable negative CYP3A5:expressor (*1/*3)	avoid/pause	UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).
colloidal curcumin (nanoparticles)	Polyphenols and other phytochemicals	UGT1A1 : IM (*1/*28) PNPLA3/Pi*MZ (liver) : unfavourable HLA-B*35:01 : probable negative CYP3A5:expressor (*1/*3)	avoid/pause	UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).
curcumin with curcuman essential oil	Polyphenols and other phytochemicals	UGT1A1 : IM (*1/*28) PNPLA3/Pi*MZ (liver) : unfavourable HLA-B*35:01 : probable negative CYP3A5:expressor (*1/*3)	avoid/pause	UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).

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curcumin phytosome	Polyphenols and other phytochemicals	UGT1A1: IM (*1/*28) PNPLA3/Pi*MZ (liver): unfavourable HLA-B*35:01: probable negative CYP3A5: expressor (*1/*3)	avoid/pause	UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).
★ solid lipid curcumin (SLCP)	Polyphenols and other phytochemicals	UGT1A1: IM (*1/*28) PNPLA3/Pi*MZ (liver): unfavourable HLA-B*35:01: probable negative CYP3A5: expressor (*1/*3)	avoid/pause	Current stack (1 g) — plan S26: PAUSE. Pause (current protocol). UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).
isoquercytrin enzymatically modified	Polyphenols and other phytochemicals	UGT1A1: IM (*1/*28) COMT: Val/Val (high) MTHFR: 677CT/1298AC CYP3A5: expressor (*1/*3)	avoid/pause	UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.
★ luteolin	Polyphenols and other phytochemicals	COMT: Val/Val (high) MTHFR: 677CT/1298AC UGT1A1: IM (*1/*28) CYP2C9: IM (*1/*3)	avoid/pause	Current stack (100 mg) — plan S27: PAUSE. Pause (current protocol). Catechol: methylated by COMT (Val/Val), spent SAM and produces homocysteine (MTHFR compound, p92 score). It inhibits UGT1A1 (*28) and CYP2C9 (*1/*3 IM) in vitro, and CYP3A4/P-gp (sirolimus). Pause; pause while ALT is high (current protocol) (suggestion, not applied).
luteolin + PEA	Polyphenols and other phytochemicals	COMT: Val/Val (high) MTHFR: 677CT/1298AC UGT1A1: IM (*1/*28) CYP2C9: IM (*1/*3)	avoid/pause	Catechol: methylated by COMT (Val/Val), spent SAM and produces homocysteine (MTHFR compound, p92 score). It inhibits UGT1A1 (*28) and CYP2C9 (*1/*3 IM) in vitro, and CYP3A4/P-gp (sirolimus). Pause; pause while ALT is high (current protocol) (suggestion, not applied).
★ quercetin	Polyphenols and other phytochemicals	UGT1A1: IM (*1/*28) COMT: Val/Val (high) MTHFR: 677CT/1298AC CYP3A5: expressor (*1/*3)	avoid/pause	Current stack (250 mg (3×/month)) — plan S29: PAUSE (suggestion, not applied). It is suggested to pause as an isolated supplement (genetic verdict); in the current protocol enters the monthly pulse with fisetin, only with normal ALT. UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.
quercetin phytosome	Polyphenols and other phytochemicals	UGT1A1: IM (*1/*28) COMT: Val/Val (high) MTHFR: 677CT/1298AC CYP3A5: expressor (*1/*3)	avoid/pause	UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
resveratrol + quercetin (combined)	Polyphenols and other phytochemicals	UGT1A1: IM (*1/*28) COMT: Val/Val (high) MTHFR: 677CT/1298AC CYP3A5: expressor (*1/*3)	avoid/pause	UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.
greater celandine	Western medicinal plants	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).
comfrey (conferred)	Western medicinal plants	PNPLA3/Pi*MZ (liver): unfavourable	avoid (pyrrolizidine alkaloids)	Pyrrolizidine alkaloids cause veno-occlusive liver disease; oral use is prohibited in several countries. Genetic liver reinforces.
germander (Teucrium)	Western medicinal plants	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).
St. John's wort	Western medicinal plants	CYP3A4: normal CYP3A5: expressor (*1/*3) CYP2C19: NM TERT/melanoma: unfavourable	avoid (strong inducer)	Strong inducer of CYP3A4, P-gp and CYP2C19: low cyclosporine/tacrolimus 30–60% → predictable descent of sirolimus, which has already reduced exposure to CYP3A5 expresser; also low tadalafil. Photosensitizing hypericin (p88 + isotretinoin) (suggestion, not applied).
pore (oil)	Western medicinal plants	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).
tussiling	Western medicinal plants	PNPLA3/Pi*MZ (liver): unfavourable	avoid (pyrrolizidine alkaloids)	Pyrrolizidine alkaloids cause veno-occlusive liver disease; oral use is prohibited in several countries. Genetic liver reinforces.
ashwagandha (full spectrum root extract)	Ayurvedic plants	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Ashwagandha: growing cholestatic cases (2017–2025).
ashwagandha (root and leaf extract)	Ayurvedic plants	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Ashwagandha: growing cholestatic cases (2017–2025).
ashwagandha (root)	Ayurvedic plants	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Ashwagandha: growing cholestatic cases (2017–2025).
babchi (Psoralea corylyfolia)	Ayurvedic plants	PNPLA3/Pi*MZ (liver): unfavourable TERT/melanoma: unfavourable	avoid (liver + photosensitization)	Babchi: photosensitizing psoralens (p88 + isotretinoin) and hepatitis (LiverTox) cases.
brahmi + ashwagdha (combined)	Ayurvedic plants	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Ashwagandha: growing cholestatic cases (2017–2025).

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giloy / guduchi (Tinospora cordifolia)	Ayurvedic plants	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).
He Shou Wu (Polygonum multiflorum, fo-ti)	Traditional Chinese medicine plants and Kampo	PNPLA3/Pi*MZ (liver): unfavourable HLA-B*35:01: probable negative	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Polygonum multiflorum: HLA-B*35:01-linked lesion (probable negative), but intrinsic hepatotoxicity remains.
Jin Bu Huan (tetra-hydropalmatin)	Traditional Chinese medicine plants and Kampo	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).
astragalosid IV	Traditional Chinese medicine plants and Kampo	TERT/melanoma: unfavourable	avoid (telomerase activator)	Used as telomerase activators. With 5 alleles TERT of oncologic risk and nevi at the 88th percentile (melanoma by chip is p6.5/p25), precaution is asymmetric (hypothetical benefit vs theoretical risk). Avoid purified extracts of astragalus, cycloastragenol and astragalus (suggestion, not applied).
astralis (root)	Traditional Chinese medicine plants and Kampo	TERT/melanoma: unfavourable	avoid (telomerase activator)	Used as telomerase activators. With 5 alleles TERT of oncologic risk and nevi at the 88th percentile (melanoma by chip is p6.5/p25), precaution is asymmetric (hypothetical benefit vs theoretical risk). Avoid purified extracts of astragalus, cycloastragenol and astragalus (suggestion, not applied).
cycloastragenol	Traditional Chinese medicine plants and Kampo	TERT/melanoma: unfavourable	avoid (telomerase activator)	Used as telomerase activators. With 5 alleles TERT of oncologic risk and nevi at the 88th percentile (melanoma by chip is p6.5/p25), precaution is asymmetric (hypothetical benefit vs theoretical risk). Avoid purified extracts of astragalus, cycloastragenol and astragalus (suggestion, not applied).
purified extract of astrogalum (telomerase activator)	Traditional Chinese medicine plants and Kampo	TERT/melanoma: unfavourable	avoid (telomerase activator)	Used as telomerase activators. With 5 alleles TERT of oncologic risk and nevi at the 88th percentile (melanoma by chip is p6.5/p25), precaution is asymmetric (hypothetical benefit vs theoretical risk). Avoid purified extracts of astragalus, cycloastragenol and astragalus (suggestion, not applied).
ma huang (ephedra)	Traditional Chinese medicine plants and Kampo	CETP/9p21/APOB: genetic vascular risk NOS3: -786CT + 298GT	avoid	Prohibited sympathomimetics (UE/EUA): Stroke/farce; 9p21 risk homozygous and NOS3 reinforce.
tripterygium (thunder Godvine)	Traditional Chinese medicine plants and Kampo	PNPLA3/Pi*MZ (liver): unfavourable	avoid	Tripterygium: hepatic, renal and reproductive toxicity; immunosuppressant.
pharmacaria (Larrea)	Plants of the Americas and Africa; superfood	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).
kava (Piper methylicum)	Plants of the Americas and Africa; superfood	PNPLA3/Pi*MZ (liver): unfavourable CYP2D6: Probable NM (without CNV)	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Kava: severe hepatitis and transplants; removal in several countries.
kratom (Mitragyna speciosa)	Plants of the Americas and Africa; superfood	PNPLA3/Pi*MZ (liver): unfavourable CYP2D6: Probable NM (without CNV) OPRM1: A/G	avoid	Atypical opioid (dependence), cholestatic liver injury (LiverTox); CYP3A4/2D6 (CNV not evaluated) and OPRM1 A/G.

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
DMHA / DMAA (prohibited stimuli)	Sport and performance	CETP/9p21/APOB : genetic vascular risk NOS3 : –786CT + 298GT	avoid	Prohibited sympathomimetics (UE/EUA): Stroke/farce; 9p21 risk homozygous and NOS3 reinforce.
prohormones/SARMs (sold as supplement)	Sport and performance	PNPLA3/Pi*MZ (liver) : unfavourable	avoid	Synthesis steroids and SARMs: severe cholestatic lesion (LiverTox) and hormonal axis; genetic liver reinforces. They are not legal supplements in UE.
desiccated liver (capsules)	Hormones	HFE : H63D het BCO1 : slightly ↓ conversion PNPLA3/Pi*MZ (liver) : unfavourable	avoid (retinol + iron)	Desiccated liver concentrates retinol (isotretinoin) and iron heme (HFE H63D).
organ glandular (mixture)	Hormones	HFE : H63D het BCO1 : slightly ↓ conversion PNPLA3/Pi*MZ (liver) : unfavourable	avoid (retinol + iron)	Desiccated liver concentrates retinol (isotretinoin) and iron heme (HFE H63D).
apigenin + magnesium (sleep)	Longevity	UGT1A1 : IM (*1/*28) CYP3A5 :expressor (*1/*3) TRPM6 : typical	avoid / pause (by apigenin)	Magnesium does not raise a problem (typical TRPM6), but for apigenin a test break (UGT1A1 *28) is suggested. CYP3A4/P-gp with sirolimus: pause while ALT is high (current protocol). Prefer magnesium without apigenin (suggestion, not applied).
senolytic combined (fisetin + quercetin)	Longevity	UGT1A1 : IM (*1/*28) COMT : Val/Val (high) MTHFR : 677CT/1298AC CYP3A5 :expressor (*1/*3)	avoid/pause	UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.
turmeric + boswellia (articular)	Articulations	UGT1A1 : IM (*1/*28) PNPLA3/Pi*MZ (liver) : unfavourable HLA-B*35:01 : probable negative CYP3A5 :expressor (*1/*3)	avoid/pause	UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).
caffeine + green tea + garcinia (combined)	Metabolism	PNPLA3/Pi*MZ (liver) : unfavourable COMT : Val/Val (high) HLA-B*35:01 : probable negative UGT1A1 : IM (*1/*28)	avoid (liver)	Concentrated green tea extract is a known cause of liver injury (LiverTox, probability B; risk in fasting and > 800 mg EGCG/day). Genetic liver (PNPLA3, MBOAT7, Pi*MZ); EGCG is catechol (COMT Val/Val). In favor: HLA-B*35:01 probable negative. Tea in drink is OK (suggestion, not applied).
thermogenic green tea (extract)	Metabolism	PNPLA3/Pi*MZ (liver) : unfavourable COMT : Val/Val (high) HLA-B*35:01 : probable negative UGT1A1 : IM (*1/*28)	avoid (liver)	Concentrated green tea extract is a known cause of liver injury (LiverTox, probability B; risk in fasting and > 800 mg EGCG/day). Genetic liver (PNPLA3, MBOAT7, Pi*MZ); EGCG is catechol (COMT Val/Val). In favor: HLA-B*35:01 probable negative. Tea in drink is OK (suggestion, not applied).
red rice yeast (monacoline K)	Metabolism	SLCO1B1 : increased function PNPLA3/Pi*MZ (liver) : unfavourable CYP3A5 :expressor (*1/*3)	avoid (unregulated statin)	Monacholine K = lovastatin with non-standardized dose and possible citrinine. SLCO1B1 *1/*14 is favourable, but it already takes pitavastatin (statin duplication) and there are genetic livers and cases of hepatitis (suggestion, not applied).

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
multi-ingredient thermogenic	Metabolism	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Multi-ingredient thermogenic formulas were associated with acute hepatitis (LiverTox); genetic liver.
Usnic acid (fat burner)	Metabolism	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).

Adjust dose, shape or time (66)

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
acerola (natural vitamin C)	Vitamins and forms	HFE: H63D het G6PD: normal SLC23A1/2 (vit. C): typical	drinking between meals (hour)	HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).
alpha-GPC (glycerophosphocholine)	Vitamins and forms	PEMT: V175M het FTO3: typical	prefer citicoline or phosphatidylcholine (form)	PEMT V175M het: a slightly smaller endogenous choline synthesis (rational to choline). Typical FTO3: bitartrate and alpha-GPC generate more TMA/TMAO than phosphatidylcholine/citicoline; with genetic vascular priority (9p21, CETP), preferring citicoline 250–500 mg or PC 1–2 g (suggestion, not applied).
calcium ascorbate (buffered vitamin C)	Vitamins and forms	HFE: H63D het G6PD: normal SLC23A1/2 (vit. C): typical	drinking between meals (hour)	HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).
sodium ascorbate	Vitamins and forms	HFE: H63D het G6PD: normal SLC23A1/2 (vit. C): typical	drinking between meals (hour)	HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).
choline (bitartrate)	Vitamins and forms	PEMT: V175M het FTO3: typical	prefer citicoline or phosphatidylcholine (form)	PEMT V175M het: a slightly smaller endogenous choline synthesis (rational to choline). Typical FTO3: bitartrate and alpha-GPC generate more TMA/TMAO than phosphatidylcholine/citicoline; with genetic vascular priority (9p21, CETP), preferring citicoline 250–500 mg or PC 1–2 g (suggestion, not applied).
B complex	Vitamins and forms	PDXK/ALPL/NBPF3 (B6): typical/high MTHFR: 677CT/1298AC DHFR: slow reduction (CC)	choose formula (low B6, without folic acid)	Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).
B complex in gums	Vitamins and forms	PDXK/ALPL/NBPF3 (B6): typical/high MTHFR: 677CT/1298AC DHFR: slow reduction (CC)	choose formula (low B6, without folic acid)	Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).
Methylated B complex	Vitamins and forms	PDXK/ALPL/NBPF3 (B6): typical/high MTHFR: 677CT/1298AC DHFR: slow reduction (CC)	choose formula (low B6, without folic acid)	Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).
iron + vitamin C (combined)	Vitamins and forms	HFE: H63D het TMPRSS6/TF: small effect	only with confirmed deficit	HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
multivitamin 50+	Vitamins and forms	MTHFR: 677CT/1298AC DHFR: slow reduction (CC) HFE: H63D het BCO1: slightly ↓ conversion PDXK/ALPL/NBPF3 (B6): typical/high	choice formula	Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).
multivitamin in gums	Vitamins and forms	MTHFR: 677CT/1298AC DHFR: slow reduction (CC) HFE: H63D het BCO1: slightly ↓ conversion PDXK/ALPL/NBPF3 (B6): typical/high	choice formula	Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).
female multivitamin	Vitamins and forms	MTHFR: 677CT/1298AC DHFR: slow reduction (CC) HFE: H63D het BCO1: slightly ↓ conversion PDXK/ALPL/NBPF3 (B6): typical/high	choice formula	Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).
General multivitamin	Vitamins and forms	MTHFR: 677CT/1298AC DHFR: slow reduction (CC) HFE: H63D het BCO1: slightly ↓ conversion PDXK/ALPL/NBPF3 (B6): typical/high	choice formula	Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).
Multivitaminic male	Vitamins and forms	MTHFR: 677CT/1298AC DHFR: slow reduction (CC) HFE: H63D het BCO1: slightly ↓ conversion PDXK/ALPL/NBPF3 (B6): typical/high	choice formula	Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).
multivitamin prenatal	Vitamins and forms	MTHFR: 677CT/1298AC DHFR: slow reduction (CC) HFE: H63D het BCO1: slightly ↓ conversion PDXK/ALPL/NBPF3 (B6): typical/high	choice formula	Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).
ascorbyl palmitate	Vitamins and forms	HFE: H63D het G6PD: normal SLC23A1/2 (vit. C): typical	drinking between meals (hour)	HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).
★ pyridoxal-5-phosphate (P5P)	Vitamins and forms	PDXK/ALPL/NBPF3 (B6): typical/high	lowering dose (≤ 5 mg/day)	Current stack (15 mg) — plan S6: ADJUST (descend). ≤ 5 mg/day or stop (above the upper limit of ingestion, without genetic indication). PDXK rs2010795 G/G (high activation) and ALPL/NBPF3 heterozygotes: no genetic need for extra B6. Chronic use of 15 mg passes the upper limit of ingestion (12 mg) and is associated with neuropathy. ≤ 5 mg/day or pause (suggestion, not applied).

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
vitamin B6 (pyridoxine hydrochloride)	Vitamins and forms	PDXK/ALPL/NBPF3 (B6): typical/high	lowering dose (≤ 5 mg/day)	PDXK rs2010795 G/G (high activation) and ALPL/NBPF3 heterozygotes: no genetic need for extra B6. Chronic use of 15 mg passes the upper limit of ingestion (12 mg) and is associated with neuropathy. ≤ 5 mg/day or pause (suggestion, not applied).
vitamin C (ascorbic acid)	Vitamins and forms	HFE: H63D het G6PD: normal SLC23A1/2 (vit. C): typical	drinking between meals (hour)	HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).
vitamin C + zinc	Vitamins and forms	HFE: H63D het G6PD: normal SLC23A1/2 (vit. C): typical	drinking between meals (hour)	HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).
vitamin C with metabolites (ascorbate + threonate)	Vitamins and forms	HFE: H63D het G6PD: normal SLC23A1/2 (vit. C): typical	drinking between meals (hour)	HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).
★ vitamin C liposomal	Vitamins and forms	HFE: H63D het G6PD: normal SLC23A1/2 (vit. C): typical	drinking between meals (hour)	Current stack (500 mg) — plan S16: ADJUST (hour). Always ≥ 2 h away from meals. HFE H63D het: vitamin C with food increases the absorption of non-heme iron. 500 mg always ≥ 2 h away from meals. Normal G6PD; Typical SLC23A1/2 (suggestion, not applied).
folic acid (synthetic)	Vitamins and forms	MTHFR: 677CT/1298AC DHFR: slow reduction (CC)	prefer 5-MTHF (form)	MTHFR 677CT + 1298AC (activity $\sim 50\text{--}60\%$) and DHFR rs408626 C/C (slow reduction of folic acid; may leave folic acid unmetabolized in circulation). Exchange by 5-MTHF or food folate; no product with folic acid (suggestion, not applied).
folic acid + B12 (combined)	Vitamins and forms	MTHFR: 677CT/1298AC DHFR: slow reduction (CC)	prefer 5-MTHF (form)	MTHFR 677CT + 1298AC (activity $\sim 50\text{--}60\%$) and DHFR rs408626 C/C (slow reduction of folic acid; may leave folic acid unmetabolized in circulation). Exchange by 5-MTHF or food folate; no product with folic acid (suggestion, not applied).
ZMA (zinc, magnesium, B6)	Minerals and trace elements	PDXK/ALPL/NBPF3 (B6): typical/high MTHFR: 677CT/1298AC DHFR: slow reduction (CC)	choose formula (low B6, without folic acid)	Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).
iron bisglycinate	Minerals and trace elements	HFE: H63D het TMPRSS6/TF: small effect	only with confirmed deficit	HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).
iron-polysaccharide complex	Minerals and trace elements	HFE: H63D het TMPRSS6/TF: small effect	only with confirmed deficit	HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).
iron carbonyl	Minerals and trace elements	HFE: H63D het TMPRSS6/TF: small effect	only with confirmed deficit	HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).
iron heme (polypeptide)	Minerals and trace elements	HFE: H63D het TMPRSS6/TF: small effect	only with confirmed deficit	HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
liposomal / sucrosomal iron	Minerals and trace elements	HFE: H63D het TMPRSS6/TF: small effect	only with confirmed deficit	HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).
ferrous fumarate	Minerals and trace elements	HFE: H63D het TMPRSS6/TF: small effect	only with confirmed deficit	HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).
ferrous gluconate	Minerals and trace elements	HFE: H63D het TMPRSS6/TF: small effect	only with confirmed deficit	HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).
★ trace minerals (multineural) ⊕	Minerals and trace elements	SLC30A8: C/C (usual zinc) GPX1: Pro198Leu het SLC39A8: normal AOC1 (DAO): slightly ↓	up zinc and copper	Current stack (Zn 11, Cu 0.9, I 0.05, Se 0.055, Mn 2.3, Cr 0.035, Mo 0.045, B 1 mg) — plan S12: ADJUST (Zn and Cu). Total zinc 20–25 mg + total copper 1.5–2 mg; remaining equal. SLC30A8 C/C + rs11558471 A/A (genetically useful zinc); GPX1 Pro198Leu (selenium already suitable, without Brazil nuts); SLC39A8 C/C (manganese without adjustment); DAO is a copper enzyme (AOC1 het). Total zinc 20–25 mg + total copper 1.5–2 mg, ≥ 2 h away from NAC, Ca-AMG and fiber (suggestion, not applied).
ferrous sulfate	Minerals and trace elements	HFE: H63D het TMPRSS6/TF: small effect	only with confirmed deficit	HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).
★ TMG (trimethylglycine, betaine anhydrous) ⊕	Amino acids	BHMT: R239Q het GNMT: TT MTHFR: 677CT/1298AC	adjust dose (plan)	Current stack (1500 mg) — plan S7: ADJUST (subir). Up to 3 g/day in 2 tomas; do not exceed 3 g (LDL). BHMT R239Q het, GNMT TT and MTHFR compound: the BHMT pathway compensates for folate. 3 g/day (≥ 4 g up to LDL) (suggestion, not applied).
★ glycine ⊕	Amino acids	CPS1: low glycine GSTM1: probable null GNMT: TT	adjust dose (plan)	Current stack (3 g) — plan S8: ADJUST (subir). Up to 5 g/day (3 g at bedtime + 2 g in the morning). CPS1 rs715 T/T + rs1047891 C/C: genetically lower plasma glycine; Probable null GSTM1 (glycine is a glutathione substrate); GNMT TT. 5 g/day in total (suggestion, not applied).
★ fish oil (EPA+DHA, TG/rTG) ⊕	Fatty acids	FADS1/FADS2: high activity APOE: ε3/ε3	adjust dose (plan)	Current stack (EPA 1125 + DHA 875 mg) — plan S1: ADJUST. Up to 3.2–4 g/day of EPA+DHA (Coordination Note), form rTG, with the meal fatter and far from psyllium. FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied).
tea catechins (drink/matcha)	Polyphenols and other phytochemicals	CYP1A2: normal to intermediate ADORA2A: C/C COMT: Val/Val (high)	count caffeine	Tea/matcha in drink does not have the liver risk of the extract; counts for the caffeine ceiling (≤ 200 mg/day, only in the morning; Normal to intermediate CYP1A2).

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
★ fisetin	Polyphenols and other phytochemicals	COMT: Val/Val (high) UGT1A1: IM (*1/*28) CYP3A5:expressor (*1/*3)	isolated monthly pulse	Current stack (2.5 g (3×/month)) — plan S29: ADJUST. Monthly pulse (2 days), ≥ 7 days away from rapamycin and FOXO4-DRI; current protocol: with quercetin, only with normal ALT. Catechol (COMT Val/Val + MTHFR) and inhibitor of CYP3A4/P-gp and UGT1A1. 9p21 homozygous (CDKN2A/B region, senescence) gives senolytic rationale. Genetic reading: 1 pulse/month (2 days), ≥ 7 days away from rapamycin and FOXO4-DRI (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.
★ lycopene	Carotenoids and pigments	BCO1: slightly ↓ conversion	lower dose (≤ 10 mg)	Current stack (35 mg) — plan S23: ADJUST (descend). 10 mg or stop. Genetic levels of lycopene tendentially high: no genetic indication for 35 mg. 10 mg or tomato cooked with olive oil (suggestion, not applied).
green coffee (extract)	Plants of the Americas and Africa; superfood	CYP1A2: normal to intermediate ADORA2A: C/C COMT: Val/Val (high) MTHFR: 677CT/1298AC	caffeine + chlorogenic acid	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied). The concentrated chlorogenic acid can rise to homocysteine (COMT/MTHFR).
yerba mate	Plants of the Americas and Africa; superfood	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).
guarana	Plants of the Americas and Africa; superfood	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).
guayusa	Plants of the Americas and Africa; superfood	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).
walnut	Plants of the Americas and Africa; superfood	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).
yaupon	Plants of the Americas and Africa; superfood	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).
★ PHGG (partially hydrolyzed guar gum) ☺	Probiotics	TCF7L2/FTO/GCKR: unfavourable CETP/9p21/APOB: genetic vascular risk FUT2: secretor	adjust dose (plan)	Current stack (6 g) — plan S35: ADJUST (subir). 10 g/day. TCF7L2/FTO/MTNR1B/GCKR (postprandial glycosis) and CETP/9p21/APOB (low to LDL). More fiber is always better.
★ psyllium (ispaghula bark) ☺	Probiotics	TCF7L2/FTO/GCKR: unfavourable CETP/9p21/APOB: genetic vascular risk FUT2: secretor	adjust dose (plan)	Current stack (10 g) — plan S35: ADJUST (subir). 15–20 g/day in 2–3 takes with water; drugs ≥ 1 h before or 2–4 h after. TCF7L2/FTO/MTNR1B/GCKR (postprandial glycosis) and CETP/9p21/APOB (low to LDL). More fiber is always better.

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
caffeine + L-theanine	Nootropics	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).
anhydrous caffeine	Nootropics	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).
prolonged release caffeine	Nootropics	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).
magnesium + B6 (cognition/stress)	Nootropics	PDXK/ALPL/NBPF3 (B6): typical/high MTHFR: 677CT/1298AC DHFR: slow reduction (CC)	choose formula (low B6, without folic acid)	Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).
melatonin (immediate release)	Nootropics	MTNR1B: C/G (carrier) CYP1A2: normal to intermediate	low dose (0.3–1 mg)	MTNR1B rs10830963 C/G: in carrier G, melatonin reduces insulin secretion (5 mg worsened glucose tolerance in carriers); Garaulet 2015). Normal CYP1A2 extends it (apigenin, with suggested pause, would add up). 0.3–1 mg, 30–60 min before bedtime and ≥ 3 h after the last meal (suggestion, not applied).
prolonged release melatonin	Nootropics	MTNR1B: C/G (carrier) CYP1A2: normal to intermediate	low dose (0.3–1 mg)	MTNR1B rs10830963 C/G: in carrier G, melatonin reduces insulin secretion (5 mg worsened glucose tolerance in carriers); Garaulet 2015). Normal CYP1A2 extends it (apigenin, with suggested pause, would add up). 0.3–1 mg, 30–60 min before bedtime and ≥ 3 h after the last meal (suggestion, not applied).
★ melatonin high dose (≥ 10 mg)	Nootropics	MTNR1B: C/G (carrier) CYP1A2: normal to intermediate	low (0.3–1 mg)	Current stack (up to 100 mg) — plan S30: ADJUST (desert much). 0.3–1 mg, 30–60 min before bedtime. MTNR1B rs10830963 C/G: in carrier G, melatonin reduces insulin secretion (5 mg worsened glucose tolerance in carriers); Garaulet 2015). Normal CYP1A2 extends it (apigenin, with suggested pause, would add up). 0.3–1 mg, 30–60 min before bedtime and ≥ 3 h after the last meal (suggestion, not applied). Doses ≥ 10 mg (up to 100 mg) have no genetic basis and prolong the effect on insulin.
melatonin in spray/sublingual	Nootropics	MTNR1B: C/G (carrier) CYP1A2: normal to intermediate	low dose (0.3–1 mg)	MTNR1B rs10830963 C/G: in carrier G, melatonin reduces insulin secretion (5 mg worsened glucose tolerance in carriers); Garaulet 2015). Normal CYP1A2 extends it (apigenin, with suggested pause, would add up). 0.3–1 mg, 30–60 min before bedtime and ≥ 3 h after the last meal (suggestion, not applied).
paraxanthine	Nootropics	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).
tryptophan + B6 (combined)	Nootropics	PDXK/ALPL/NBPF3 (B6): typical/high MTHFR: 677CT/1298AC DHFR: slow reduction (CC)	choose formula (low B6, without folic acid)	Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
electrolytes with caffeine	Sport and performance	CYP1A2 : normal to intermediate ADORA2A : C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).
energy gel with caffeine	Sport and performance	CYP1A2 : normal to intermediate ADORA2A : C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).
pre-training with caffeine	Sport and performance	CYP1A2 : normal to intermediate ADORA2A : C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).
tauro-caffeine (energy drink powder)	Sport and performance	CYP1A2 : normal to intermediate ADORA2A : C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).
energy formula (B + caffeine)	Others	PDXK/ALPL/NBPF3 (B6) : typical/high MTHFR : 677CT/1298AC DHFR : slow reduction (CC)	choose formula (low B6, without folic acid)	Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).
homocysteine formula with folic acid	Others	MTHFR : 677CT/1298AC DHFR : slow reduction (CC)	prefer 5-MTHF (form)	MTHFR 677CT + 1298AC (activity ~50–60%) and DHFR rs408626 C/C (slow reduction of folic acid; may leave folic acid unmetabolized in circulation). Exchange by 5-MTHF or food folate; no product with folic acid (suggestion, not applied).
methylation formula (B12 + folate + B6 + TMG)	Others	PDXK/ALPL/NBPF3 (B6) : typical/high MTHFR : 677CT/1298AC DHFR : slow reduction (CC)	choose formula (low B6, without folic acid)	Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).
lysine + vitamin C (Pauling protocol)	Others	HFE : H63D het G6PD : normal SLC23A1/2 (vit. C) : typical	drinking between meals (hour)	HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).

Monitor / confirm before using (111)

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
formula AREDS2 (eyes)	Vitamins and forms	CFH/ARMS2 : CFH het SLC30A8 : C/C (usual zinc)	without indication; zinc × CFH controversy	CFH Y402H het and ARMS2 G/G.
potassium bicarbonate	Minerals and trace elements	Sal (AGT/ADD1/GNB3/CYP11B2) : sensitive (modest)	potassium only for food	Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control.
potassium citrate	Minerals and trace elements	Sal (AGT/ADD1/GNB3/CYP11B2) : sensitive (modest)	potassium only for food	Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control.

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
potassium chloride	Minerals and trace elements	Sal (AGT/ADD1/GNB3/CYP11B2): sensitive (modest)	potassium only for food	Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control.
salt tablets (sodium chloride)	Minerals and trace elements	Sal (AGT/ADD1/GNB3/CYP11B2): sensitive (modest)	avoid routine use	Genetic sensitivity to salt: salt tablets only for intense and prolonged sweat.
potassium gluconate	Minerals and trace elements	Sal (AGT/ADD1/GNB3/CYP11B2): sensitive (modest)	potassium only for food	Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control.
salt with potassium chloride (salt substitute)	Minerals and trace elements	Sal (AGT/ADD1/GNB3/CYP11B2): sensitive (modest)	potassium only for food	Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control.
BCAA (leucine, isoleucine, valine)	Amino acids	TCF7L2/FTO/GCKR: unfavourable	the genetic plan advises to add	High BCAA are associated with insulin resistance;
L-isoleucine	Amino acids	TCF7L2/FTO/GCKR: unfavourable	the genetic plan advises to add	High BCAA are associated with insulin resistance;
L-leucine	Amino acids	TCF7L2/FTO/GCKR: unfavourable	the genetic plan advises to add	High BCAA are associated with insulin resistance;
L-methionine	Amino acids	COMT: Val/Val (high) GNMT: TT MTHFR: 677CT/1298AC	avoiding isolated supplement	Extra methionine turns SAM and then homocysteine;
L-valine	Amino acids	TCF7L2/FTO/GCKR: unfavourable	the genetic plan advises to add	High BCAA are associated with insulin resistance;
SAMe (S-adenosylmethionine)	Amino acids	COMT: Val/Val (high) GNMT: TT MTHFR: 677CT/1298AC	the genetic plan advises to add	COMT Val/Val + GNMT TT: plus SAM tends to be transformed into homocysteine (genetic score p92).
CLA (conjugated linoleic acid)	Fatty acids	PNPLA3/Pi*MZ (liver): unfavourable TCF7L2/FTO/GCKR: unfavourable	without benefit; possible metabolic worsening	CLA t10,c12 increased insulin resistance and hepatic fat in trials.
Sewer oil (GLA)	Fatty acids	FADS1/FADS2: high activity PNPLA3/Pi*MZ (liver): unfavourable	GLA→AA + alkaloids	High FADS1 (GLA→AA) and possible presence of hepatotoxic pyrrolizidine alkaloids (genetic liver): only certified products without PA.
coconut oil	Fatty acids	CETP/9p21/APOB: genetic vascular risk	saturated fat (LDL)	Genetic priority to low LDL/ApoB (high CETP, 9p21, APOB): coconut oil rises to LDL;
safflower oil	Fatty acids	FADS1/FADS2: high activity	Omega-6 unnecessary	The extra LA is converted effectively to AA (high FADS);
shark liver oil (alkylglycerols)	Fatty acids	BCO1: slightly ↓ conversion PNPLA3/Pi*MZ (liver): unfavourable	confirm retinol content	It can contain vitamin A preformed in variable quantity;
Grapeseed oil	Fatty acids	FADS1/FADS2: high activity	Omega-6 unnecessary	The extra LA is converted effectively to AA (high FADS);
Blackcurrant oil (GLA)	Fatty acids	FADS1/FADS2: high activity	GLA can turn AA	With high activity FADS1, part of GLA→DGLA goes to arachidonic acid (pro-inflammatory), reducing the benefit.

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
Primule oil (GLA)	Fatty acids	FADS1/FADS2 : high activity	GLA can turn AA	With high activity FADS1, part of GLA→DGLA goes to arachidonic acid (pro-inflammatory), reducing the benefit.
Omega-3-6-9	Fatty acids	FADS1/FADS2 : high activity	Omega-6 unnecessary	The extra LA is converted effectively to AA (high FADS);
(–)-epicatechin	Polyphenols and other phytochemicals	COMT : Val/Val (high)	catechol	Epicatechin isolated at high dose is methylated by COMT (Val/Val);
allicin (stabilized garlic extract)	Polyphenols and other phytochemicals	CYP3A5 :expressor (*1/*3)	P-gp/CYP3A4 inducer	Allicin of fresh extracts induces P-gp/CYP3A4 (lowered saquinavir ~50%): it can lower sirolimus.
Saffron powder	Polyphenols and other phytochemicals	UGT1A1 : IM (*1/*28) PNPLA3/Pi*MZ (liver) : unfavourable HLA-B*35:01 : probable negative	only cooking doses	Less curcumin than extracts, but cases of liver injury include root powder.
baicalein	Polyphenols and other phytochemicals	PNPLA3/Pi*MZ (liver) : unfavourable	cases of liver injury	Scutellaria baicalensis was associated with liver injury (LiverTox);
baicalin (Scutellaria baicalensis)	Polyphenols and other phytochemicals	PNPLA3/Pi*MZ (liver) : unfavourable	cases of liver injury	Scutellaria baicalensis was associated with liver injury (LiverTox);
berberine	Polyphenols and other phytochemicals	CYP2D6 : Probable NM (without CNV) CYP2C9 : IM (*1/*3) CYP3A5 :expressor (*1/*3)	the genetic plan advises to add	It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures;
dihydroberberine	Polyphenols and other phytochemicals	CYP2D6 : Probable NM (without CNV) CYP2C9 : IM (*1/*3) CYP3A5 :expressor (*1/*3)	the genetic plan advises to add	It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures;
glabridine (liquorice)	Polyphenols and other phytochemicals	CYP3A5 :expressor (*1/*3)	inhibits CYP3A4	Glabridin inhibits CYP3A4 in vitro (liquorice).
myricetin	Polyphenols and other phytochemicals	COMT : Val/Val (high) MTHFR : 677CT/1298AC	catechol: prefer low doses	Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score).
naringin/naringenin	Polyphenols and other phytochemicals	CYP3A5 :expressor (*1/*3)	weak inhibition of CYP3A4	Naringenin contributes to the orange effect (weakly isolated).
piceatannol	Polyphenols and other phytochemicals	COMT : Val/Val (high) MTHFR : 677CT/1298AC	catechol: prefer low doses	Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score).
piperine (black pepper extract)	Polyphenols and other phytochemicals	CYP3A5 :expressor (*1/*3) UGT1A1 : IM (*1/*28)	increases drug exposure	Piperine inhibits CYP3A4, P-gp and glucuronidation: up to levels of drugs and other supplements.
polydatin	Polyphenols and other phytochemicals	UGT1A1 : IM (*1/*28)	precursor of resveratrol	Converted into resveratrol: the same logic (UGT1A1 *28, CYP3A4).
pterostilben	Polyphenols and other phytochemicals	CETP/9p21/APOB : genetic vascular risk UGT1A1 : IM (*1/*28)	can rise to LDL	In trial, pterostilbene rose to LDL (Riche 2014);
resveratrol	Polyphenols and other phytochemicals	UGT1A1 : IM (*1/*28) CYP2C9 : IM (*1/*3) CYP3A5 :expressor (*1/*3)	inhibits CYP3A4/2C9	Glucuronidated by UGT1A1 (*28: a little higher exposure).

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
rutin	Polyphenols and other phytochemicals	COMT: Val/Val (high) UGT1A1: IM (*1/*28)	catechol (via quercetin)	Rutin releases quercetin in the intestine: the same logic (COMT Val/Val, UGT1A1 *28) but lower exposure.
silibinin phytosome	Polyphenols and other phytochemicals	UGT1A1: IM (*1/*28) CYP2C9: IM (*1/*3) PNPLA3/Pi*MZ (liver): unfavourable	without proven gain; inhibits UGT1A1/CYP2C9	Silibinin is glucuronidated by UGT1A1 and inhibits it (bilirubin can rise in *28) and inhibits CYP2C9 (effect on losartan greater in *1/*1 than in *3;
silymarin (marian card)	Polyphenols and other phytochemicals	UGT1A1: IM (*1/*28) CYP2C9: IM (*1/*3) PNPLA3/Pi*MZ (liver): unfavourable	without proven gain; inhibits UGT1A1/CYP2C9	Silibinin is glucuronidated by UGT1A1 and inhibits it (bilirubin can rise in *28) and inhibits CYP2C9 (effect on losartan greater in *1/*1 than in *3;
taxifolin (dihydroquercetin)	Polyphenols and other phytochemicals	COMT: Val/Val (high) MTHFR: 677CT/1298AC	catechol: prefer low doses	Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score).
caffeic acid	Polyphenols and other phytochemicals	COMT: Val/Val (high) MTHFR: 677CT/1298AC	catechol: prefer low doses	Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score).
chlorogenic acid (green coffee)	Polyphenols and other phytochemicals	COMT: Val/Val (high) MTHFR: 677CT/1298AC	can rise homocysteine	In trial, 2 g/day chlorogenic acid increased homocysteine (Olthof 2001);
gallic acid	Polyphenols and other phytochemicals	COMT: Val/Val (high) MTHFR: 677CT/1298AC	catechol: prefer low doses	Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score).
rosmarinic acid	Polyphenols and other phytochemicals	COMT: Val/Val (high) MTHFR: 677CT/1298AC	catechol: prefer low doses	Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score).
garlic/oil	Western medicinal plants	CYP3A5: expressor (*1/*3)	P-gp/CYP3A4 inducer	Allicin of fresh extracts induces P-gp/CYP3A4 (lowered saquinavir ~50%): it can lower sirolimus.
aloe vera (oral)	Western medicinal plants	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver	Cases of liver injury described (LiverTox);
chinese angelic (quang quai)	Western medicinal plants	CYP3A5: expressor (*1/*3) TERT/melanoma: unfavourable	furanocoumarins	Furanocoumarins: inhibition of CYP3A4 and photosensitization (p88 + isotretinoin nevi).
pocket	Western medicinal plants	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver	Cases of liver injury described (LiverTox);
willow bark (salicin)	Western medicinal plants	UGT1A6: *2 het CYP2C9: IM (*1/*3)	salicylate	Salicylate is metabolized by UGT1A6 (*2 het) and CYP2C9 (*1/*3);
cimicifiga (black cohosh)	Western medicinal plants	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver	Cases of liver injury described (LiverTox);
cascara-sagrada	Western medicinal plants	PNPLA3/Pi*MZ (liver): unfavourable	only short term	Anthraquinones: chronic use associated with rare liver injury and melanotic colon;
yerba (Melissa) + valerian	Western medicinal plants	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver (valerian)	Cases of liver injury described (LiverTox);
American listener (skullcap)	Western medicinal plants	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver	Cases of liver injury described (LiverTox);
Cerval thorn (frangula)	Western medicinal plants	PNPLA3/Pi*MZ (liver): unfavourable	only short term	Anthraquinones: chronic use associated with rare liver injury and melanotic colon;

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
hydrate (goldenseal)	Western medicinal plants	CYP2D6 : Probable NM (without CNV) CYP3A5 :expressor (*1/*3)	strong inhibitor CYP3A4/2D6	Hydrastin/berberine: clinical inhibition of CYP3A4 and CYP2D6 (Gurley 2008).
mahonia (Oregon grape)	Western medicinal plants	CYP2D6 : Probable NM (without CNV) CYP2C9 : IM (*1/*3) CYP3A5 :expressor (*1/*3)	the genetic plan advises to add	It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures;
pelargonium sidoides	Western medicinal plants	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring liver	Cases of liver injury described (LiverTox);
petasitis (butterbur, without PA)	Western medicinal plants	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring liver	Even extracts without PA had cases of hepatitis (LiverTox);
angelica root (Angelica arcangelica)	Western medicinal plants	CYP3A5 :expressor (*1/*3) TERT/melanoma : unfavourable	furanocoumarins	Furanocoumarins: inhibition of CYP3A4 and photosensitization (p88 + isotretinoin nevi).
rhubarb (root)	Western medicinal plants	PNPLA3/Pi*MZ (liver) : unfavourable	only short term	Anthraquinones: chronic use associated with rare liver injury and melanotic colon;
ruta (ruda)	Western medicinal plants	CYP3A5 :expressor (*1/*3) TERT/melanoma : unfavourable	furanocoumarins	Furanocoumarins: inhibition of CYP3A4 and photosensitization (p88 + isotretinoin nevi).
sene	Western medicinal plants	PNPLA3/Pi*MZ (liver) : unfavourable	only short term	Anthraquinones: chronic use associated with rare liver injury and melanotic colon;
meadowsweet	Western medicinal plants	UGT1A6 : *2 het CYP2C9 : IM (*1/*3)	salicylate	Salicylate is metabolized by UGT1A6 (*2 het) and CYP2C9 (*1/*3);
grape (brerberis)	Western medicinal plants	CYP2D6 : Probable NM (without CNV) CYP2C9 : IM (*1/*3) CYP3A5 :expressor (*1/*3)	the genetic plan advises to add	It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures;
valerian	Western medicinal plants	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring liver	Cases of liver injury described (LiverTox);
gotu kola (centella)	Ayurvedic plants	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring liver	Cases of liver injury described (LiverTox);
guggul	Ayurvedic plants	PNPLA3/Pi*MZ (liver) : unfavourable CYP3A5 :expressor (*1/*3)	CYP3A4 + liver inducer	Guggulsterone activates PXR (induces CYP3A4: it can lower sirolimus) and there are cases of hepatitis.
mucuna pruriens (L-dopa)	Ayurvedic plants	COMT : Val/Val (high) MTHFR : 677CT/1298AC	L-dopa: COMT/ homocysteine	The L-dopa is O-methylated by COMT (Val/Val, high) spending SAM and up to homocysteine;
pippali (long pepper)	Ayurvedic plants	CYP3A5 :expressor (*1/*3) UGT1A1 : IM (*1/*28)	increases drug exposure	Piperine inhibits CYP3A4, P-gp and glucuronidation: up to levels of drugs and other supplements.
trikatu (peppers + ginger)	Ayurvedic plants	CYP3A5 :expressor (*1/*3) UGT1A1 : IM (*1/*28)	increases drug exposure	Piperine inhibits CYP3A4, P-gp and glucuronidation: up to levels of drugs and other supplements.
Sho-saiko-to / Xiao Chai Hu Tang	Traditional Chinese medicine plants and Kampo	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring liver	Cases of liver injury described (LiverTox);

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
angelica dahurica (bai zhi)	Traditional Chinese medicine plants and Kampo	CYP3A5 :expressor (*1/*3) TERT/melanoma : unfavourable	furanocoumarins	Furanocoumarins: inhibition of CYP3A4 and photosensitization (p88 + isotretinoin nevi).
bupleur	Traditional Chinese medicine plants and Kampo	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring liver	Cases of liver injury described (LiverTox);
coptis (huang lian)	Traditional Chinese medicine plants and Kampo	CYP2D6 : Probable NM (without CNV) CYP2C9 : IM (*1/*3) CYP3A5 :expressor (*1/*3)	the genetic plan advises to add	It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures;
corydalis (yan hu suo)	Traditional Chinese medicine plants and Kampo	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring liver	Cases of liver injury described (LiverTox);
Chinese mantle (Scutellaria baicalensis)	Traditional Chinese medicine plants and Kampo	PNPLA3/Pi*MZ (liver) : unfavourable	cases of liver injury	Scutellaria baicalensis was associated with liver injury (LiverTox);
schisandra (Schisandra chinensis)	Traditional Chinese medicine plants and Kampo	CYP3A5 :expressor (*1/*3) PNPLA3/Pi*MZ (liver) : unfavourable	inhibits CYP3A4/P-gp	Schisandra rises the levels of tacrolimus (used for this in China);
hu zhang (Polygonum cuspidatum, knotweed)	Traditional Chinese medicine plants and Kampo	PNPLA3/Pi*MZ (liver) : unfavourable	only short term	Anthraquinones: chronic use associated with rare liver injury and melanotic colon;
phellodendron	Traditional Chinese medicine plants and Kampo	CYP2D6 : Probable NM (without CNV) CYP2C9 : IM (*1/*3) CYP3A5 :expressor (*1/*3)	the genetic plan advises to add	It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures;
cassava seed (jue ming zi)	Traditional Chinese medicine plants and Kampo	PNPLA3/Pi*MZ (liver) : unfavourable	only short term	Anthraquinones: chronic use associated with rare liver injury and melanotic colon;
cinnamon-cassava (coumarin)	Plants of the Americas and Africa; superfood	CYP2A6 : probable normal PNPLA3/Pi*MZ (liver) : unfavourable	hepatotoxic coumarin	Cassia cinnamon has coumarin (EFSA fixed a low tolerable daily intake);
yohimbine HCl	Plants of the Americas and Africa; superfood	CYP2D6 : Probable NM (without CNV)	CYP2D6 (confirm CNV)	Yohimbine depends on CYP2D6: slow metabolizers have often higher exposure.
noni	Plants of the Americas and Africa; superfood	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring liver	Cases of liver injury described (LiverTox);
concentrated beet juice (nitrate)	Plants of the Americas and Africa; superfood	NOS3 : -786CT + 298GT	avoid concentrated nitrate (tadalafil)	NOS3 gives rational, but concentrated inorganic nitrate is added to the tadalafil (hypotension).
yohimbe (shell)	Plants of the Americas and Africa; superfood	CYP2D6 : Probable NM (without CNV)	CYP2D6 (confirm CNV)	Yohimbine depends on CYP2D6: slow metabolizers have often higher exposure.
reishi (Ganoderma lucidum)	Medicinal mushrooms	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring liver	Cases of liver injury described (LiverTox);

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
CBD (cannabidiol)	Nootropics	CYP2C19 : NM CYP2C9 : IM (*1/*3) PNPLA3/Pi*MZ (liver) : unfavourable CYP3A5 :expressor (*1/*3)	inhibits CYP; transaminase	CBD is metabolized by CYP2C19 (*1/*1, normal) and inhibits CYP2C19/2C9/3A4;
galantamine (winter bell)	Nootropics	CYP2D6 : Probable NM (without CNV)	CYP2D6 (confirm CNV)	Galantamine is a substrate of CYP2D6/3A4;
Hemp oil (total spectrum)	Nootropics	CYP2C19 : NM CYP2C9 : IM (*1/*3) PNPLA3/Pi*MZ (liver) : unfavourable CYP3A5 :expressor (*1/*3)	inhibits CYP; transaminase	CBD is metabolized by CYP2C19 (*1/*1, normal) and inhibits CYP2C19/2C9/3A4;
BCAA + electrolytes	Sport and performance	TCF7L2/FTO/GCKR : unfavourable	the genetic plan advises to add	High BCAA are associated with insulin resistance;
waxy corn starch	Sport and performance	TCF7L2/FTO/GCKR : unfavourable	prefer intact hydrates	TCF7L2/FTO/GCKR risk + 4 drugs that move on glucose: rapid absorption hydrates only during long training.
gainer (hydrates + protein)	Sport and performance	TCF7L2/FTO/GCKR : unfavourable	prefer intact hydrates	TCF7L2/FTO/GCKR risk + 4 drugs that move on glucose: rapid absorption hydrates only during long training.
leucine + HMB (combined)	Sport and performance	TCF7L2/FTO/GCKR : unfavourable	the genetic plan advises to add	High BCAA are associated with insulin resistance;
maltodextrin/cyclic dextrin	Sport and performance	TCF7L2/FTO/GCKR : unfavourable	prefer intact hydrates	TCF7L2/FTO/GCKR risk + 4 drugs that move on glucose: rapid absorption hydrates only during long training.
synephrine (Orange-yellow)	Sport and performance	CYP3A5 :expressor (*1/*3) CETP/9p21/APOB : genetic vascular risk	stimulant + CYP3A4	Yellow orange furanocoumarins inhibit CYP3A4 (grapefruit effect);
deer horn velvet (IGF-1)	Sport and performance	TERT/melanoma : unfavourable	IGF-1 axis	Sold by IGF-1: the plan avoids the GH/IGF-1 axis (risk TERT alleles).
Garcinia + green coffee (combined)	Metabolism	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring liver	Cases of liver injury described (LiverTox);
sine detox tea	Metabolism	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring liver	Cases of liver injury described (LiverTox);
chromium + cinnamon (glucose)	Metabolism	CYP2A6 : probable normal PNPLA3/Pi*MZ (liver) : unfavourable	hepatotoxic coumarin	Cassia cinnamon has coumarin (EFSA fixed a low tolerable daily intake);
garcinia cambogia (HCA)	Metabolism	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring liver	Cases of liver injury described (LiverTox);
sodium nitrate/potassium	Metabolism	NOS3 : -786CT + 298GT	avoid concentrated nitrate (tadalafil)	NOS3 gives rational, but concentrated inorganic nitrate is added to the tadalafil (hypotension).
black pepper (extract)	Metabolism	CYP3A5 :expressor (*1/*3) UGT1A1 : IM (*1/*28)	increases drug exposure	Piperine inhibits CYP3A4, P-gp and glucuronidation: up to levels of drugs and other supplements.
thermogenic (caffeine + synephrine)	Metabolism	CYP3A5 :expressor (*1/*3) CETP/9p21/APOB : genetic vascular risk	stimulant + CYP3A4	Yellow orange furanocoumarins inhibit CYP3A4 (grapefruit effect);

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
extract of fava (Vicia faba, L-dopa natural)	Others	G6PD : normal COMT : Val/Val (high) MTHFR : 677CT/1298AC	Normal G6PD; L-dopa (COMT)	Normal G6PD (probable favism);
anti-stress formula (adaptogen)	Others	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring liver	Cases of liver injury described (LiverTox);
hepatic support formula (detox)	Others	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring liver	Cases of liver injury described (LiverTox);
hangover formula (DHM, dihydromyricetin)	Others	COMT : Val/Val (high) MTHFR : 677CT/1298AC	catechol: prefer low doses	Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score).
hanging formula with NAC + cardo	Others	UGT1A1 : IM (*1/*28) CYP2C9 : IM (*1/*3) PNPLA3/Pi*MZ (liver) : unfavourable	without proven gain; inhibits UGT1A1/CYP2C9	Silibinin is glucuronidated by UGT1A1 and inhibits it (bilirubin can rise in *28) and inhibits CYP2C9 (effect on losartan greater in *1/*1 than in *3;
Sleep formula (mixture of herbs)	Others	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring liver (valerian)	Cases of liver injury described (LiverTox);
nutritional yeast	Others	MTHFR : 677CT/1298AC DHFR : slow reduction (CC)	check fortification	Many brands are fortified with synthetic folic acid (DHFR CC): choosing not fortified or with 5-MTHF.

Undetermined (11)

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
★ L-carnosine	Amino acids	CNDP1 : indeterminate	indeterminate (carnosinase)	Current stack (2 g) — plan S21: KEEP. 1 g × 2 (non-evaluable carnosinase CNDP1). Serum carnosinase (CNDP1) is regulated by a repetition (CTG) _n that the chip does not read: the fraction of oral carnosine that arrives intact is not predictable. The plan maintains 2 g (1 g × 2).
anserine	Amino acids	CNDP1 : indeterminate	indeterminate (carnosinase)	Serum carnosinase (CNDP1) is regulated by a repetition (CTG) _n that the chip does not read: the fraction of oral carnosine that arrives intact is not predictable. The plan maintains 2 g (1 g × 2).
licorice (root)	Western medicinal plants	HSD11B2 : indeterminate	non-valuable individual sensitivity	Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).
Hochuekkito (Kampo)	Traditional Chinese medicine plants and Kampo	HSD11B2 : indeterminate	non-valuable individual sensitivity	Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).
Juzentaiho (Kampo)	Traditional Chinese medicine plants and Kampo	HSD11B2 : indeterminate	non-valuable individual sensitivity	Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).
Kakkonto (Kampo)	Traditional Chinese medicine plants and Kampo	HSD11B2 : indeterminate	non-valuable individual sensitivity	Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).
Shakuyakukanzoto (Kampo)	Traditional Chinese medicine plants and Kampo	HSD11B2 : indeterminate	non-valuable individual sensitivity	Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).

Supplement	Category	Genes (state)	Predicted response	What to do (summer)
Yokukansan (Kampo)	Traditional Chinese medicine plants and Kampo	HSD11B2 : indeterminate	non-valuable individual sensitivity	Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).
amylase	Enzymes	AMY1 : indeterminate	non-valuable AMY1 copies	The number of copies of AMY1 (salivary amylase) is not visible in the chip.
nicotine (chilets/gums)	Nootropics	CYP2A6 : probable normal	debugging not fully evaluated	Nicotine is eliminated by CYP2A6 (*2/*9 absent; non-visible *4 deletions): non-determinable clearance rate. Dependence; no indication.
carnosine + ALCAR (combined)	Longevity	CNDP1 : indeterminate	indeterminate (carnosinase)	Serum carnosinase (CNDP1) is regulated by a repetition (CTG)n that the chip does not read: the fraction of oral carnosine that arrives intact is not predictable. The plan maintains 2 g (1 g × 2).

Nutrigenetic card (summary to take to the consultation)

Axis	Genotype	Indication (suggestion, not applied)
Liver × hepatotoxic plants	PNPLA3 I148M C/G; MBOAT7 C/T; SERPINA1 Pi* ^{MZ} probable; HSD17B13 protector; HLA-B*35:probable negative 01	Avoid green tea extract, kava, ashwagandha, red rice yeast, pyrrolizidine alkaloids (consolidation, tussilage), He Shou Wu, germander, celandine, chaparral, usnic acid, kratom, Tinospora, Psoralea and multi-ingredient fat burners. Plants with sporadic cases (black cohosh, valerian, garcinia, oral aloe, noni, gotu kola, reishi, CBD): only short term.
Gilbert × polyphenols	UGT1A1 *1/*28	Suggestion (not applied): pause curcumin, quercetin, luteolin and apigenin (test-pause), because they inhibit UGT1A1 (concentrate bilirubin and liver) and CYP3A4/P-gp (sirolimus); pause while ALT is high (current protocol); curcumin until ALT normalizes. Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed. Resveratrol and silymarin: monitoring.
Methylation	MTHFR 677CT/1298AC DHFR CC; BHMT R239Q; GNMT TT; COMT Val/Val; CPS1 (low glycine)	5-MTHF instead of folic acid (multivitamins, complexes B, fortified yeasts). TMG ≤ 3 g, glycine 5 g, creatine. Avoid SAME, methionine and concentrated catechols (luteolin, quercetin, EGCG, chlorogenic acid, L-dopa). B6 ≤ 5 mg.
Iron	HFE H63D het (C282Y absent)	No iron or liver dried by routine; ferritin target below 100; vitamin C ≥ 2 h away from meals; calcium at meals.
Caffeine and melatonin	CYP1A2 *1F het + rs2472297 C/C; ADORA2A C/C; MTNR1B C/G	Total caffeine ≤ 200 mg/day to 12h (including guarana, mate, glue, green coffee, pre-training). Melatonin 0.3–1 mg, ≥ 3 h after the last meal. The suggested apigenin (pause) would prolong both (inhibit CYP1A2).
Telomerase × cancer	5 TERT risk alleles; p88 nevi	Avoid purified extracts of astragalus, cycloastragenol, astragaloside IV and astragalus; photoprotection (St John's wort, psoralens and angelica are photosensitizers; Oral polypodium can help).
Sirolimus × CYP3A4/P-gp	CYP3A5 *1/*3 (expressor); CYP3A4 *1/*1	Avoid St John's wort (strong inducer). Schistandra, berberine/hydraste, piperine, allicin, guggul, resveratrol, CBD, blackberry orange and furanocoumarin never in the rapamycin window.
Omega 3	FADS1/2 of high activity; APOE ε3/ε3	EPA/DHA useful preformed; total 3.2–4 g/day (Coordination Note). GLA (primary, blurring) and extra Omega-6 without advantage.
Zinc	SLC30A8 C/C + rs11558471 A/A	Total zinc 20–25 mg + copper 1.5–2 mg.
Phase II / NRF2	Probable null GSTM1; GSTA1 *B/*B; NRF2 –617 C/C; NQO1 *1/*1	Sulforaphane and NAC coherent; CoQ10 without genetic indication.
Hemolysis / lactose / gluten	Normal G6PD; Persistent LCT; without HLA-DQ2/DQ8	High-dose vitamin C, methylene blue, quinine, yellow melon and fava without genetic restriction of hemolysis; lactose and gluten tolerated.

Current x genetic stack

Supplement (current dose)	Genes (state)	Interpretation	Plan	Comment
★ solid lipid curcumin (SLCP) 1 g	UGT1A1: IM (*1/*28) PNPLA3/Pi*MZ (liver): unfavourable HLA-B*35:01: probable negative CYP3A5:expressor (*1/*3)	avoid/pause	S26: PAUSE	Pause (current protocol). UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).
★ luteolin 100 mg	COMT: Val/Val (high) MTHFR: 677CT/1298AC UGT1A1: IM (*1/*28) CYP2C9: IM (*1/*3)	avoid/pause	S27: PAUSE	Pause (current protocol). Catechol: methylated by COMT (Val/Val), spent SAM and produces homocysteine (MTHFR compound, p92 score). It inhibits UGT1A1 (*28) and CYP2C9 (*1/*3 IM) in vitro, and CYP3A4/P-gp (sirolimus). Pause; pause while ALT is high (current protocol) (suggestion, not applied).
★ apigenin 100 mg	UGT1A1: IM (*1/*28) CYP3A5:expressor (*1/*3) CYP1A2: normal to intermediate CYP2C9: IM (*1/*3)	avoid / pause (test- pause)	S28: PAUSE (test- pause)	Test-pause (current protocol). Test pause for the same reasons of luteolin: inhibits UGT1A1 (*1/*28: up to bilirubin and confuses liver reading) and CYP3A4/P-gp (sirolimus; CYP3A5 expressor). It is not catechol, but also inhibits CYP1A2 (already normal to intermediate: *1F het + rs2472297 C/C), which rises melatonin and caffeine, and CYP2C9 (*1/*3). Pause; pause while ALT is high (current protocol). Coordination note: in the current protocol, the pause lasts while ALT is high; if it returns, 25-50 mg at night, outside the rapamycin window (suggestion, not applied).
★ quercetin 250 mg (3×/month)	UGT1A1: IM (*1/*28) COMT: Val/Val (high) MTHFR: 677CT/1298AC CYP3A5:expressor (*1/*3)	avoid/pause	S29: PAUSE	It is suggested to pause as an isolated supplement (genetic verdict); in the current protocol enters the monthly pulse with fisetin, only with normal ALT. UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.
★ fish oil (EPA+DHA, TG/rTG) ⊕ EPA 1125 + DHA 875 mg	FADS1/FADS2: high activity APOE: ε3/ε3	adjust dose (plan)	S1: ADJUST	Up to 3.2-4 g/day of EPA+DHA (Coordination Note), form rTG, with the meal fatter and far from psyllium. FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2-4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied).
★ trace minerals (multineural) ⊕ Zn 11, Cu 0,9, I 0,05, Se 0,055, Mn 2,3, Cr 0,035, Mo 0,045, B 1 mg	SLC30A8: C/C (usual zinc) GPX1: Pro198Leu het SLC39A8: normal AOC1 (DAO): slightly ↓	up zinc and copper	S12: ADJUST (Zn and Cu)	Total zinc 20-25 mg + total copper 1.5-2 mg; remaining equal. SLC30A8 C/C + rs11558471 A/A (genetically useful zinc); GPX1 Pro198Leu (selenium already suitable, without Brazil nuts); SLC39A8 C/C (manganese without adjustment); DAO is a copper enzyme (AOC1 het). Total zinc 20-25 mg + total copper 1.5-2 mg, ≥ 2 h away from NAC, Ca-AGK and fiber (suggestion, not applied).
★ vitamin C liposomal 500 mg	HFE: H63D het G6PD: normal SLC23A1/2 (vit. C): typical	drinking between meals (hour)	S16: ADJUST (hour)	Always ≥ 2 h away from meals. HFE H63D het: vitamin C with food increases the absorption of non-heme iron. 500 mg always ≥ 2 h away from meals. Normal G6PD; Typical SLC23A1/2 (suggestion, not applied).
★ lycopene 35 mg	BCO1: slightly ↓ conversion	lower dose (≤ 10 mg)	S23: ADJUST (descend)	10 mg or stop. Genetic levels of lycopene tendentially high: no genetic indication for 35 mg. 10 mg or tomato cooked with olive oil (suggestion, not applied).

Supplement (current dose)	Genes (state)	Interpretation	Plan	Comment
★ fisetin 2.5 g (3×/month)	COMT: Val/Val (high) UGT1A1: IM (*1/*28) CYP3A5:expressor (*1/*3)	isolated monthly pulse	S29: ADJUST	Monthly pulse (2 days), ≥ 7 days away from rapamycin and FOXO4-DRI; current protocol: with quercetin, only with normal ALT. Catechol (COMT Val/Val + MTHFR) and inhibitor of CYP3A4/P-gp and UGT1A1. 9p21 homozygous (CDKN2A/B region, senescence) gives senolytic rationale. Genetic reading: 1 pulse/month (2 days), ≥ 7 days away from rapamycin and FOXO4-DRI (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.
★ melatonin high dose (≥ 10 mg) up to 100 mg	MTNR1B: C/G (carrier) CYP1A2: normal to intermediate	low (0.3–1 mg)	S30: ADJUST (descent)	0.3–1 mg, 30–60 min before bedtime. MTNR1B rs10830963 C/G: in carrier G, melatonin reduces insulin secretion (5 mg worsened glucose tolerance in carriers; Garaulet 2015). Normal CYP1A2 extends it (apigenin, with suggested pause, would add up). 0.3–1 mg, 30–60 min before bedtime and ≥ 3 h after the last meal (suggestion, not applied). Doses ≥ 10 mg (up to 100 mg) have no genetic basis and prolong the effect on insulin.
★ PHGG (partially hydrolyzed guar gum) ⊕ 6 g	TCF7L2/FTO/GCKR: unfavourable CETP/9p21/APOB: genetic vascular risk FUT2: secretor	adjust dose (plan)	S35: ADJUST (subir)	10 g/day. TCF7L2/FTO/MTNR1B/GCKR (postprandial glycosis) and CETP/9p21/APOB (low to LDL). More fiber is always better.
★ psyllium (ispaghula bark) ⊕ 10 g	TCF7L2/FTO/GCKR: unfavourable CETP/9p21/APOB: genetic vascular risk FUT2: secretor	adjust dose (plan)	S35: ADJUST (subir)	15–20 g/day in 2–3 takes with water; drugs ≥ 1 h before or 2–4 h after. TCF7L2/FTO/MTNR1B/GCKR (postprandial glycosis) and CETP/9p21/APOB (low to LDL). More fiber is always better.
★ pyridoxal-5-phosphate (P5P) 15 mg	PDXK/ALPL/NBPF3 (B6): typical/high	lowering dose (≤ 5 mg/day)	S6: ADJUST (descend)	≤ 5 mg/day or stop (above the upper limit of ingestion, without genetic indication). PDXK rs2010795 G/G (high activation) and ALPL/NBPF3 heterozygotes: no genetic need for extra B6. Chronic use of 15 mg passes the upper limit of ingestion (12 mg) and is associated with neuropathy. ≤ 5 mg/day or pause (suggestion, not applied).
★ TMG (trimethylglycine, betaine anhydrous) ⊕ 1500 mg	BHMT: R239Q het GNMT: TT MTHFR: 677CT/1298AC	adjust dose (plan)	S7: ADJUST (subir)	Up to 3 g/day in 2 tomas; do not exceed 3 g (LDL). BHMT R239Q het, GNMT TT and MTHFR compound: the BHMT pathway compensates for folate. 3 g/day (≥ 4 g up to LDL) (suggestion, not applied).
★ glycine ⊕ 3 g	CPS1: low glycine GSTM1: probable null GNMT: TT	adjust dose (plan)	S8: ADJUST (subir)	Up to 5 g/day (3 g at bedtime + 2 g in the morning). CPS1 rs715 T/T + rs1047891 C/C: genetically lower plasma glycine; Probable null GSTM1 (glycine is a glutathione substrate); GNMT TT. 5 g/day in total (suggestion, not applied).
★ L-carnosine 2 g	CNDP1: indeterminate	indeterminate (carnosinase)	S21: KEEP	1 g × 2 (non-valuable carnosinase CNDP1). Serum carnosinase (CNDP1) is regulated by a repetition (CTG) _n that the chip does not read: the fraction of oral carnosine that arrives intact is not predictable. The plan maintains 2 g (1 g × 2).
★ creatine monohydrate ⊕ 5 g	AMPD1: normal ACTN3: R/X GNMT: TT	useful (muscle + save SAM)	S10: KEEP	The same. Normal AMPD1; ACTN3 R/X; high fragility genetic score. Exogenous creatine saves SAM spent on endogenous synthesis (GAMT), useful with compound MTHFR/GNMT TT. Monohydrate 3–5 g is the form with evidence.
★ magnesium acetyl-taurate 750 mg	TRPM6: typical	typical absorption	S13: KEEP	The same. TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.

Supplement (current dose)	Genes (state)	Interpretation	Plan	Comment
★ magnesium glycinate 500 mg	TRPM6: typical	typical absorption	S13: KEEP	The same. TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.
★ lithium orotate (low dose) 5 mg	APOE: ε3/ε3	without a safety gene	S15: KEEP	Similarly; pause in days of vomiting/diarrhea (telmisartan/dapagliflozin). No PGx for lithium in microdose; APOE ε3/ε3; Alzheimer's PGS of 39 variants p91 (genomic p52 to p85): weak rational. With telmisartan (sodium lithium) and dapagliflozin (low-o): pause in days of vomiting/diarrhea.
★ NAC (N-acetylcysteine) ☼ 1800 mg	GSTM1: probable null GSTA1: *B/*B GPX1: Pro198Leu het	useful support for glutathione	S17: KEEP	1200–1800 mg, far from training. Probable null GSTM1, GSTA1 *B/*B, GCLM –588 C/T and GPX1 Pro198Leu: weak II → NAC phase is coherent. Far from the training; ≥ 2 h away from zinc/copper.
★ acetyl-L-carnitine (ALCAR) 500 mg	SLC22A4/5 (OCTN1/2): typical FMO3: typical	Typical transport	S19: KEEP	500 mg in the morning. OCTN2 (SLC22A5) typical; CPT2 V368I het; FMO3 typical (some TMAO). Keep 500 mg in the morning.
★ vitamin D3 (cholecalciferol) 10 000 IU	GC (DBP): favourable VDR: typical CYP2R1/DHCR7: typical	metabolism of vitamin D favourable/typical	S2: KEEP (coordination)	Maintain 10 000 IU (Coordination note: measurement-based decision). GC without Gc2 (DBP favourable), typical CYP2R1/DHCR7, typical VDR: genetics does not predict increased need. The dose is defined by the analysis; by coordination, it remains 10 000 IU of D3 (measurement-based decision) (suggestion, not applied).
★ vitamin K2 MK-7 240 µg	VKORC1: TT (sensitive) CYP4F2: normal GGCX: C/T	normal (relevant only with coumarins)	S2: KEEP	The same; it only weighs if you take coumarins (VKORC1 TT). VKORC1 –1639 T/T + CYP4F2 C/C + GGCX C/T: no effect in the absence of anticoagulant. If you ever take warfarin/acenocoumarol, keep K intake stable and warn the doctor (VKORC1 TT makes INR very sensitive).
★ ergothioneine 10 mg	SLC22A4/5 (OCTN1/2): typical	typical capture	S20: KEEP	The same. OCTN1 (SLC22A4) L503 C/C: typical ergothioneine uptake. Maintain.
★ lutein 20 mg	CFH/ARMS2: CFH het BCO1: slightly ↓ conversion	useful (retina)	S24: KEEP	The same, with fat. CFH Y402H het, ARMS2 G/G; lutein/zeaxanthin do not depend on BCO1. Keeping with fat.
★ zeaxanthin 4 mg	CFH/ARMS2: CFH het BCO1: slightly ↓ conversion	useful (retina)	S24: KEEP	The same, with fat. CFH Y402H het, ARMS2 G/G; lutein/zeaxanthin do not depend on BCO1. Keeping with fat.
★ sulforaphane (glucoraphanine + myrosinase) ☼ glucoraphanin 100 mg	GSTM1: probable null NFE2L2 (NRF2): normal	favourable response	S25: KEEP	The same in the morning. Probable null GSTM1: isothiocyanates are more in circulation (associated with more effect); NRF2 –617 C/C: KEAP1–NRF2 intact. Maintain.
★ 5-MTHF (calcium L-methylfolate) ☼ 800 µg	MTHFR: 677CT/1298AC DHFR: slow reduction (CC)	ideal form for the genotype	S3: KEEP	Keep 800 µg; no product with folic acid. 5-MTHF enters directly into the cycle without DHFR or MTHFR. Keep 800 µg (not climbing) (suggestion, not applied).
★ pine bark extract (marine pine hull) ☼ 100 mg	NOS3: –786CT + 298GT	useful (NO)	S31: KEEP	In the morning, outside the rapamycin window. NOS3 –786 C/T + Glu298Asp: pine bark extract improves endothelial function. Keep in the morning, outside the rapamycin window for prudence.
★ olive leaf extract (oleuropein) ☼ oleuropein 250 mg	PON1: low expression	useful (PON1)	S32: KEEP	The same, with meal. Low expression PON1: olive oil/oliveira polyphenols increase the activity of PON1.
★ hydroxytyrosol ☼ 25 mg	PON1: low expression COMT: Val/Val (high)	useful in low dose	S32: KEEP	The same, with meal. Low expression PON1: olive oil polyphenols rise to PON1. Low dose (25 mg) : small impact on COMT. Maintain.
★ black garlic 10 g	NOS3: –786CT + 298GT CYP3A5:expressor (*1/*3)	low allicin	S33: optional	Redundant with AGE. NOS3 gives vascular rationale; AGE and black garlic have little allicin → minor effect on sirolimus. Maintain AGE; black garlic is redundant.

Supplement (current dose)	Genes (state)	Interpretation	Plan	Comment
★ extract of aged garlic (AGE) 1.2 g	NOS3: –786CT + 298GT CYP3A5:expressor (*1/*3)	low allicin	S33: KEEP	The same. NOS3 gives vascular rationale; AGE and black garlic have little allicin → minor effect on sirolimus. Maintain AGE; black garlic is redundant.
★ multistrain (generic) multistrain	FUT2: secretor	typical colonization (secretor)	S36: KEEP	The same. FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.
★ methylcobalamine 1000 µg (not specified)	TCN2: 776CG (light) FUT2: secretor MTR: normal (AA) MTRR: normal (AA)	Transport and use of typical B12	S4: KEEP	1000 µg (500 µg if acne appears); methyl- or adenosyl-cobalamine. MTR A/A and MTRR A/A (common); TCN2 776 C/G and TCN1 G/A with small effect FUT2 secretor. No genetic requirement of high dose or special; methyl/adenosyl are options, cyanocobalamin also works (relevant cyanide wear). 1000 µg arrives (500 µg if acne appears).
★ riboflavin-5-phosphate (R5P) 25 mg	MTHFR: 677CT/1298AC	modest benefit in 677CT	S5: ADJUST (optimal)	5–10 mg arrives; maintaining 25 mg is at low risk. Riboflavin stabilizes thermolabile MTHFR; the benefit is clear in 677TT and only modest in heterozygotes. Low dose (5–10 mg) arrives; 25 mg is safe.
★ bovine hydrolyzed collagen 20 g	COL1A1/COL5A1: normal	without risk alleles	S9: KEEP	The same (glycine source). COL1A1 Sp1 C/C and COL5A1 T/T: without risk alleles; useful source of glycine (CPS1).
★ taurine 6 g	—	without known genetic relevance	S11: KEEP	The same (no specific gene). No known gene-supplement interaction or relevant security gene.
★ calcium alpha-ketoglutarate (Ca-AKG) 2 g	—	without known genetic relevance	S14: KEEP	1 g + 1 g at meals (calcium at meals reduces iron absorption); HFE H63D). No known gene-supplement interaction or relevant security gene.
★ R-ALA (R-lipoic acid) 240 mg	—	without known genetic relevance	S18: KEEP (optimal)	240 mg fasting. No known gene-supplement interaction or relevant security gene.
★ astaxanthin 12 mg	—	without known genetic relevance	S22: KEEP	12 mg with fat (not pro-vitamin A). No known gene-supplement interaction or relevant security gene.
★ ginger 400 mg	—	without known genetic relevance	S34: KEEP	The same (useful for the nausea of the retatrutide). No known gene-supplement interaction or relevant security gene.
★ tributyrine 500 mg	—	without known genetic relevance	S37: KEEP	The same. No known gene-supplement interaction or relevant security gene.
Not yet taken, with genetic indication: choline (citicoline or phosphatidylcholine)	PEMT: V175M het FMO3: typical	preferred form	S38: ADD	250–500 mg/day of choline; prefer citicolin/PC to bitartrate (TMAO) (suggestion, not applied).

What this work adds to what was already known

- **AMPD1 Q12X absent** (rs17602729 G/G): no myoadenylate deaminase deficiency; D-ribose has no genetic indication. *New.*
- **CYP2E1 *5 absent** (rs2031920 C/C, rs3813867 G/G): typical metabolism of solvents, ethanol and some essential oils constituents. *New.*
- **UGT1A6 *2 heterozygous** (rs2070959 A/G, rs6759892 T/G): small effect on salicylate glucuronidation (sallow bark, almaria), added to CYP2C9 *1/*3. *New.*
- **EPHX1 Tyr113His T/T + His139Arg A/G**: intermediate activity of epoxide-hydrolase (informative). *New.*
- **Normal G6PD applied to supplements**: high-dose vitamin C, methylene blue, quinine (quine peel), yellow melon and fava extract without genetic risk of hemolysis. *New application.*
- **PAH without pathogenic variants tested**: phenylalanine, DLPA and spirulina without genetic restriction. *New application.*

- **Gene-nutrient interactions applied for the first time:** FADS1 high × GLA (primary/borage tend to give arachidonic acid); COMT/MTHFR × chlorogenic acid and L-dopa (homocysteine); priority LDL × pterostilbene and coconut oil FUT2 secretor × bifidobacteria and 2'-FL; SLC01B1 *14 × red rice yeast (favourable, but doubles statin and has liver risk); CYP2D6 without CNV × yohimbine and galantamine; CYP2A6 × coumarin of cinnamon-cassava and nicotine. *New*.
- **Telomerase activators:** the precaution of epithelon extends to the purified extracts of astragalum, cycloaestrageenol, astagaloside IV and astragalum. *Coherent with the plan*.
- **Not determinable in the chip** and therefore in gray: CNDP1 (carnosinase → carnosine, anserin), HSD11B2 (sensitivity to liquor and formulas Kampo with liquor), AMY1 (amylase), CYP2A6 deletions (nicotin). *Clarification*.
- Confirmed without change and applied to each supplement: MTHFR compound, DHFR CC, COMT Val/Val, UGT1A1 *1/*28, HFE H63D, SLC30A8 C/C, FADS high, GSTM1 probable null, CYP1A2 *1F, MTNR1B C/G, PNPLA3/MBOAT7/Pi*MZ, HLA-B*35:01 probable negative, TERT risk.

2. Phenotype per gene

Gene	Genotype	Phenotype	Trust / origin
PNPLA3/Pi*MZ (liver)	PNPLA3 I148M (rs738409) C/G; MBOAT7 rs641738 C/T; SERPINA1 Pi*Z (rs28929474) probable C/T; HSD17B13 rs72613567 T/TA (protector, imputation DR2 1.00)	Hepatic genetic vulnerability (attenuated by HSD17B13)	Moderate - chip + imputation (Pi*Z to clinically confirm)
TERT/melanoma	TERT rs2736100 C/C, rs10069690 T/T, rs7734992 C/C, rs2242652 G/G, rs2736098 C/C; nevus at the 88th percentile (melanoma by chip is p6.5/p25)	TERT alleles of cancer risk + nevi p88	high (genotypes) · moderate (score)
COMT	Val158Met (rs4680) G/G (Val/Val); rs4633 C/C	high activity (methyl catechols fast; spend SAM)	high - direct chip
CYP1A2	*1F heterozygous (rs762551 A/C); rs2472297 C/C; AHR rs4410790 C/C	clearance of normal to intermediate caffeine	high - direct chip
CYP2C9	*1/*3 (rs1057910 A/C)	intermediate metabolizer (AS 1.0)	high - direct chip
CYP3A5	*1/*3 (rs776746 C/T)	expresser (intermediate metabolizer)	high - direct chip
DHFR	rs408626 C/C	slow reduction of synthetic folic acid (small effect)	Moderate - direct chip (functional effect discussed)
HFE	H63D (rs1799945) C/G heterozygous; C282Y (rs1800562) G/G (absent); S65C absent	H63D carrier (iron absorption slightly ↑)	high - direct chip
MTHFR	677 C/T (rs1801133 A/G) + 1298 A/C (rs1801131 T/G); probable compound heterozygous (undetermined phase)	activity ~50-60%	high - direct chip (undetermined phase)
MTNR1B	rs10830963 C/G	carrier G: high melatonin reduces insulin secretion	high - direct chip
CETP/9p21/APOB	CETP rs708272 G/G, rs3764261 C/C; 9p21 rs1333049/rs10757278/rs4977574 homozygous risk; APOB rs693 G/A	genetic priority to low LDL/ApoB	high - direct chip
CPS1	rs715 T/T + rs1047891 C/C	Genetically lower plasma glycine	high - direct chip
CYP2D6	Probable *1/*2 (rs16947 A/G, rs1080985 C/G; without *3/*4/*6/*9/*10/*41)	Probable normal metabolizer (AS 2)	moderate - chip + imputation; CNV (deletion *5, duplications)
GNMT	rs9296404 T/T	higher conversion of SAM to homocysteine (hypothesis)	high - Imputed DR2 ≥ 0.97 + chip
GSTM1	null *0/*0 (inferred by rs366631)	without GSTM1 activity (probable)	moderate - inferred deletion
NOS3	-786 T/C (rs2070744 C/T); Glu298Asp (rs1799983) G/T	production of slightly reduced NO	high - chip + imputed
PON1	Q192R (rs662) T/T (QQ); L55M M/M; -108 T/T (rs705379)	low expression	high - direct chip
Sal (AGT/ADD1/GNB3/CYP11B2)	AGT M235T (rs699) A/G; ADD1 G460W (rs4961) G/T; GNB3 825 C/T (rs5443); CYP11B2 -344 (rs1799998) A/A; ACE rs4343 G/A	sensitivity to modest salt (polygenic)	high - direct chip (small effect)
TCF7L2/FTO/GCKR	TCF7L2 rs7903146 C/T; FTO rs9939609 T/A; GCKR rs1260326 T/C	Glucemic and hepatic response to less favourable sugars/fructose	high - direct chip
UGT1A1	*1/*28 ((TA)7 heterozygous; rs887829 C/T)	intermediate metabolizer (Gilbert heterozygote)	high - DR2 1.00 + chip imputed

Gene	Genotype	Phenotype	Trust / origin
VKORC1	–1639 T/T (rs9923231); rs9934438 A/A	high sensitivity to coumarins (only relevant if taking warfarin/acenocoumarol)	high – DR2 0.99 + chip imputed
AMY1	number of non-visible copies in the chip	indeterminate	not determinable
CNDP1	repeat (CTG)n D18S880 not testable	indeterminate	not determinable
CYP2A6	without *2 or *9; deletions (*4) not visible	probable normal	low – deletions not evaluated
HSD11B2	Functional variants not tested in the chip	indeterminate (sensitivity to liquor)	not determinable
ABCG5/ABCG8	ABCG8 D19H (rs11887534) G/G; rs4148217 C/C; ABCG5/8 rs6544713 C/C	without hyperabsorption of plant sterols	high - direct chip
ACTN3	R577X (rs1815739) T/C (R/X)	mixed strength/resistance	high - direct chip
ADORA2A	rs5751876 C/C	genotype with less anxiety sensitivity, but caffeine-sensitive sleep (small studies)	high - direct chip (weak effect)
AMPD1	Q12X (rs17602729) G/G	normal (without myoadenylate deaminase deficiency)	high - direct chip
AOC1 (DAO)	Thr16Met (rs10156191) C/T; His645Asp (rs1049793) C/G; rs1049742 C/C	DAO activity slightly reduced	high - direct chip (small effect)
APOE	ε3/ε3 (rs429358 T/T, rs7412 C/C)	common genotype	high - chip + imputation DR2 ≥ 0.97
BCO1	R267S (rs12934922) A/T; A379V (rs7501331) C/C; rs6564851 G/T	conversion of slightly reduced β-carotene	high - direct chip (small effect)
BHMT	rs3733890 G/A (R239Q het)	betaine pathway with common variant	high - direct chip (small effect)
CFH/ARMS2	CFH Y402H (rs1061170) C/T; ARMS2 A69S (rs10490924) G/G	Mild DMI risk (1 allele CFH)	high - direct chip
COL1A1/COL5A1	COL1A1 Sp1 (rs1800012) C/C; COL5A1 rs12722 T/T	without risk alleles	high - direct chip
CYP2C19	*1/*1 (without *2, *3, *4, *17)	normal metabolizer	high - direct chip
CYP2R1/DHCR7	CYP2R1 rs10741657 G/A, rs10766197 G/G (imputed); DHCR7 rs12785878 G/G	typical	high – chip + imputation
CYP3A4	*1/*1 (without *22, rs35599367 G/G)	normal	high - direct chip
CYP4F2	V433M (rs2108622) C/C	normal (metabolism of vitamins E and normal K)	high - direct chip
FADS1/FADS2	FADS1 rs174546 C/C; rs174537 G/G; FADS2 rs1535 A/A; ELOVL2 rs953413 G/G	High activity desaturases (plus AA from LA; good conversion ALA→EPA)	high - direct chip
FMO3	E158K (rs2266782) G/G; rs2266780 A/A	FMO3 typical (TMAO form from choline/ carnitine)	high - direct chip
FUT2	rs601338 G/A (W143X heterozygous); rs602662 G/A	secretor (1 functional allele)	high - direct chip
G6PD	B (normal), hemizygous; without c.202G>A (rs1050828) or c.563C>T (rs5030868)	normal	high for the tested variants - chip
GC (DBP)	rs2282679 T/T; rs4588 G/G; rs7041 C/C (without Gc2)	vitamin D binding protein	high - direct chip
GGCX	rs699664 C/T	common variant	high - direct chip (small effect)
GPX1	Pro198Leu (rs1050450) A/G	slightly reduced activity	high - direct chip

Gene	Genotype	Phenotype	Trust / origin
GSTA1	*B/*B (rs3957357 A/A)	reduced expression	high – chip + imputation
HLA-B*35:01	non-probable carrier (imputed tags; VPN 0.97)	without the HLA of liver injury by curcuma/ green tea/Polygonum	moderate - imputed tags
HLA-DQ2/DQ8	rs2187668 C/C; rs7454108 T/T; rs7775228 T/T (without DQ2.5 or DQ8)	without HLA of celiac disease	high - chip tags
HNMT	Thr105Ile (rs11558538) T/C	slightly reduced activity	high - direct chip (small effect)
LCT	–13910 C/T (rs4988235 A/G); rs182549 C/T	lactase persistence	high - direct chip
MTR	A2756G (rs1805087) A/A	normal	high - direct chip
MTRR	A66G (rs1801394) A/A; rs1532268 C/C	normal	high - direct chip
NAT2	Probable *5/*5 (341C, 481T, 803G homozygotes)	slow acetylator	high - direct chip (7 SNPs)
NFE2L2 (NRF2)	–617 C/C (rs6721961)	Normal promoter: intact NRF2 response	high - direct chip
NQO1	P187S (rs1800566) G/G (*1/*1)	normal (without null allele)	high - direct chip
OPRM1	118 A/G (rs1799971)	G allele carrier	high – chip (inconsistent clinical effect)
PAH	without pathogenic variants among the tested ones (chip ClinVar tracking)	normal for the tested variants	partial – chip (only known variants)
PDXK/ALPL/NBPF3 (B6)	PDXK rs2010795 G/G; ALPL rs1256335 G/A; NBPF3 rs4654748 C/T	activation of typical to high B6	high - direct chip
PEMT	rs7946 C/T (V175M het); rs12325817 C/C	endogenous synthesis of reduced choline (mild)	high - direct chip
SLC22A4/5 (OCTN1/2)	SLC22A4 L503F (rs1050152) C/C; SLC22A5 rs2631367 G/G	Ergothioneine and carnitine transport	high - direct chip
SLC23A1/2 (vit. C)	SLC23A1 rs33972313 C/C, rs10063949 C/C; SLC23A2 rs6053005 C/C	typical vitamin C transport	high - direct chip
SLC30A8	rs13266634 C/C (R325); rs11558471 A/A	Alleles that most benefit from zinc intake (glucose)	high - direct chip
SLC39A8	A391T (rs13107325) C/C; rs233804 C/C	Normal manganese transport	high - direct chip
SLCO1B1	*1/*14 (521 T/T; 388G and 463A heterozygotes)	increased function	high - direct chip
SOD2	Val16Ala (rs4880) G/G (Ala/Ala)	Efficient mitochondrial import of MnSOD	high - direct chip
TCN2	776 C/G (rs1801198 C/G); rs9606756 A/G	slightly less efficient B12 transport (small effect)	high - direct chip
TMPRSS6/TF	TMPRSS6 rs855791 G/A; TF rs3811647 G/G, rs1799852 C/T	small effects on iron	high - direct chip
TRPM6	rs3750425 C/C; rs2274924 T/T	typical magnesium absorption	high - direct chip
UGT1A6	*2 heterozygous (rs2070959 A/G, rs6759892 T/G)	Glucuronidation of salicylate/phenols slightly different	high - direct chip (small effect)
VDR	FokI (rs2228570) G/A; TaqI (rs731236) G/A; BsmI (rs1544410) T/C; ApaI (rs7975232) A/A	typical (heterozygous in the main)	high - direct chip (weak effect)

71 genes/axes used in this report. Colors of this table: red = vulnerability with consequence of avoiding; orange = phenotype that changes dose or shape; yellow = attention or small effect; gray = non-determinable; green = normal/favourable.

3. Supplements by category

In each category, the table shows the supplements with relevant gene, ordered by gravity. Below comes the compact list of those that do not have known genetic relevance, with the main pathway (when relevant) and, if it is a evaluated gene, its state in Sérgio F.

3.1 Vitamins and forms — 70 supplements

prevention: 3 · dose adjustment: 24 · Monitoring: 1 · normal: 33 · without known genetic relevance: 9

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
vitamin A (retinyl acetate) Avoid	BCO1 : slightly ↓ conversion PNPLA3/Pi*MZ (liver) : unfavourable	avoid preformed retinol	The conversion of β-carotene is only slightly reduced (BCO1 R267S het; without A379V), so there is no genetic need for retinol. With the isotretinoin of the stack, retinol supplemental sum retinoids (toxicity, transaminases); the genetic liver (PNPLA3/Pi*MZ) reinforces. Obtain vitamin A from carotenoids (suggestion, not applied).	NIH ODS (vitamin A); genetic plan (b.3); isotretinoin label <i>strong (interaction) · weak (BCO1)</i>	BCO1: high PNPLA3/Pi*MZ (liver): moderate
vitamin A (retinol, retinyl palmitate) Avoid	BCO1 : slightly ↓ conversion PNPLA3/Pi*MZ (liver) : unfavourable	avoid preformed retinol	The conversion of β-carotene is only slightly reduced (BCO1 R267S het; without A379V), so there is no genetic need for retinol. With the isotretinoin of the stack, retinol supplemental sum retinoids (toxicity, transaminases); the genetic liver (PNPLA3/Pi*MZ) reinforces. Obtain vitamin A from carotenoids (suggestion, not applied).	NIH ODS (vitamin A); genetic plan (b.3); isotretinoin label <i>strong (interaction) · weak (BCO1)</i>	BCO1: high PNPLA3/Pi*MZ (liver): moderate
vitamin A + D (oil) Avoid	BCO1 : slightly ↓ conversion PNPLA3/Pi*MZ (liver) : unfavourable	avoid preformed retinol	The conversion of β-carotene is only slightly reduced (BCO1 R267S het; without A379V), so there is no genetic need for retinol. With the isotretinoin of the stack, retinol supplemental sum retinoids (toxicity, transaminases); the genetic liver (PNPLA3/Pi*MZ) reinforces. Obtain vitamin A from carotenoids (suggestion, not applied).	NIH ODS (vitamin A); genetic plan (b.3); isotretinoin label <i>strong (interaction) · weak (BCO1)</i>	BCO1: high PNPLA3/Pi*MZ (liver): moderate
acerola (natural vitamin C) Adjust dose	HFE : H63D het G6PD : normal SLC23A1/2 (vit. C) : typical	drinking between meals (hour)	HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).	Genetic plan (S16); CPIC G6PD 2022 (analogy) <i>moderate (iron interaction)</i>	HFE: high G6PD: high for the tested variants SLC23A1/2 (vit. C): high
alpha-GPC (glycerophosphocholine) Adjust dose	PEMT : V175M het FMO3 : typical	prefer citicoline or phosphatidylcholine (form)	PEMT V175M het: a slightly smaller endogenous choline synthesis (rational to choline). Typical FMO3: bitartrate and alpha-GPC generate more TMAO than phosphatidylcholine/citicoline; with genetic vascular priority (9p21, CETP), preferring citicoline 250–500 mg or PC 1–2 g (suggestion, not applied).	Genetic plan (S38); TMAO studies by choline <i>weak-moderate</i>	PEMT: high FMO3: high
calcium ascorbate (buffered vitamin C) Adjust dose	HFE : H63D het G6PD : normal SLC23A1/2 (vit. C) : typical	drinking between meals (hour)	HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).	Genetic plan (S16); CPIC G6PD 2022 (analogy) <i>moderate (iron interaction)</i>	HFE: high G6PD: high for the tested variants SLC23A1/2 (vit. C): high
sodium ascorbate Adjust dose	HFE : H63D het G6PD : normal SLC23A1/2 (vit. C) : typical	drinking between meals (hour)	HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).	Genetic plan (S16); CPIC G6PD 2022 (analogy) <i>moderate (iron interaction)</i>	HFE: high G6PD: high for the tested variants SLC23A1/2 (vit. C): high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
choline (bitartrate) Adjust dose	PEMT : V175M het FMO3 : typical	prefer citicoline or phosphatidylcholine (form)	PEMT V175M het: a slightly smaller endogenous choline synthesis (rational to choline). Typical FMO3: bitartrate and alpha-GPC generate more TMA/TMAO than phosphatidylcholine/citicoline; with genetic vascular priority (9p21, CETP), preferring citicoline 250–500 mg or PC 1–2 g (suggestion, not applied).	Genetic plan (S38); TMAO studies by choline <i>weak-moderate</i>	PEMT: high FMO3: high
B complex Adjust dose	PDXK/ALPL/NBPF3 (B6) : typical/high MTHFR : 677CT/1298AC DHFR : slow reduction (CC)	choose formula (low B6, without folic acid)	Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).	Genetic plan (S3, S6); EFSA UL B6 <i>moderate</i>	PDXK/ALPL/NBPF3 (B6): high MTHFR: high DHFR: moderate
B complex in gums Adjust dose	PDXK/ALPL/NBPF3 (B6) : typical/high MTHFR : 677CT/1298AC DHFR : slow reduction (CC)	choose formula (low B6, without folic acid)	Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).	Genetic plan (S3, S6); EFSA UL B6 <i>moderate</i>	PDXK/ALPL/NBPF3 (B6): high MTHFR: high DHFR: moderate
Methylated B complex Adjust dose	PDXK/ALPL/NBPF3 (B6) : typical/high MTHFR : 677CT/1298AC DHFR : slow reduction (CC)	choose formula (low B6, without folic acid)	Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).	Genetic plan (S3, S6); EFSA UL B6 <i>moderate</i>	PDXK/ALPL/NBPF3 (B6): high MTHFR: high DHFR: moderate
iron + vitamin C (combined) Adjust dose	HFE : H63D het TMPRSS6/TF : small effect	only with confirmed deficit	HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).	Genetic plan (b.3); HFE guidelines (EASL 2022) <i>moderate</i>	HFE: high TMPRSS6/TF: high
multivitamin 50+ Adjust dose	MTHFR : 677CT/1298AC DHFR : slow reduction (CC) HFE : H63D het BCO1 : slightly ↓ conversion PDXK/ALPL/NBPF3 (B6) : typical/high	choice formula	Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).	Genetic plan (b.3) <i>moderate</i>	MTHFR: high DHFR: moderate HFE: high BCO1: high PDXK/ALPL/NBPF3 (B6): high
multivitamin in gums Adjust dose	MTHFR : 677CT/1298AC DHFR : slow reduction (CC) HFE : H63D het BCO1 : slightly ↓ conversion PDXK/ALPL/NBPF3 (B6) : typical/high	choice formula	Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).	Genetic plan (b.3) <i>moderate</i>	MTHFR: high DHFR: moderate HFE: high BCO1: high PDXK/ALPL/NBPF3 (B6): high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
female multivitamin Adjust dose	MTHFR : 677CT/1298AC DHFR : slow reduction (CC) HFE : H63D het BCO1 : slightly ↓ conversion PDXK/ALPL/NBPF3 (B6) : typical/high	choice formula	Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).	Genetic plan (b.3) <i>moderate</i>	MTHFR: high DHFR: moderate HFE: high BCO1: high PDXK/ALPL/NBPF3 (B6): high
General multivitamin Adjust dose	MTHFR : 677CT/1298AC DHFR : slow reduction (CC) HFE : H63D het BCO1 : slightly ↓ conversion PDXK/ALPL/NBPF3 (B6) : typical/high	choice formula	Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).	Genetic plan (b.3) <i>moderate</i>	MTHFR: high DHFR: moderate HFE: high BCO1: high PDXK/ALPL/NBPF3 (B6): high
Multivitaminic male Adjust dose	MTHFR : 677CT/1298AC DHFR : slow reduction (CC) HFE : H63D het BCO1 : slightly ↓ conversion PDXK/ALPL/NBPF3 (B6) : typical/high	choice formula	Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).	Genetic plan (b.3) <i>moderate</i>	MTHFR: high DHFR: moderate HFE: high BCO1: high PDXK/ALPL/NBPF3 (B6): high
multivitamin prenatal Adjust dose	MTHFR : 677CT/1298AC DHFR : slow reduction (CC) HFE : H63D het BCO1 : slightly ↓ conversion PDXK/ALPL/NBPF3 (B6) : typical/high	choice formula	Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).	Genetic plan (b.3) <i>moderate</i>	MTHFR: high DHFR: moderate HFE: high BCO1: high PDXK/ALPL/NBPF3 (B6): high
ascorbyl palmitate Adjust dose	HFE : H63D het G6PD : normal SLC23A1/2 (vit. C) : typical	drinking between meals (hour)	HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).	Genetic plan (S16); CPIC G6PD 2022 (analogy) <i>moderate (iron interaction)</i>	HFE: high G6PD: high for the tested variants SLC23A1/2 (vit. C): high
★ pyridoxal-5-phosphate (P5P) Adjust dose	PDXK/ALPL/NBPF3 (B6) : typical/high	lowering dose (≤ 5 mg/day)	Current stack (15 mg) — plan S6: ADJUST (descend). ≤ 5 mg/day or stop (above the upper limit of ingestion, without genetic indication). PDXK rs2010795 G/G (high activation) and ALPL/NBPF3 heterozygotes: no genetic need for extra B6. Chronic use of 15 mg passes the upper limit of ingestion (12 mg) and is associated with neuropathy. ≤ 5 mg/day or pause (suggestion, not applied).	Genetic plan (S6); EFSA UL B6 (2023) <i>moderate (safety limit)</i>	PDXK/ALPL/NBPF3 (B6): high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
vitamin B6 (pyridoxine hydrochloride) Adjust dose	PDXK/ALPL/NBPF3 (B6): typical/high	lowering dose (≤ 5 mg/day)	PDXK rs2010795 G/G (high activation) and ALPL/NBPF3 heterozygotes: no genetic need for extra B6. Chronic use of 15 mg passes the upper limit of ingestion (12 mg) and is associated with neuropathy. ≤ 5 mg/day or pause (suggestion, not applied).	Genetic plan (S6); EFSA UL B6 (2023) <i>moderate (safety limit)</i>	PDXK/ALPL/NBPF3 (B6): high
vitamin C (ascorbic acid) Adjust dose	HFE: H63D het G6PD: normal SLC23A1/2 (vit. C): typical	drinking between meals (hour)	HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).	Genetic plan (S16); CPIC G6PD 2022 (analogy) <i>moderate (iron interaction)</i>	HFE: high G6PD: high for the tested variants SLC23A1/2 (vit. C): high
vitamin C + zinc Adjust dose	HFE: H63D het G6PD: normal SLC23A1/2 (vit. C): typical	drinking between meals (hour)	HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).	Genetic plan (S16); CPIC G6PD 2022 (analogy) <i>moderate (iron interaction)</i>	HFE: high G6PD: high for the tested variants SLC23A1/2 (vit. C): high
vitamin C with metabolites (ascorbate + threonate) Adjust dose	HFE: H63D het G6PD: normal SLC23A1/2 (vit. C): typical	drinking between meals (hour)	HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).	Genetic plan (S16); CPIC G6PD 2022 (analogy) <i>moderate (iron interaction)</i>	HFE: high G6PD: high for the tested variants SLC23A1/2 (vit. C): high
★ vitamin C liposomal Adjust dose	HFE: H63D het G6PD: normal SLC23A1/2 (vit. C): typical	drinking between meals (hour)	Current stack (500 mg) — plan S16: ADJUST (hour). Always ≥ 2 h away from meals. HFE H63D het: vitamin C with food increases the absorption of non-heme iron. 500 mg always ≥ 2 h away from meals. Normal G6PD; Typical SLC23A1/2 (suggestion, not applied).	Genetic plan (S16) <i>moderate (iron interaction)</i>	HFE: high G6PD: high for the tested variants SLC23A1/2 (vit. C): high
folic acid (synthetic) Adjust dose	MTHFR: 677CT/1298AC DHFR: slow reduction (CC)	prefer 5-MTHF (form)	MTHFR 677CT + 1298AC (activity $\sim 50\text{--}60\%$) and DHFR rs408626 C/C (slow reduction of folic acid; may leave folic acid unmetabolized in circulation). Exchange by 5-MTHF or food folate; no product with folic acid (suggestion, not applied).	Genetic plan (S3, b.3); MTHFR/DHFR literature <i>moderate</i>	MTHFR: high DHFR: moderate
folic acid + B12 (combined) Adjust dose	MTHFR: 677CT/1298AC DHFR: slow reduction (CC)	prefer 5-MTHF (form)	MTHFR 677CT + 1298AC (activity $\sim 50\text{--}60\%$) and DHFR rs408626 C/C (slow reduction of folic acid; may leave folic acid unmetabolized in circulation). Exchange by 5-MTHF or food folate; no product with folic acid (suggestion, not applied).	Genetic plan (S3, b.3); MTHFR/DHFR literature <i>moderate</i>	MTHFR: high DHFR: moderate
formula AREDS2 (eyes) Monitoring	CFH/ARMS2: CFH het SLC30A8: C/C (usual zinc)	without indication; zinc $\times$ CFH controversy	CFH Y402H het and ARMS2 G/G. One group (Awh et al.) suggested that, in this pattern, high dose zinc (80 mg) of AREDS worsens DMI; the reanalysis of AREDS (Chew 2014) did not confirm. Without DMI, the formula is not indicated; lutein/zeaxanthin alone arrives.	AREDS/AREDS2; Awh 2013/2015 vs Chew 2014 <i>weak (controversy)</i>	CFH/ARMS2: high SLC30A8: high
★ 5-MTHF (calcium L-methylfolate) ☺ Normal	MTHFR: 677CT/1298AC DHFR: slow reduction (CC)	ideal form for the genotype	Current stack (800 μg) — S3 plan: KEEP. Keep 800 μg ; no product with folic acid. 5-MTHF enters directly into the cycle without DHFR or MTHFR. Keep 800 μg (not climbing) (suggestion, not applied).	Genetic plan (S3) <i>moderate</i>	MTHFR: high DHFR: moderate
5-MTHF glucosamine salt ☺ Normal	MTHFR: 677CT/1298AC DHFR: slow reduction (CC)	ideal form for the genotype	The same logic of the calcium 5-MTHF does not depend on DHFR or MTHFR. Do not add to the 5-MTHF already taken (teto ~ 800 μg).	Genetic plan (S3) <i>moderate</i>	MTHFR: high DHFR: moderate
PABA (paraaminobenzoic acid) Normal	NAT2: slow	slow acetylation, with no relevance in nutritional dose	PABA is acetylated by NAT2 (slow $\ast 5/\ast 5$); with no known clinical consequence in supplement doses.	Pharmacology of PABA <i>weak</i>	NAT2: high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
adenosylcobalamine Normal	TCN2 : 776CG (light) FUT2 : secretor MTR : normal (AA) MTRR : normal (AA)	Transport and use of typical B12	MTR A/A and MTRR A/A (common); TCN2 776 C/G and TCN1 G/A with small effect FUT2 secretor. No genetic requirement of high dose or special; methyl/adenosyl are options, cyanocobalamin also works (relevant cyanide wear). 1000 µg arrives (500 µg if acne appears).	Genetic plan (S4); GWAS of B12 (Hazra 2008) <i>weak</i>	TCN2: high FUT2: high MTR: high MTRR: high
betacarotene Normal	BCO1 : slightly ↓ conversion	slightly reduced conversion, without practical relevance	BCO1 R267S het + rs6564851 G/T: conversion to retinol slightly smaller; A379V absent. It is the preferred form of vitamin A with isotretinoin (regulated conversion). Food or low dose; β-carotene at a high dose does not bring benefits (and is not advised in smokers).	BCO1 studies (Leung 2009; Lietz 2012); NIH ODS <i>moderate (genotype) · weak (practice)</i>	BCO1: high
biotin + zinc (hair) ⊕ Normal	SLC30A8 : C/C (usual zinc)	genetically useful zinc	SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied).	Kanoni 2011 (CHARGE); genetic plan (S12) <i>moderate</i>	SLC30A8: high
calcifiediol (25-OH-D3) Normal	GC (DBP) : favourable VDR : typical CYP2R1/DHCR7 : typical	metabolism of vitamin D favourable/typical	GC without Gc2 (DBP favourable), typical CYP2R1/DHCR7, typical VDR: genetics does not predict increased need. The dose is defined by the analysis; by coordination, it remains 10 000 IU of D3 (measurement-based decision) (suggestion, not applied).	Genetic plan (S2 + coordination note); GWAS vitamin D (Wang 2010) <i>moderate (genotype) · weak (dose)</i>	GC (DBP): high VDR: high CYP2R1/DHCR7: high
mixed carotenoids Normal	BCO1 : slightly ↓ conversion	slightly reduced conversion, without practical relevance	BCO1 R267S het + rs6564851 G/T: conversion to retinol slightly smaller; A379V absent. It is the preferred form of vitamin A with isotretinoin (regulated conversion). Food or low dose; β-carotene at a high dose does not bring benefits (and is not advised in smokers).	BCO1 studies (Leung 2009; Lietz 2012); NIH ODS <i>moderate (genotype) · weak (practice)</i>	BCO1: high
citicoline (CDP-choline) ⊕ Normal	PEMT : V175M het FMO3 : typical PNPLA3/Pi*MZ (liver) : unfavourable	Choline preferred form	PEMT V175M het + genetic liver give rational to choline; this form generates little TMAO. 250–500 mg of choline/day in total (suggestion, not applied).	Genetic plan (S38) <i>weak-moderate</i>	PEMT: high FMO3: high PNPLA3/Pi*MZ (liver): moderate
calcium + vitamin D3 Normal	VDR : typical GC (DBP) : favourable HFE : H63D het	typical absorption	VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.	NIH ODS (calcium); genetic plan (S14) <i>weak</i>	VDR: high GC (DBP): high HFE: high
natural folate (food extract) ⊕ Normal	MTHFR : 677CT/1298AC DHFR : slow reduction (CC)	normal	Food folate (leaves, legumes) is always suitable for the genotype.	Genetic plan (e.6) <i>moderate</i>	MTHFR: high DHFR: moderate
phosphatidylcholine ⊕ Normal	PEMT : V175M het FMO3 : typical PNPLA3/Pi*MZ (liver) : unfavourable	Choline preferred form	PEMT V175M het + genetic liver give rational to choline; this form generates little TMAO. 250–500 mg of choline/day in total (suggestion, not applied).	Genetic plan (S38) <i>weak-moderate</i>	PEMT: high FMO3: high PNPLA3/Pi*MZ (liver): moderate
hydroxocobalamin Normal	TCN2 : 776CG (light) FUT2 : secretor MTR : normal (AA) MTRR : normal (AA)	Transport and use of typical B12	MTR A/A and MTRR A/A (common); TCN2 776 C/G and TCN1 G/A with small effect FUT2 secretor. No genetic requirement of high dose or special; methyl/adenosyl are options, cyanocobalamin also works (relevant cyanide wear). 1000 µg arrives (500 µg if acne appears).	Genetic plan (S4); GWAS of B12 (Hazra 2008) <i>weak</i>	TCN2: high FUT2: high MTR: high MTRR: high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
soybean lecithin/ gyrassol  Normal	PEMT : V175M het FMO3 : typical PNPLA3/Pi*MZ (liver) : unfavourable	Choline preferred form	PEMT V175M het + genetic liver give rational to choline; this form generates little TMAO. 250–500 mg of choline/day in total (suggestion, not applied).	Genetic plan (S38) <i>weak-moderate</i>	PEMT: high FMO3: high PNPLA3/Pi*MZ (liver): moderate
★ methylcobalamine Normal	TCN2 : 776CG (light) FUT2 : secretor MTR : normal (AA) MTRR : normal (AA)	Transport and use of typical B12	Current stack (1000 µg (unspecified form)) — S4 plan: KEEP. 1000 µg (500 µg if acne appears); methyl- or adenosyl-cobalamine. MTR A/A and MTRR A/A (common); TCN2 776 C/G and TCN1 G/A with small effect FUT2 secretor. No genetic requirement of high dose or special; methyl/adenosyl are options, cyanocobalamin also works (relevant cyanide wear). 1000 µg arrives (500 µg if acne appears).	Genetic plan (S4); GWAS of B12 (Hazra 2008) <i>weak</i>	TCN2: high FUT2: high MTR: high MTRR: high
★ riboflavin-5-phosphate (R5P) Normal	MTHFR : 677CT/ 1298AC	modest benefit in 677CT	Current stack (25 mg) — plan S5: ADJUST (optimal). 5–10 mg arrives; maintaining 25 mg is at low risk. Riboflavin stabilizes thermolabile MTHFR; the benefit is clear in 677TT and only modest in heterozygotes. Low dose (5–10 mg) arrives; 25 mg is safe.	Riboflavin × MTHFR 677 (McNulty 2006) trials <i>moderate (TT) · weak (CT)</i>	MTHFR: high
mixed tocopherols Normal	CYP4F2 : normal	normal metabolism	CYP4F2 V433M C/C (normal): vitamin E ω-hydroxylation is typical. Nutritional dose; no genetic benefit for high doses.	CYP4F2 × tocopherol (Bardowell 2012) studies <i>weak</i>	CYP4F2: high
tocotrienols Normal	CYP4F2 : normal	normal metabolism	CYP4F2 V433M C/C (normal): vitamin E ω-hydroxylation is typical. Nutritional dose; no genetic benefit for high doses.	CYP4F2 × tocopherol (Bardowell 2012) studies <i>weak</i>	CYP4F2: high
vitamin B12 (cyanocobalamine) Normal	TCN2 : 776CG (light) FUT2 : secretor MTR : normal (AA) MTRR : normal (AA)	Transport and use of typical B12	MTR A/A and MTRR A/A (common); TCN2 776 C/G and TCN1 G/A with small effect FUT2 secretor. No genetic requirement of high dose or special; methyl/adenosyl are options, cyanocobalamin also works (relevant cyanide wear). 1000 µg arrives (500 µg if acne appears).	Genetic plan (S4); GWAS of B12 (Hazra 2008) <i>weak</i>	TCN2: high FUT2: high MTR: high MTRR: high
vitamin B12 sublingual/ spray Normal	TCN2 : 776CG (light) FUT2 : secretor MTR : normal (AA) MTRR : normal (AA)	Transport and use of typical B12	MTR A/A and MTRR A/A (common); TCN2 776 C/G and TCN1 G/A with small effect FUT2 secretor. No genetic requirement of high dose or special; methyl/adenosyl are options, cyanocobalamin also works (relevant cyanide wear). 1000 µg arrives (500 µg if acne appears).	Genetic plan (S4); GWAS of B12 (Hazra 2008) <i>weak</i>	TCN2: high FUT2: high MTR: high MTRR: high
vitamin B2 (riboflavin) Normal	MTHFR : 677CT/ 1298AC	modest benefit in 677CT	Riboflavin stabilizes thermolabile MTHFR; the benefit is clear in 677TT and only modest in heterozygotes. Low dose (5–10 mg) arrives; 25 mg is safe.	Riboflavin × MTHFR 677 (McNulty 2006) trials <i>moderate (TT) · weak (CT)</i>	MTHFR: high
vitamin D2 (ergocalciferol) Normal	GC (DBP) : favourable CYP2R1/DHCR7 : typical	typical (D3 preferable)	Favourable genetics; D2 binds less to DBP and rises less to 25(OH)D than D3 (form question, not genes).	NIH ODS (vitamin D) <i>moderate</i>	GC (DBP): high CYP2R1/DHCR7: high
★ vitamin D3 (cholecalciferol) Normal	GC (DBP) : favourable VDR : typical CYP2R1/DHCR7 : typical	metabolism of vitamin D favourable/typical	Current stack (10 000 IU) — plan S2: KEEP (coordination). Maintain 10 000 IU (Coordination note: measurement-based decision). GC without Gc2 (DBP favourable), typical CYP2R1/DHCR7, typical VDR: genetics does not predict increased need. The dose is defined by the analysis; by coordination, it remains 10 000 IU of D3 (measurement-based decision) (suggestion, not applied).	Genetic plan (S2 + coordination note); GWAS vitamin D (Wang 2010) <i>moderate (genotype) · weak (dose)</i>	GC (DBP): high VDR: high CYP2R1/DHCR7: high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
vitamin D3 + K2 Normal	GC (DBP): favourable VDR: typical CYP2R1/DHCR7: typical	metabolism of vitamin D favourable/typical	GC without Gc2 (DBP favourable), typical CYP2R1/DHCR7, typical VDR: genetics does not predict increased need. The dose is defined by the analysis; by coordination, it remains 10 000 IU of D3 (measurement-based decision) (suggestion, not applied).	Genetic plan (S2 + coordination note); GWAS vitamin D (Wang 2010) <i>moderate (genotype) · weak (dose)</i>	GC (DBP): high VDR: high CYP2R1/DHCR7: high
vitamin D3 weekly high dose (50 000 IU) Normal	GC (DBP): favourable VDR: typical CYP2R1/DHCR7: typical	metabolism of vitamin D favourable/typical	GC without Gc2 (DBP favourable), typical CYP2R1/DHCR7, typical VDR: genetics does not predict increased need. The dose is defined by the analysis; by coordination, it remains 10 000 IU of D3 (measurement-based decision) (suggestion, not applied).	Genetic plan (S2 + coordination note); GWAS vitamin D (Wang 2010) <i>moderate (genotype) · weak (dose)</i>	GC (DBP): high VDR: high CYP2R1/DHCR7: high
vitamin D3 vegan (lichen) Normal	GC (DBP): favourable VDR: typical CYP2R1/DHCR7: typical	metabolism of vitamin D favourable/typical	GC without Gc2 (DBP favourable), typical CYP2R1/DHCR7, typical VDR: genetics does not predict increased need. The dose is defined by the analysis; by coordination, it remains 10 000 IU of D3 (measurement-based decision) (suggestion, not applied).	Genetic plan (S2 + coordination note); GWAS vitamin D (Wang 2010) <i>moderate (genotype) · weak (dose)</i>	GC (DBP): high VDR: high CYP2R1/DHCR7: high
vitamin E (tocopheryl acetate) Normal	CYP4F2: normal	normal metabolism	CYP4F2 V433M C/C (normal): vitamin E ω -hydroxylation is typical. Nutritional dose; no genetic benefit for high doses.	CYP4F2 × tocopherol (Bardowell 2012) studies <i>weak</i>	CYP4F2: high
vitamin E (d-alpha-tocopherol) Normal	CYP4F2: normal	normal metabolism	CYP4F2 V433M C/C (normal): vitamin E ω -hydroxylation is typical. Nutritional dose; no genetic benefit for high doses.	CYP4F2 × tocopherol (Bardowell 2012) studies <i>weak</i>	CYP4F2: high
synthetic vitamin E (dl-alpha-tocopherol) Normal	CYP4F2: normal	normal metabolism	CYP4F2 V433M C/C (normal): vitamin E ω -hydroxylation is typical. Nutritional dose; no genetic benefit for high doses.	CYP4F2 × tocopherol (Bardowell 2012) studies <i>weak</i>	CYP4F2: high
vitamin K1 (phyloquinone) Normal	VKORC1: TT (sensitive) CYP4F2: normal GGCX: C/T	normal (relevant only with coumarins)	VKORC1 –1639 T/T + CYP4F2 C/C + GGCX C/T: no effect in the absence of anticoagulant. If you ever take warfarin/acenocoumarol, keep K intake stable and warn the doctor (VKORC1 TT makes INR very sensitive).	CPIC warfarin 2017 (context); genetic plan (S2) <i>strong (coumarinic context)</i>	VKORC1: high CYP4F2: high GGCX: high
vitamin K2 MK-4 Normal	VKORC1: TT (sensitive) CYP4F2: normal GGCX: C/T	normal (relevant only with coumarins)	VKORC1 –1639 T/T + CYP4F2 C/C + GGCX C/T: no effect in the absence of anticoagulant. If you ever take warfarin/acenocoumarol, keep K intake stable and warn the doctor (VKORC1 TT makes INR very sensitive).	CPIC warfarin 2017 (context); genetic plan (S2) <i>strong (coumarinic context)</i>	VKORC1: high CYP4F2: high GGCX: high
★ vitamin K2 MK-7 Normal	VKORC1: TT (sensitive) CYP4F2: normal GGCX: C/T	normal (relevant only with coumarins)	Current stack (240 µg) — S2 plan: KEEP. The same; it only weighs if you take coumarins (VKORC1 TT). VKORC1 –1639 T/T + CYP4F2 C/C + GGCX C/T: no effect in the absence of anticoagulant. If you ever take warfarin/acenocoumarol, keep K intake stable and warn the doctor (VKORC1 TT makes INR very sensitive).	CPIC warfarin 2017 (context); genetic plan (S2) <i>strong (coumarinic context)</i>	VKORC1: high CYP4F2: high GGCX: high
folinic acid (calcium folinate) Normal	MTHFR: 677CT/1298AC DHFR: slow reduction (CC)	avoids DHFR	The folinic jumps to DHFR, but still needs the MTHFR to give 5-MTHF; with the 5-MTHF already in the stack is redundant.	Biochemistry of folate cycle <i>weak</i>	MTHFR: high DHFR: moderate

No known genetic relevance (9):

benfotiamine (SLC19A2/3)
biotin (B7) (renal excretion)
D-quirositol

inositol (mio-inositol)
pantetin (CoA)
sulbutiamine (liposoluble)

TTFD (fursultiamine) (liposoluble)
vitamin B1 (tiamine HCl) (SLC19A2/3)
pantoténico ácido (B5) (CoA)

3.2 Minerals and trace elements — 74 supplements

dose adjustment: 10 · Monitoring: 6 · normal: 41 · without known genetic relevance: 17

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
ZMA (zinc, magnesium, B6) Adjust dose	PDXK/ALPL/NBPF3 (B6) : typical/high MTHFR : 677CT/1298AC DHFR : slow reduction (CC)	choose formula (low B6, without folic acid)	Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).	Genetic plan (S3, S6); EFSA UL B6 <i>moderate</i>	PDXK/ALPL/NBPF3 (B6): high MTHFR: high DHFR: moderate
iron bisglycinate Adjust dose	HFE : H63D het TMPRSS6/TF : small effect	only with confirmed deficit	HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).	Genetic plan (b.3); HFE guidelines (EASL 2022) <i>moderate</i>	HFE: high TMPRSS6/TF: high
iron-polysaccharide complex Adjust dose	HFE : H63D het TMPRSS6/TF : small effect	only with confirmed deficit	HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).	Genetic plan (b.3); HFE guidelines (EASL 2022) <i>moderate</i>	HFE: high TMPRSS6/TF: high
iron carbonyl Adjust dose	HFE : H63D het TMPRSS6/TF : small effect	only with confirmed deficit	HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).	Genetic plan (b.3); HFE guidelines (EASL 2022) <i>moderate</i>	HFE: high TMPRSS6/TF: high
iron heme (polypeptide) Adjust dose	HFE : H63D het TMPRSS6/TF : small effect	only with confirmed deficit	HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).	Genetic plan (b.3); HFE guidelines (EASL 2022) <i>moderate</i>	HFE: high TMPRSS6/TF: high
liposomal / sucrosomal iron Adjust dose	HFE : H63D het TMPRSS6/TF : small effect	only with confirmed deficit	HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).	Genetic plan (b.3); HFE guidelines (EASL 2022) <i>moderate</i>	HFE: high TMPRSS6/TF: high
ferrous fumarate Adjust dose	HFE : H63D het TMPRSS6/TF : small effect	only with confirmed deficit	HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).	Genetic plan (b.3); HFE guidelines (EASL 2022) <i>moderate</i>	HFE: high TMPRSS6/TF: high
ferrous gluconate Adjust dose	HFE : H63D het TMPRSS6/TF : small effect	only with confirmed deficit	HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).	Genetic plan (b.3); HFE guidelines (EASL 2022) <i>moderate</i>	HFE: high TMPRSS6/TF: high
★ trace minerals (multineural) ⓘ Adjust dose	SLC30A8 : C/C (usual zinc) GPX1 : Pro198Leu het SLC39A8 : normal AOC1 (DAO) : slightly ↓	up zinc and copper	Current stack (Zn 11, Cu 0.9, I 0.05, Se 0.055, Mn 2.3, Cr 0.035, Mo 0.045, B 1 mg) — plan S12: ADJUST (Zn and Cu). Total zinc 20–25 mg + total copper 1.5–2 mg; remaining equal. SLC30A8 C/C + rs11558471 A/A (genetically useful zinc); GPX1 Pro198Leu (selenium already suitable, without Brazil nuts); SLC39A8 C/C (manganese without adjustment); DAO is a copper enzyme (AOC1 het). Total zinc 20–25 mg + total copper 1.5–2 mg, ≥ 2 h away from NAC, Ca-AMG and fiber (suggestion, not applied).	Genetic plan (S12) <i>moderate</i>	SLC30A8: high GPX1: high SLC39A8: high AOC1 (DAO): high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
ferrous sulfate Adjust dose	HFE: H63D het TMPRSS6/TF: small effect	only with confirmed deficit	HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).	Genetic plan (b.3); HFE guidelines (EASL 2022) <i>moderate</i>	HFE: high TMPRSS6/TF: high
potassium bicarbonate Monitoring	Sal (AGT/ADD1/GNB3/CYP11B2): sensitive (modest)	potassium only for food	Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control. To prefer vegetables and legumes (suggestion, not applied).	Genetic plan (b.3, f.12) <i>moderate (interaction)</i>	Salt (AGT/ADD1/GNB3/CYP11B2): high
potassium citrate Monitoring	Sal (AGT/ADD1/GNB3/CYP11B2): sensitive (modest)	potassium only for food	Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control. To prefer vegetables and legumes (suggestion, not applied).	Genetic plan (b.3, f.12) <i>moderate (interaction)</i>	Salt (AGT/ADD1/GNB3/CYP11B2): high
potassium chloride Monitoring	Sal (AGT/ADD1/GNB3/CYP11B2): sensitive (modest)	potassium only for food	Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control. To prefer vegetables and legumes (suggestion, not applied).	Genetic plan (b.3, f.12) <i>moderate (interaction)</i>	Salt (AGT/ADD1/GNB3/CYP11B2): high
salt tablets (sodium chloride) Monitoring	Sal (AGT/ADD1/GNB3/CYP11B2): sensitive (modest)	avoid routine use	Genetic sensitivity to salt: salt tablets only for intense and prolonged sweat.	Genetic plan (e.3) <i>moderate</i>	Salt (AGT/ADD1/GNB3/CYP11B2): high
potassium gluconate Monitoring	Sal (AGT/ADD1/GNB3/CYP11B2): sensitive (modest)	potassium only for food	Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control. To prefer vegetables and legumes (suggestion, not applied).	Genetic plan (b.3, f.12) <i>moderate (interaction)</i>	Salt (AGT/ADD1/GNB3/CYP11B2): high
salt with potassium chloride (salt substitute) Monitoring	Sal (AGT/ADD1/GNB3/CYP11B2): sensitive (modest)	potassium only for food	Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control. To prefer vegetables and legumes (suggestion, not applied).	Genetic plan (b.3, f.12) <i>moderate (interaction)</i>	Salt (AGT/ADD1/GNB3/CYP11B2): high
Magnesium aspartate Normal	TRPM6: typical	typical absorption	TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.	TRPM6 (Song 2009) studies; genetic plan (S13) <i>weak</i>	TRPM6: high
Magnesium l-threonate Normal	TRPM6: typical	typical absorption	TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.	TRPM6 (Song 2009) studies; genetic plan (S13) <i>weak</i>	TRPM6: high
zinc acetate (chipelets) ☯ Normal	SLC30A8: C/C (usual zinc)	genetically useful zinc	SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied).	Kanoni 2011 (CHARGE); genetic plan (S12) <i>moderate</i>	SLC30A8: high
★ magnesium acetyl-taurate Normal	TRPM6: typical	typical absorption	Current stack (750 mg) — plan S13: KEEP. The same. TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.	TRPM6 (Song 2009) studies; genetic plan (S13) <i>weak</i>	TRPM6: high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
lithium aspartate Normal	APOE: ε3/ε3	without a safety gene	No PGx for lithium in microdose; APOE ε3/ε3; Alzheimer's PGS of 39 variants p91 (genomic p52 to p85): weak rational. With telmisartan (sodium lithium) and dapagliflozin (low-o): pause in days of vomiting/diarrhea.	Genetic plan (S15, f.19) <i>weak</i>	APOE: high
copper bisglycinate Normal	SLC30A8: C/C (usual zinc) AOC1 (DAO): slightly ↓	useful if zinc rises	Necessary to maintain Zn:Cu ≈ 10–15:1 if zinc rises (SLC30A8); DAO is a copper enzyme (AOC1 het). Total 1.5–2 mg/day (suggestion, not applied).	Genetic plan (S12) <i>moderate</i>	SLC30A8: high AOC1 (DAO): high
zinc bisglycinate ☺ Normal	SLC30A8: C/C (usual zinc)	genetically useful zinc	SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied).	Kanoni 2011 (CHARGE); genetic plan (S12) <i>moderate</i>	SLC30A8: high
calcium carbonate Normal	VDR: typical GC (DBP): favourable HFE: H63D het	typical absorption	VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.	NIH ODS (calcium); genetic plan (S14) <i>weak</i>	VDR: high GC (DBP): high HFE: high
calcium citrate Normal	VDR: typical GC (DBP): favourable HFE: H63D het	typical absorption	VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.	NIH ODS (calcium); genetic plan (S14) <i>weak</i>	VDR: high GC (DBP): high HFE: high
magnesium citrate Normal	TRPM6: typical	typical absorption	TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.	TRPM6 (Song 2009) studies; genetic plan (S13) <i>weak</i>	TRPM6: high
zinc citrate ☺ Normal	SLC30A8: C/C (usual zinc)	genetically useful zinc	SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied).	Kanoni 2011 (CHARGE); genetic plan (S12) <i>moderate</i>	SLC30A8: high
magnesium chloride Normal	TRPM6: typical	typical absorption	TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.	TRPM6 (Song 2009) studies; genetic plan (S13) <i>weak</i>	TRPM6: high
calcium + magnesium Normal	VDR: typical GC (DBP): favourable HFE: H63D het	typical absorption	VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.	NIH ODS (calcium); genetic plan (S14) <i>weak</i>	VDR: high GC (DBP): high HFE: high
Algal calcium (Lithothamnion) Normal	VDR: typical GC (DBP): favourable HFE: H63D het	typical absorption	VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.	NIH ODS (calcium); genetic plan (S14) <i>weak</i>	VDR: high GC (DBP): high HFE: high
choral calcium Normal	VDR: typical GC (DBP): favourable HFE: H63D het	typical absorption	VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.	NIH ODS (calcium); genetic plan (S14) <i>weak</i>	VDR: high GC (DBP): high HFE: high
calcium-magnesium-zinc Normal	VDR: typical GC (DBP): favourable HFE: H63D het	typical absorption	VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.	NIH ODS (calcium); genetic plan (S14) <i>weak</i>	VDR: high GC (DBP): high HFE: high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
★ magnesium glycinate Normal	TRPM6: typical	typical absorption	Current stack (500 mg) — plan S13: KEEP. The same. TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.	TRPM6 (Song 2009) studies; genetic plan (S13) <i>weak</i>	TRPM6: high
copper gluconate Normal	SLC30A8: C/C (usual zinc) AOC1 (DAO): slightly ↓	useful if zinc rises	Necessary to maintain Zn:Cu ≈ 10–15:1 if zinc rises (SLC30A8); DAO is a copper enzyme (AOC1 het). Total 1.5–2 mg/day (suggestion, not applied).	Genetic plan (S12) <i>moderate</i>	SLC30A8: high AOC1 (DAO): high
zinc gluconate ☺ Normal	SLC30A8: C/C (usual zinc)	genetically useful zinc	SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied).	Kanoni 2011 (CHARGE); genetic plan (S12) <i>moderate</i>	SLC30A8: high
hydroxyapatite (MCHC) Normal	VDR: typical GC (DBP): favourable HFE: H63D het	typical absorption	VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.	NIH ODS (calcium); genetic plan (S14) <i>weak</i>	VDR: high GC (DBP): high HFE: high
iodine + selenium (thyroid) Normal	GPX1: Pro198Leu het	Nutritional dose	GPX1 Pro198Leu het: activity slightly lower, but without benefit of high doses; selenium > 200 µg/day is associated with 2 type diabetes. The stack already gives 55 µg; without Brazil nuts.	SELECT; genetic plan (S12) <i>weak</i>	GPX1: high
calcium lactate/ gluconate Normal	VDR: typical GC (DBP): favourable HFE: H63D het	typical absorption	VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.	NIH ODS (calcium); genetic plan (S14) <i>weak</i>	VDR: high GC (DBP): high HFE: high
selenium yeast Normal	GPX1: Pro198Leu het	Nutritional dose	GPX1 Pro198Leu het: activity slightly lower, but without benefit of high doses; selenium > 200 µg/day is associated with 2 type diabetes. The stack already gives 55 µg; without Brazil nuts.	SELECT; genetic plan (S12) <i>weak</i>	GPX1: high
magnesium + zinc Normal	TRPM6: typical	typical absorption	TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.	TRPM6 (Song 2009) studies; genetic plan (S13) <i>weak</i>	TRPM6: high
magnesium malate Normal	TRPM6: typical	typical absorption	TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.	TRPM6 (Song 2009) studies; genetic plan (S13) <i>weak</i>	TRPM6: high
manganese (bisglycine/ gluconate) Normal	SLC39A8: normal	Normal transport	SLC39A8 A391T absent (C/C): manganese without adjustment; not exceeding 11 mg/day.	SLC39A8 studies (Park 2015) <i>moderate</i>	SLC39A8: high
zinc monomethionine ☺ Normal	SLC30A8: C/C (usual zinc)	genetically useful zinc	SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied).	Kanoni 2011 (CHARGE); genetic plan (S12) <i>moderate</i>	SLC30A8: high
★ lithium orotate (low dose) Normal	APOE: ε3/ε3	without a safety gene	Current stack (5 mg) — plan S15: KEEP. Similarly; pause in days of vomiting/ diarrhea (telmisartan/dapagliflozin). No PGx for lithium in microdose; APOE ε3/ε3; Alzheimer's PGS of 39 variants p91 (genomic p52 to p85): weak rational. With telmisartan (sodium lithium) and dapagliflozin (low-o): pause in days of vomiting/diarrhea.	Genetic plan (S15, f.19) <i>weak</i>	APOE: high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
magnesium orotato Normal	TRPM6 : typical	typical absorption	TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.	TRPM6 (Song 2009) studies; genetic plan (S13) <i>weak</i>	TRPM6: high
zinc picolinate ☺ Normal	SLC30A8 : C/C (usual zinc)	genetically useful zinc	SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied).	Kanoni 2011 (CHARGE); genetic plan (S12) <i>moderate</i>	SLC30A8: high
sodium selenate Normal	GPX1 : Pro198Leu het	Nutritional dose	GPX1 Pro198Leu het: activity slightly lower, but without benefit of high doses; selenium > 200 µg/day is associated with 2 type diabetes. The stack already gives 55 µg; without Brazil nuts.	SELECT; genetic plan (S12) <i>weak</i>	GPX1: high
sodium selenite Normal	GPX1 : Pro198Leu het	Nutritional dose	GPX1 Pro198Leu het: activity slightly lower, but without benefit of high doses; selenium > 200 µg/day is associated with 2 type diabetes. The stack already gives 55 µg; without Brazil nuts.	SELECT; genetic plan (S12) <i>weak</i>	GPX1: high
selenomethionine Normal	GPX1 : Pro198Leu het	Nutritional dose	GPX1 Pro198Leu het: activity slightly lower, but without benefit of high doses; selenium > 200 µg/day is associated with 2 type diabetes. The stack already gives 55 µg; without Brazil nuts.	SELECT; genetic plan (S12) <i>weak</i>	GPX1: high
copper sulfate Normal	SLC30A8 : C/C (usual zinc) AOC1 (DAO) : slightly ↓	useful if zinc rises	Necessary to maintain Zn:Cu ≈ 10–15:1 if zinc rises (SLC30A8); DAO is a copper enzyme (AOC1 het). Total 1.5–2 mg/day (suggestion, not applied).	Genetic plan (S12) <i>moderate</i>	SLC30A8: high AOC1 (DAO): high
magnesium sulfate (epsom salt) Normal	TRPM6 : typical	typical absorption	TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.	TRPM6 (Song 2009) studies; genetic plan (S13) <i>weak</i>	TRPM6: high
zinc sulfate ☺ Normal	SLC30A8 : C/C (usual zinc)	genetically useful zinc	SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied).	Kanoni 2011 (CHARGE); genetic plan (S12) <i>moderate</i>	SLC30A8: high
magnesium taurate Normal	TRPM6 : typical	typical absorption	TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.	TRPM6 (Song 2009) studies; genetic plan (S13) <i>weak</i>	TRPM6: high
zinc + copper (combined) ☺ Normal	SLC30A8 : C/C (usual zinc)	genetically useful zinc	SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied).	Kanoni 2011 (CHARGE); genetic plan (S12) <i>moderate</i>	SLC30A8: high
zinc carnosine (polaprezinc) ☺ Normal	SLC30A8 : C/C (usual zinc) CNDP1 : indeterminate	useful zinc	Zinc (SLC30A8 C/C) with local carnosine in the stomach; carnosinase (CNDP1) is not evaluated, but here the effect is topic.	Kanoni 2011; polaprezinc trials <i>weak</i>	SLC30A8: high CNDP1: non-determinable
magnesium oxide Normal	TRPM6 : typical	typical absorption	TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.	TRPM6 (Song 2009) studies; genetic plan (S13) <i>weak</i>	TRPM6: high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
zinc oxide ⓘ Normal	SLC30A8: C/C (usual zinc)	genetically useful zinc	SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied).	Kanoni 2011 (CHARGE); genetic plan (S12) moderate	SLC30A8: high

No known genetic relevance (17):

boron (citrate/glycinate)	ionic mineral droplets (various)	colloidal silver
strontium citrate (bone)	potassium iodide (thyroid)	Lugol solution (thyroid)
chromium GTF (light)	algal iodine (kelp) (thyroid)	silica / orthosilic acid (renal)
organic sulfur (MSM)	iodine (thyroid)	sodium/electrolytes (renal)
Phosphorus (phosphate) (renal)	molybdenum (glycinate)	vanadium (vanadyl sulfate)
organic germanium (Ge-132)	chromium picolinate	

3.3 Amino acids, proteins and derivatives — 80 supplements

dose adjustment: 2 · Monitoring: 6 · indeterminate: 2 · normal: 30 · without known genetic relevance: 40

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
★ TMG (trimethylglycine, betaine anhydrous) ⓘ Adjust dose	BHMT: R239Q het GNMT: TT MTHFR: 677CT/1298AC	adjust dose (plan)	Current stack (1500 mg) — plan S7: ADJUST (subir). Up to 3 g/day in 2 tomas; do not exceed 3 g (LDL). BHMT R239Q het, GNMT TT and MTHFR compound: the BHMT pathway compensates for folate. 3 g/day (≥ 4 g up to LDL) (suggestion, not applied).	Genetic plan (S7) <i>moderate</i>	BHMT: high GNMT: high MTHFR: high
★ glycine ⓘ Adjust dose	CPS1: low glycine GSTM1: probable null GNMT: TT	adjust dose (plan)	Current stack (3 g) — plan S8: ADJUST (subir). Up to 5 g/day (3 g at bedtime + 2 g in the morning). CPS1 rs715 T/T + rs1047891 C/C: genetically lower plasma glycine; Probable null GSTM1 (glycine is a glutathione substrate); GNMT TT. 5 g/day in total (suggestion, not applied).	GWAS glycine (CPS1); genetic plan (S8) <i>moderate</i>	CPS1: high GSTM1: moderate GNMT: high
BCAA (leucine, isoleucine, valine) Monitoring	TCF7L2/FTO/GCKR: unfavourable	the genetic plan advises to add	High BCAA are associated with insulin resistance; with risk TCF7L2/FTO and genetic levels of leucine/valine tendentially high, prefer complete protein (suggestion, not applied).	Genetic plan (b.3) <i>weak</i>	TCF7L2/FTO/GCKR: high
L-isoleucine Monitoring	TCF7L2/FTO/GCKR: unfavourable	the genetic plan advises to add	High BCAA are associated with insulin resistance; with risk TCF7L2/FTO and genetic levels of leucine/valine tendentially high, prefer complete protein (suggestion, not applied).	Genetic plan (b.3) <i>weak</i>	TCF7L2/FTO/GCKR: high
L-leucine Monitoring	TCF7L2/FTO/GCKR: unfavourable	the genetic plan advises to add	High BCAA are associated with insulin resistance; with risk TCF7L2/FTO and genetic levels of leucine/valine tendentially high, prefer complete protein (suggestion, not applied).	Genetic plan (b.3) <i>weak</i>	TCF7L2/FTO/GCKR: high
L-methionine Monitoring	COMT: Val/Val (high) GNMT: TT MTHFR: 677CT/1298AC	avoiding isolated supplement	Extra methionine turns SAM and then homocysteine; with compound MTHFR, GNMT TT and COMT Val/Val (uses up SAM) the genetic score of homocysteine is at the 92nd percentile. No indication.	Genetic plan (f.2) <i>weak</i>	COMT: high GNMT: high MTHFR: high
L-valine Monitoring	TCF7L2/FTO/GCKR: unfavourable	the genetic plan advises to add	High BCAA are associated with insulin resistance; with risk TCF7L2/FTO and genetic levels of leucine/valine tendentially high, prefer complete protein (suggestion, not applied).	Genetic plan (b.3) <i>weak</i>	TCF7L2/FTO/GCKR: high
SAMe (S-adenosylmethionine) Monitoring	COMT: Val/Val (high) GNMT: TT MTHFR: 677CT/1298AC	the genetic plan advises to add	COMT Val/Val + GNMT TT: plus SAM tends to be transformed into homocysteine (genetic score p92). For methylation, prefer TMG + 5-MTHF + creatine (suggestion, not applied).	Genetic plan (b.3) <i>weak</i>	COMT: high GNMT: high MTHFR: high
★ L-carnosine Undetermined	CNDP1: indeterminate	indeterminate (carnosinase)	Current stack (2 g) — plan S21: KEEP. 1 g × 2 (non-evaluable carnosinase CNDP1). Serum carnosinase (CNDP1) is regulated by a repetition (CTG) _n that the chip does not read: the fraction of oral carnosine that arrives intact is not predictable. The plan maintains 2 g (1 g × 2).	CNDP1 Literature (Janssen 2005) <i>not determinable</i>	CNDP1: non-determinable
anserine Undetermined	CNDP1: indeterminate	indeterminate (carnosinase)	Serum carnosinase (CNDP1) is regulated by a repetition (CTG) _n that the chip does not read: the fraction of oral carnosine that arrives intact is not predictable. The plan maintains 2 g (1 g × 2).	CNDP1 Literature (Janssen 2005) <i>not determinable</i>	CNDP1: non-determinable
AAKG (alpha-ketoglutarate arginine) Normal	NOS3: –786CT + 298GT	rational NO (small effect)	NOS3 –786 T/C + Glu298Asp G/T: a slightly smaller production of NO. Oral arginine has low bioavailability; with tadalafil the vasodilator effect is added (mild).	NOS3 studies; NIH ODS (arginine) <i>weak</i>	NOS3: high
DL-phenylalanine (DLPA) Normal	PAH: no variants	normal for the tested variants	No pathogenic variants of PAH among those tested (improbable phenylketonuria).	ClinVar (chip) <i>moderate</i>	PAH: partial

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
GlyNAC (glycine + NAC) ⊕ Normal	CPS1 : low glycine GSTM1 : probable null GNMT : TT	genetically useful glycine	CPS1 rs715 T/T + rs1047891 C/C: genetically lower plasma glycine; Probable null GSTM1 (glycine is a glutathione substrate); GNMT TT. 5 g/day in total (suggestion, not applied).	GWAS glycine (CPS1); genetic plan (S8) <i>moderate</i>	CPS1: high GSTM1: moderate GNMT: high
L-arginine Normal	NOS3 : –786CT + 298GT	rational NO (small effect)	NOS3 –786 T/C + Glu298Asp G/T: a slightly smaller production of NO. Oral arginine has low bioavailability; with tadalafil the vasodilator effect is added (mild).	NOS3 studies; NIH ODS (arginine) <i>weak</i>	NOS3: high
L-carnitine (tartrate) Normal	SLC22A4/5 (OCTN1/2) : typical FMO3 : typical	Typical transport; TMAO form	Typical OCTN2. Typical FMO3: oral carnitine generates TMAO by the microbiota; with genetic vascular priority, moderate and non-chronic dose at a high dose.	Genetic plan (S19); Koeth 2013 <i>weak</i>	SLC22A4/5 (OCTN1/2): high FMO3: high
L-carnitine base Normal	SLC22A4/5 (OCTN1/2) : typical FMO3 : typical	Typical transport; TMAO form	Typical OCTN2. Typical FMO3: oral carnitine generates TMAO by the microbiota; with genetic vascular priority, moderate and non-chronic dose at a high dose.	Genetic plan (S19); Koeth 2013 <i>weak</i>	SLC22A4/5 (OCTN1/2): high FMO3: high
L-citrulline ⊕ Normal	NOS3 : –786CT + 298GT	rational NO	NOS3 –786 T/C + Glu298Asp: citrulline rises arginine better than arginine itself. With tadalafil, slight vasodilator sum.	NOS3 studies; citrulline trials <i>weak</i>	NOS3: high
L-phenylalanine Normal	PAH : no variants	normal for the tested variants	No pathogenic variants of PAH among those tested (improbable phenylketonuria).	ClinVar (chip) <i>moderate</i>	PAH: partial
L-tyrosine Normal	COMT : Val/Val (high)	without robust interaction	COMT Val/Val: rapid dopamine degradation; tyrosine is safe, without robust evidence of genotype-dependent effect.	COMT × tyrosine studies (small) <i>weak</i>	COMT: high
N-acetyl-L-tyrosine (NALT) Normal	COMT : Val/Val (high)	without robust interaction	COMT Val/Val: rapid dopamine degradation; tyrosine is safe, without robust evidence of genotype-dependent effect.	COMT × tyrosine studies (small) <i>weak</i>	COMT: high
★ NAC (N-acetylcysteine) ⊕ Normal	GSTM1 : probable null GSTA1 : *B/*B GPX1 : Pro198Leu het	useful support for glutathione	Current stack (1800 mg) — plan S17: KEEP. 1200–1800 mg, far from training. Probable null GSTM1, GSTA1 *B/*B, GCLM –588 C/T and GPX1 Pro198Leu: weak II → NAC phase is coherent. Far from the training; ≥ 2 h away from zinc/copper.	Genetic plan (S17) <i>moderate</i>	GSTM1: moderate GSTA1: high GPX1: high
S-acetyl-glutathione Normal	GSTM1 : probable null GSTA1 : *B/*B	rational (redundant with NAC + glycine)	Probable null GSTM1/GSTA1 *B/*B gives rationale; with NAC + glycine is redundant.	Genetic plan (d) <i>weak</i>	GSTM1: moderate GSTA1: high
★ acetyl-L-carnitine (ALCAR) Normal	SLC22A4/5 (OCTN1/2) : typical FMO3 : typical	Typical transport	Current stack (500 mg) — plan S19: KEEP. 500 mg in the morning. OCTN2 (SLC22A5) typical; CPT2 V368I het; FMO3 typical (some TMAO). Keep 500 mg in the morning.	Genetic plan (S19) <i>weak</i>	SLC22A4/5 (OCTN1/2): high FMO3: high
betaine HCl (with pepsin) ⊕ Normal	BHMT : R239Q het GNMT : TT MTHFR : 677CT/1298AC	genetically useful betaine pathway	BHMT R239Q het, GNMT TT and MTHFR compound: the BHMT pathway compensates for folate. 3 g/day (≥ 4 g up to LDL) (suggestion, not applied).	Genetic plan (S7) <i>moderate</i>	BHMT: high GNMT: high MTHFR: high
citrulline malate ⊕ Normal	NOS3 : –786CT + 298GT	rational NO	NOS3 –786 T/C + Glu298Asp: citrulline rises arginine better than arginine itself. With tadalafil, slight vasodilator sum.	NOS3 studies; citrulline trials <i>weak</i>	NOS3: high
★ bovine hydrolyzed collagen Normal	COL1A1/COL5A1 : normal	without risk alleles	Current stack (20 g) — plan S9: KEEP. The same (glycine source). COL1A1 Sp1 C/C and COL5A1 T/T: without risk alleles; useful source of glycine (CPS1).	Genetic plan (S9) <i>weak</i>	COL1A1/COL5A1: high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
marine collagen Normal	COL1A1/COL5A1: normal	without risk alleles	COL1A1 Sp1 C/C and COL5A1 T/T: without risk alleles; useful source of glycine (CPS1).	Genetic plan (S9) <i>weak</i>	COL1A1/COL5A1: high
creatine HCl ☺ Normal	AMPD1: normal ACTN3: R/X GNMT: TT	useful (muscle + save SAM)	Normal AMPD1; ACTN3 R/X; high fragility genetic score. Exogenous creatine saves SAM spent on endogenous synthesis (GAMT), useful with compound MTHFR/GNMT TT. Monohydrate 3–5 g is the form with evidence.	Genetic plan (S10); ISSN (<i>muscle</i>) · <i>moderate</i> (<i>methylation</i>)	AMPD1: high ACTN3: high GNMT: high
creatine ethyl ester ☺ Normal	AMPD1: normal ACTN3: R/X GNMT: TT	useful (muscle + save SAM)	Normal AMPD1; ACTN3 R/X; high fragility genetic score. Exogenous creatine saves SAM spent on endogenous synthesis (GAMT), useful with compound MTHFR/GNMT TT. Monohydrate 3–5 g is the form with evidence.	Genetic plan (S10); ISSN (<i>muscle</i>) · <i>moderate</i> (<i>methylation</i>)	AMPD1: high ACTN3: high GNMT: high
★ creatine monohydrate ☺ Normal	AMPD1: normal ACTN3: R/X GNMT: TT	useful (muscle + save SAM)	Current stack (5 g) — plan S10: KEEP. The same. Normal AMPD1; ACTN3 R/X; high fragility genetic score. Exogenous creatine saves SAM spent on endogenous synthesis (GAMT), useful with compound MTHFR/GNMT TT. Monohydrate 3–5 g is the form with evidence.	Genetic plan (S10); ISSN (<i>muscle</i>) · <i>moderate</i> (<i>methylation</i>)	AMPD1: high ACTN3: high GNMT: high
buffered creatine ☺ Normal	AMPD1: normal ACTN3: R/X GNMT: TT	useful (muscle + save SAM)	Normal AMPD1; ACTN3 R/X; high fragility genetic score. Exogenous creatine saves SAM spent on endogenous synthesis (GAMT), useful with compound MTHFR/GNMT TT. Monohydrate 3–5 g is the form with evidence.	Genetic plan (S10); ISSN (<i>muscle</i>) · <i>moderate</i> (<i>methylation</i>)	AMPD1: high ACTN3: high GNMT: high
glycine-propionyl-L-carnitine (GPLC) Normal	SLC22A4/5 (OCTN1/2): typical FMO3: typical	Typical transport; TMAO form	Typical OCTN2. Typical FMO3: oral carnitine generates TMAO by the microbiota; with genetic vascular priority, moderate and non-chronic dose at a high dose.	Genetic plan (S19); Koeth 2013 <i>weak</i>	SLC22A4/5 (OCTN1/2): high FMO3: high
glutathione liposomal Normal	GSTM1: probable null GSTA1: *B/*B	rational (redundant with NAC + glycine)	Probable null GSTM1/GSTA1 *B/*B gives rationale; with NAC + glycine is redundant.	Genetic plan (d) <i>weak</i>	GSTM1: moderate GSTA1: high
reduced oral glutathione Normal	GSTM1: probable null GSTA1: *B/*B	rational (redundant with NAC + glycine)	Probable null GSTM1/GSTA1 *B/*B gives rationale; with NAC + glycine is redundant.	Genetic plan (d) <i>weak</i>	GSTM1: moderate GSTA1: high
creatine nitrate ☺ Normal	AMPD1: normal ACTN3: R/X GNMT: TT	useful (muscle + save SAM)	Normal AMPD1; ACTN3 R/X; high fragility genetic score. Exogenous creatine saves SAM spent on endogenous synthesis (GAMT), useful with compound MTHFR/GNMT TT. Monohydrate 3–5 g is the form with evidence.	Genetic plan (S10); ISSN (<i>muscle</i>) · <i>moderate</i> (<i>methylation</i>)	AMPD1: high ACTN3: high GNMT: high
propionyl-L-carnitine Normal	SLC22A4/5 (OCTN1/2): typical FMO3: typical	Typical transport; TMAO form	Typical OCTN2. Typical FMO3: oral carnitine generates TMAO by the microbiota; with genetic vascular priority, moderate and non-chronic dose at a high dose.	Genetic plan (S19); Koeth 2013 <i>weak</i>	SLC22A4/5 (OCTN1/2): high FMO3: high
whey protein Normal	LCT: persistent	Lactose tolerated	LCT –13910 C/T: persistence of lactase, without genetic lactose intolerance.	Genetic plan (e.6) <i>strong</i>	LCT: high
specific bioactive collagen peptides Normal	COL1A1/COL5A1: normal	without risk alleles	COL1A1 Sp1 C/C and COL5A1 T/T: without risk alleles; useful source of glycine (CPS1).	Genetic plan (S9) <i>weak</i>	COL1A1/COL5A1: high
whey hydrolyzed Normal	LCT: persistent	Lactose tolerated	LCT –13910 C/T: persistence of lactase, without genetic lactose intolerance.	Genetic plan (e.6) <i>strong</i>	LCT: high
whey isolated Normal	LCT: persistent	Lactose tolerated	LCT –13910 C/T: persistence of lactase, without genetic lactose intolerance.	Genetic plan (e.6) <i>strong</i>	LCT: high

No known genetic relevance (40):

5-HTP (AADC)
agmatin

beta-alanine (CNDP1) — *CNDP1: undetermined*
Bone juice powder

casein micellar
collagen type II undenatured

DMG (dimethylglycine)
EAA (essential amino acids) (various)
GABA
gelatin
glutamine peptide
HICA (alpha-hydroxyisocaproic acid)
hydroxyproline
HMB (beta-hydroxy-betamethylbutyrate)
L-alanine
L-aspartate
L-cysteine (CBS/GSH)

L-glutamine
L-histidine (HDC)
L-lysine
L-ornithine (OTC)
L-proline
L-serine (PHGDH)
L-theanine
L-threonine
L-tryptophan (TPH/IDO)
OKG (ornithine alpha-ketoglutarate)
ornithine-aspartate (LOLA) (urea cycle)

rice protein
meat protein (beef insulator)
hemp protein
pea protein
cricket protein (insects)
egg protein (albumin)
soy protein isolated
sarcosin
★ **taurine**
D-aspartic acid
fermented gamma-aminobutyric acid
glutamic acid

3.4 Fatty acids, oils and phospholipids — 45 supplements

prevention: 1 · dose adjustment: 1 · Monitoring: 9 · normal: 19 · without known genetic relevance: 15

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
cod liver oil Avoid	BCO1 : slightly ↓ conversion PNPLA3/Pi*MZ (liver) : unfavourable	avoid preformed retinol	The conversion of β-carotene is only slightly reduced (BCO1 R267S het; without A379V), so there is no genetic need for retinol. With the isotretinoin of the stack, retinol supplemental sum retinoids (toxicity, transaminases); the genetic liver (PNPLA3/Pi*MZ) reinforces. Obtain vitamin A from carotenoids (suggestion, not applied).	NIH ODS (vitamin A); genetic plan (b.3); isotretinoin label <i>strong (interaction) · weak (BCO1)</i>	BCO1: high PNPLA3/Pi*MZ (liver): moderate
★ fish oil (EPA+DHA, TG/rTG) ⊕ Adjust dose	FADS1/FADS2 : high activity APOE : ε3/ε3	adjust dose (plan)	Current stack (EPA 1125 + DHA 875 mg) — plan S1: ADJUST. Up to 3.2–4 g/day of EPA+DHA (Coordination Note), form rTG, with the meal fatter and far from psyllium. FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied).	Genetic plan (S1 + coordination note); FADS studies <i>moderate</i>	FADS1/FADS2: high APOE: high
CLA (conjugated linoleic acid) Monitoring	PNPLA3/Pi*MZ (liver) : unfavourable TCF7L2/FTO/GCKR : unfavourable	without benefit; possible metabolic worsening	CLA t10,c12 increased insulin resistance and hepatic fat in trials; with PNPLA3/TCF7L2 at risk, avoid.	CLA (Risérus 2002) trials <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate TCF7L2/FTO/GCKR: high
Sewer oil (GLA) Monitoring	FADS1/FADS2 : high activity PNPLA3/Pi*MZ (liver) : unfavourable	GLA→AA + alkaloids	High FADS1 (GLA→AA) and possible presence of hepatotoxic pyrrolizidine alkaloids (genetic liver): only certified products without PA.	FADS1 studies; EMA (PA) <i>weak</i>	FADS1/FADS2: high PNPLA3/Pi*MZ (liver): moderate
coconut oil Monitoring	CETP/9p21/APOB : genetic vascular risk	saturated fat (LDL)	Genetic priority to low LDL/ApoB (high CETP, 9p21, APOB): coconut oil rises to LDL; not use as a supplement.	Genetic plan (e.3) <i>moderate</i>	CETP/9p21/APOB: high
safflower oil Monitoring	FADS1/FADS2 : high activity	Omega-6 unnecessary	The extra LA is converted effectively into AA (high FADS); the diet already has enough omega-6. Prefer omega-3 isolated.	FADS × LA studies <i>weak</i>	FADS1/FADS2: high
shark liver oil (alkylglycerols) Monitoring	BCO1 : slightly ↓ conversion PNPLA3/Pi*MZ (liver) : unfavourable	confirm retinol content	It can contain vitamin A preformed in variable quantity; with isotretinoin and genetic liver, only one product without retinol (suggestion, not applied).	NIH ODS (vitamin A) <i>weak</i>	BCO1: high PNPLA3/Pi*MZ (liver): moderate
Grapeseed oil Monitoring	FADS1/FADS2 : high activity	Omega-6 unnecessary	The extra LA is converted effectively into AA (high FADS); the diet already has enough omega-6. Prefer omega-3 isolated.	FADS × LA studies <i>weak</i>	FADS1/FADS2: high
Blackcurrant oil (GLA) Monitoring	FADS1/FADS2 : high activity	GLA can turn AA	With high activity FADS1, part of GLA→DGLA goes to arachidonic acid (pro-inflammatory), reducing the benefit. No genetic indication.	FADS1 × GLA (Gillingham 2013) studies <i>weak</i>	FADS1/FADS2: high
Primule oil (GLA) Monitoring	FADS1/FADS2 : high activity	GLA can turn AA	With high activity FADS1, part of GLA→DGLA goes to arachidonic acid (pro-inflammatory), reducing the benefit. No genetic indication.	FADS1 × GLA (Gillingham 2013) studies <i>weak</i>	FADS1/FADS2: high
Omega-3-6-9 Monitoring	FADS1/FADS2 : high activity	Omega-6 unnecessary	The extra LA is converted effectively into AA (high FADS); the diet already has enough omega-6. Prefer omega-3 isolated.	FADS × LA studies <i>weak</i>	FADS1/FADS2: high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
DHA concentrate ⊕ Normal	FADS1/FADS2: high activity APOE: ε3/ε3	EPA/DHA useful preformed	FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied).	Genetic plan (S1 + coordination note); FADS studies <i>moderate</i>	FADS1/FADS2: high APOE: high
EPA concentrate ⊕ Normal	FADS1/FADS2: high activity APOE: ε3/ε3	EPA/DHA useful preformed	FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied).	Genetic plan (S1 + coordination note); FADS studies <i>moderate</i>	FADS1/FADS2: high APOE: high
Plant stanols ⊕ Normal	ABCG5/ABCG8: normal CETP/9p21/APOB: genetic vascular risk	useful for LDL (without hyperabsorption)	ABCG5/8 without hyperabsorption variants (safe); low genetic priority to LDL (CETP/9p21/APOB). 2 g/day lower LDL ~8–10%; compatible with ezetimibe (suggestion, not applied).	Genetic plan (1 annex); consensus EAS phytosterols <i>moderate</i>	ABCG5/ABCG8: high CETP/9p21/APOB: high
phytosterol (beta-sitosterol) ⊕ Normal	ABCG5/ABCG8: normal CETP/9p21/APOB: genetic vascular risk	useful for LDL (without hyperabsorption)	ABCG5/8 without hyperabsorption variants (safe); low genetic priority to LDL (CETP/9p21/APOB). 2 g/day lower LDL ~8–10%; compatible with ezetimibe (suggestion, not applied).	Genetic plan (1 annex); consensus EAS phytosterols <i>moderate</i>	ABCG5/ABCG8: high CETP/9p21/APOB: high
krill phospholipids ⊕ Normal	FADS1/FADS2: high activity APOE: ε3/ε3	EPA/DHA useful preformed	FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied).	Genetic plan (S1 + coordination note); FADS studies <i>moderate</i>	FADS1/FADS2: high APOE: high
caprylic acid (C8) Normal	APOE: ε3/ε3	typical response	APOE ε3/ε3: typical ketone metabolism (the ketogenic advantage described in ε4 does not apply).	APOE × ketones studies <i>weak</i>	APOE: high
oleic acid / olive oil in capsules ⊕ Normal	PON1: low expression	useful (PON1)	Low expression PON1: olive oil polyphenols increase the activity of PON1; extra virgin olive oil in food is better than capsules.	Genetic plan (S32, e.2) <i>weak</i>	PON1: high
MCT oil Normal	APOE: ε3/ε3	typical response	APOE ε3/ε3: typical ketone metabolism (the ketogenic advantage described in ε4 does not apply).	APOE × ketones studies <i>weak</i>	APOE: high
Algal oil (DHA vegan) ⊕ Normal	FADS1/FADS2: high activity APOE: ε3/ε3	EPA/DHA useful preformed	FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied).	Genetic plan (S1 + coordination note); FADS studies <i>moderate</i>	FADS1/FADS2: high APOE: high
chia oil Normal	FADS1/FADS2: high activity	good conversion ALA→EPA	With high activity FADS, the conversion of ALA/SDA to EPA is one of the best; however, it does not replace the preformed DHA.	FADS × ALA studies <i>moderate</i>	FADS1/FADS2: high
Equium oil (SDA) Normal	FADS1/FADS2: high activity	good conversion ALA→EPA	With high activity FADS, the conversion of ALA/SDA to EPA is one of the best; however, it does not replace the preformed DHA.	FADS × ALA studies <i>moderate</i>	FADS1/FADS2: high
Focus oil ⊕ Normal	FADS1/FADS2: high activity APOE: ε3/ε3	EPA/DHA useful preformed	FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied).	Genetic plan (S1 + coordination note); FADS studies <i>moderate</i>	FADS1/FADS2: high APOE: high
krill oil ⊕ Normal	FADS1/FADS2: high activity APOE: ε3/ε3	EPA/DHA useful preformed	FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied).	Genetic plan (S1 + coordination note); FADS studies <i>moderate</i>	FADS1/FADS2: high APOE: high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
linseed oil (ALA) Normal	FADS1/FADS2: high activity	good conversion ALA→EPA	With high activity FADS, the conversion of ALA/SDA to EPA is one of the best; however, it does not replace the preformed DHA.	FADS × ALA studies <i>moderate</i>	FADS1/FADS2: high
Squid oil ⊕ Normal	FADS1/FADS2: high activity APOE: ε3/ε3	EPA/DHA useful preformed	FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied).	Genetic plan (S1 + coordination note); FADS studies <i>moderate</i>	FADS1/FADS2: high APOE: high
Green lips mussel oil ⊕ Normal	FADS1/FADS2: high activity APOE: ε3/ε3	EPA/DHA useful preformed	FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied).	Genetic plan (S1 + coordination note); FADS studies <i>moderate</i>	FADS1/FADS2: high APOE: high
Concentrated fish oil (ethyl esters) ⊕ Normal	FADS1/FADS2: high activity APOE: ε3/ε3	EPA/DHA useful preformed	FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied).	Genetic plan (S1 + coordination note); FADS studies <i>moderate</i>	FADS1/FADS2: high APOE: high
perila oil Normal	FADS1/FADS2: high activity	good conversion ALA→EPA	With high activity FADS, the conversion of ALA/SDA to EPA is one of the best; however, it does not replace the preformed DHA.	FADS × ALA studies <i>moderate</i>	FADS1/FADS2: high
Salmon oil ⊕ Normal	FADS1/FADS2: high activity APOE: ε3/ε3	EPA/DHA useful preformed	FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied).	Genetic plan (S1 + coordination note); FADS studies <i>moderate</i>	FADS1/FADS2: high APOE: high

No known genetic relevance (15):

plant ceramides (phytoceramides)	PEA (palmitoylethanolamide) (FAAH/NAAA)	Nigela oil (black cumin)
squalene	plasmalogens	pine nut oil (pinenic acid)
phosphatidylethanolamine	polycosanol	pumpkin seed oil
phosphatidylinositol	phosphatidic acid	Hemp seed oil (FADS1/2) — <i>FADS1/FADS2: high activity</i>
phosphatidylserine	avocado oil (capsule)	Omega-7 (palmitoleic acid, seabream)

3.5 Polyphenols and other phytochemicals — 91 supplements

prevention: 14 · dose adjustment: 2 · Monitoring: 23 · normal: 17 · without known genetic relevance: 35

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
EGCG / green tea extract Avoid	PNPLA3/Pi*MZ (liver): unfavourable COMT: Val/Val (high) HLA-B*35:01: probable negative UGT1A1: IM (*1/*28)	avoid (liver)	Concentrated green tea extract is a known cause of liver injury (LiverTox, probability B; risk in fasting and > 800 mg EGCG/day). Genetic liver (PNPLA3, MBOAT7, Pi*MZ); EGCG is catechol (COMT Val/Val). In favor: HLA-B*35:01 probable negative. Tea in drink is OK (suggestion, not applied).	LiverTox; EFSA 2018 (catechins); genetic plan (b.3) <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate COMT: high HLA-B*35:01: moderate UGT1A1: high
★ apigenin Avoid	UGT1A1: IM (*1/*28) CYP3A5:expressor (*1/*3) CYP1A2: normal to intermediate CYP2C9: IM (*1/*3)	avoid / pause (test-pause)	Current stack (100 mg) — plan S28: PAUSE (test-pause). Test-pause (current protocol). Test pause for the same reasons of luteolin: inhibits UGT1A1 (*1/*28: up to bilirubin and confuses liver reading) and CYP3A4/P-gp (sirolimus; CYP3A5 expressor). It is not catechol, but also inhibits CYP1A2 (already normal to intermediate: *1F het + rs2472297 C/C), which rises melatonin and caffeine, and CYP2C9 (*1/*3). Pause; pause while ALT is high (current protocol). Coordination note: in the current protocol, the pause lasts while ALT is high; if it returns, 25–50 mg at night, outside the rapamycin window (suggestion, not applied).	Genetic plan (S28 + coordination note) <i>weak</i>	UGT1A1: high CYP3A5: high CYP1A2: high CYP2C9: high
curcumin (95% extract) Avoid	UGT1A1: IM (*1/*28) PNPLA3/Pi*MZ (liver): unfavourable HLA-B*35:01: probable negative CYP3A5:expressor (*1/*3)	avoid/pause	UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).	Genetic plan (S26); LiverTox (curcuma, HLA-B*35:01) <i>weak-moderate</i>	UGT1A1: high PNPLA3/Pi*MZ (liver): moderate HLA-B*35:01: moderate CYP3A5: high
curcumin + piperine Avoid	UGT1A1: IM (*1/*28) PNPLA3/Pi*MZ (liver): unfavourable HLA-B*35:01: probable negative CYP3A5:expressor (*1/*3)	avoid/pause	UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).	Genetic plan (S26); LiverTox (curcuma, HLA-B*35:01) <i>weak-moderate</i>	UGT1A1: high PNPLA3/Pi*MZ (liver): moderate HLA-B*35:01: moderate CYP3A5: high
colloidal curcumin (nanoparticles) Avoid	UGT1A1: IM (*1/*28) PNPLA3/Pi*MZ (liver): unfavourable HLA-B*35:01: probable negative CYP3A5:expressor (*1/*3)	avoid/pause	UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).	Genetic plan (S26); LiverTox (curcuma, HLA-B*35:01) <i>weak-moderate</i>	UGT1A1: high PNPLA3/Pi*MZ (liver): moderate HLA-B*35:01: moderate CYP3A5: high
curcumin with curcuman essential oil Avoid	UGT1A1: IM (*1/*28) PNPLA3/Pi*MZ (liver): unfavourable HLA-B*35:01: probable negative CYP3A5:expressor (*1/*3)	avoid/pause	UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).	Genetic plan (S26); LiverTox (curcuma, HLA-B*35:01) <i>weak-moderate</i>	UGT1A1: high PNPLA3/Pi*MZ (liver): moderate HLA-B*35:01: moderate CYP3A5: high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
curcumin phytosome Avoid	UGT1A1: IM (*1/*28) PNPLA3/Pi*MZ (liver): unfavourable HLA-B*35:01: probable negative CYP3A5: expressor (*1/*3)	avoid/pause	UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).	Genetic plan (S26); LiverTox (curcuma, HLA-B*35:01) <i>weak-moderate</i>	UGT1A1: high PNPLA3/Pi*MZ (liver): moderate HLA-B*35:01: moderate CYP3A5: high
★ solid lipid curcumin (SLCP) Avoid	UGT1A1: IM (*1/*28) PNPLA3/Pi*MZ (liver): unfavourable HLA-B*35:01: probable negative CYP3A5: expressor (*1/*3)	avoid/pause	Current stack (1 g) — plan S26: PAUSE. Pause (current protocol). UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).	Genetic plan (S26); LiverTox (curcuma, HLA-B*35:01) <i>weak-moderate</i>	UGT1A1: high PNPLA3/Pi*MZ (liver): moderate HLA-B*35:01: moderate CYP3A5: high
isoquercytrin enzymatically modified Avoid	UGT1A1: IM (*1/*28) COMT: Val/Val (high) MTHFR: 677CT/1298AC CYP3A5: expressor (*1/*3)	avoid/pause	UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.	Genetic plan (S29 + coordination note) <i>weak</i>	UGT1A1: high COMT: high MTHFR: high CYP3A5: high
★ luteolin Avoid	COMT: Val/Val (high) MTHFR: 677CT/1298AC UGT1A1: IM (*1/*28) CYP2C9: IM (*1/*3)	avoid/pause	Current stack (100 mg) — plan S27: PAUSE. Pause (current protocol). Catechol: methylated by COMT (Val/Val), spent SAM and produces homocysteine (MTHFR compound, p92 score). It inhibits UGT1A1 (*28) and CYP2C9 (*1/*3 IM) in vitro, and CYP3A4/P-gp (sirolimus). Pause; pause while ALT is high (current protocol) (suggestion, not applied).	Genetic plan (S27) <i>weak</i>	COMT: high MTHFR: high UGT1A1: high CYP2C9: high
luteolin + PEA Avoid	COMT: Val/Val (high) MTHFR: 677CT/1298AC UGT1A1: IM (*1/*28) CYP2C9: IM (*1/*3)	avoid/pause	Catechol: methylated by COMT (Val/Val), spent SAM and produces homocysteine (MTHFR compound, p92 score). It inhibits UGT1A1 (*28) and CYP2C9 (*1/*3 IM) in vitro, and CYP3A4/P-gp (sirolimus). Pause; pause while ALT is high (current protocol) (suggestion, not applied).	Genetic plan (S27) <i>weak</i>	COMT: high MTHFR: high UGT1A1: high CYP2C9: high
★ quercetin Avoid	UGT1A1: IM (*1/*28) COMT: Val/Val (high) MTHFR: 677CT/1298AC CYP3A5: expressor (*1/*3)	avoid/pause	Current stack (250 mg (3x/month)) — plan S29: PAUSE (suggestion, not applied). It is suggested to pause as an isolated supplement (genetic verdict); in the current protocol enters the monthly pulse with fisetin, only with normal ALT. UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.	Genetic plan (S29 + coordination note) <i>weak</i>	UGT1A1: high COMT: high MTHFR: high CYP3A5: high

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quercetin phytosome Avoid	UGT1A1: IM (*1/*28) COMT: Val/Val (high) MTHFR: 677CT/1298AC CYP3A5: expressor (*1/*3)	avoid/pause	UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.	Genetic plan (S29 + coordination note) <i>weak</i>	UGT1A1: high COMT: high MTHFR: high CYP3A5: high
resveratrol + quercetin (combined) Avoid	UGT1A1: IM (*1/*28) COMT: Val/Val (high) MTHFR: 677CT/1298AC CYP3A5: expressor (*1/*3)	avoid/pause	UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.	Genetic plan (S29 + coordination note) <i>weak</i>	UGT1A1: high COMT: high MTHFR: high CYP3A5: high
tea catechins (drink/matcha) Adjust dose	CYP1A2: normal to intermediate ADORA2A: C/C COMT: Val/Val (high)	count caffeine	Tea/matcha in drink does not have the liver risk of the extract; counts for the caffeine ceiling (≤ 200 mg/day, only in the morning; Normal to intermediate CYP1A2).	Genetic plan (e.4) <i>moderate</i>	CYP1A2: high ADORA2A: high COMT: high
★ fisetin Adjust dose	COMT: Val/Val (high) UGT1A1: IM (*1/*28) CYP3A5: expressor (*1/*3)	isolated monthly pulse	Current stack (2.5 g (3×/month)) — plan S29: ADJUST. Monthly pulse (2 days), ≥ 7 days away from rapamycin and FOXO4-DRI; current protocol: with quercetin, only with normal ALT. Catechol (COMT Val/Val + MTHFR) and inhibitor of CYP3A4/P-gp and UGT1A1. 9p21 homozygous (CDKN2A/B region, senescence) gives senolytic rationale. Genetic reading: 1 pulse/month (2 days), ≥ 7 days away from rapamycin and FOXO4-DRI (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.	Genetic plan (S29) <i>weak</i>	COMT: high UGT1A1: high CYP3A5: high
(–)-epicatechin Monitoring	COMT: Val/Val (high)	catechol	Epicatechin isolated at high dose is methylated by COMT (Val/Val); prefer cocoa.	Genetic plan (f.2) <i>weak</i>	COMT: high
allicin (stabilized garlic extract) Monitoring	CYP3A5: expressor (*1/*3)	P-gp/CYP3A4 inducer	Allicin of fresh extracts induces P-gp/CYP3A4 (lowered saquinavir ~50%): it can lower sirolimus. Never in the rapamycin window (3 days before 7 days after each dose).	Piscitelli 2002 <i>moderate (interaction)</i>	CYP3A5: high
Saffron powder Monitoring	UGT1A1: IM (*1/*28) PNPLA3/Pi*MZ (liver): unfavourable HLA-B*35:01: probable negative	only cooking doses	Less curcumin than extracts, but cases of liver injury include root powder. Cooking doses without problem; capsules such as extracts.	LiverTox (curcuma) <i>weak</i>	UGT1A1: high PNPLA3/Pi*MZ (liver): moderate HLA-B*35:01: moderate
baicalein Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	cases of liver injury	Scutellaria baicalensis was associated with liver injury (LiverTox); with genetic liver, avoiding prolonged use.	LiverTox (hearing) <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
baicalin (Scutellaria baicalensis) Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	cases of liver injury	Scutellaria baicalensis was associated with liver injury (LiverTox); with genetic liver, avoiding prolonged use.	LiverTox (hearing) <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate

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berberine Monitoring	CYP2D6 : Probable NM (without CNV) CYP2C9 : IM (*1/*3) CYP3A5 :expressor (*1/*3)	the genetic plan advises to add	It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures; it is added to 4 drugs that lower glucose (suggestion, not applied). Never in the rapamycin window (3 days before 7 days after each dose).	Guo 2012; genetic plan (b.3) <i>moderate (interaction)</i>	CYP2D6: moderate CYP2C9: high CYP3A5: high
dihydroberberine Monitoring	CYP2D6 : Probable NM (without CNV) CYP2C9 : IM (*1/*3) CYP3A5 :expressor (*1/*3)	the genetic plan advises to add	It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures; it is added to 4 drugs that lower glucose (suggestion, not applied). Never in the rapamycin window (3 days before 7 days after each dose).	Guo 2012; genetic plan (b.3) <i>moderate (interaction)</i>	CYP2D6: moderate CYP2C9: high CYP3A5: high
glabridine (liquorice) Monitoring	CYP3A5 :expressor (*1/*3)	inhibits CYP3A4	Glabridin inhibits CYP3A4 in vitro (liquorice). Never in the rapamycin window (3 days before 7 days after each dose).	In vitro <i>weak</i>	CYP3A5: high
myricetin Monitoring	COMT : Val/Val (high) MTHFR : 677CT/1298AC	catechol: prefer low doses	Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). Food doses without a problem.	Genetic plan (f.2) <i>weak</i>	COMT: high MTHFR: high
naringin/naringenin Monitoring	CYP3A5 :expressor (*1/*3)	weak inhibition of CYP3A4	Naringenin contributes to the orange effect (weakly isolated). Never in the rapamycin window (3 days before 7 days after each dose).	Grapevine pharmacology <i>weak</i>	CYP3A5: high
piceatannol Monitoring	COMT : Val/Val (high) MTHFR : 677CT/1298AC	catechol: prefer low doses	Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). Food doses without a problem.	Genetic plan (f.2) <i>weak</i>	COMT: high MTHFR: high
piperine (black pepper extract) Monitoring	CYP3A5 :expressor (*1/*3) UGT1A1 : IM (*1/*28)	increases drug exposure	Piperine inhibits CYP3A4, P-gp and glucuronidation: up to levels of drugs and other supplements. Never in the rapamycin window (3 days before 7 days after each dose).	Bhardwaj 2002 <i>moderate (interaction)</i>	CYP3A5: high UGT1A1: high
polydatin Monitoring	UGT1A1 : IM (*1/*28)	precursor of resveratrol	Converted into resveratrol: the same logic (UGT1A1 *28, CYP3A4).	Pharmacokinetics <i>weak</i>	UGT1A1: high
pterostilben Monitoring	CETP/9p21/APOB : genetic vascular risk UGT1A1 : IM (*1/*28)	can rise to LDL	In the trial, pterostilbene rose to LDL (Riche 2014); with low genetic priority to LDL/ApoB, avoid or monitor.	2014 Rich <i>weak-moderate</i>	CETP/9p21/APOB: high UGT1A1: high
resveratrol Monitoring	UGT1A1 : IM (*1/*28) CYP2C9 : IM (*1/*3) CYP3A5 :expressor (*1/*3)	inhibits CYP3A4/2C9	Glucuronidated by UGT1A1 (*28: a little higher exposure). With 1 g/day inhibits CYP3A4, CYP2C9 and CYP2D6 in humans (Chow 2010): added to CYP2C9 *1/*3, increases exposure to NSAIDs; and moves on the sirolimus. Never in the rapamycin window (3 days before 7 days after each dose).	Chow 2010; genetic plan (f.1) <i>moderate (interaction)</i>	UGT1A1: high CYP2C9: high CYP3A5: high
rutin Monitoring	COMT : Val/Val (high) UGT1A1 : IM (*1/*28)	catechol (via quercetin)	Rutin releases quercetin in the intestine: the same logic (COMT Val/Val, UGT1A1 *28) but lower exposure. Food doses OK.	Pharmacokinetics of rutin <i>weak</i>	COMT: high UGT1A1: high
silibinin phytosome Monitoring	UGT1A1 : IM (*1/*28) CYP2C9 : IM (*1/*3) PNPLA3/Pi*MZ (liver) : unfavourable	without proven gain; inhibits UGT1A1/ CYP2C9	Silibinin is glucuronidated by UGT1A1 and inhibits it (bilirubin can rise in *28) and inhibits CYP2C9 (effect on losartan greater in *1/*1 than in *3; Han 2009). It does not protect the liver in a proven way in PNPLA3/Pi*MZ.	Han 2009; silymarin NASH trials <i>weak</i>	UGT1A1: high CYP2C9: high PNPLA3/Pi*MZ (liver): moderate
silymarin (marian card) Monitoring	UGT1A1 : IM (*1/*28) CYP2C9 : IM (*1/*3) PNPLA3/Pi*MZ (liver) : unfavourable	without proven gain; inhibits UGT1A1/ CYP2C9	Silibinin is glucuronidated by UGT1A1 and inhibits it (bilirubin can rise in *28) and inhibits CYP2C9 (effect on losartan greater in *1/*1 than in *3; Han 2009). It does not protect the liver in a proven way in PNPLA3/Pi*MZ.	Han 2009; silymarin NASH trials <i>weak</i>	UGT1A1: high CYP2C9: high PNPLA3/Pi*MZ (liver): moderate

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taxifolin (dihydroquercetin) Monitoring	COMT: Val/Val (high) MTHFR: 677CT/ 1298AC	catechol: prefer low doses	Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). Food doses without a problem.	Genetic plan (f.2) <i>weak</i>	COMT: high MTHFR: high
caffeic acid Monitoring	COMT: Val/Val (high) MTHFR: 677CT/ 1298AC	catechol: prefer low doses	Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). Food doses without a problem.	Genetic plan (f.2) <i>weak</i>	COMT: high MTHFR: high
chlorogenic acid (green coffee) Monitoring	COMT: Val/Val (high) MTHFR: 677CT/ 1298AC	can rise homocysteine	In the trial, 2 g/day of chlorogenic acid rose to homocysteine (Olthof 2001); with COMT Val/Val + MTHFR compound, avoiding concentrated extracts.	Olthof 2001 <i>weak-moderate</i>	COMT: high MTHFR: high
gallic acid Monitoring	COMT: Val/Val (high) MTHFR: 677CT/ 1298AC	catechol: prefer low doses	Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). Food doses without a problem.	Genetic plan (f.2) <i>weak</i>	COMT: high MTHFR: high
rosmarinic acid Monitoring	COMT: Val/Val (high) MTHFR: 677CT/ 1298AC	catechol: prefer low doses	Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). Food doses without a problem.	Genetic plan (f.2) <i>weak</i>	COMT: high MTHFR: high
DIM (di-indolmethane) Normal	CYP1A2: normal to intermediate GSTM1: probable null	induces CYP1A2	CYP1A2 inducers: accelerate caffeine and melatonin clearance (which is normal to intermediate). Null GSTM1: no known negative effect.	Pharmacology DIM/I3C <i>weak</i>	CYP1A2: high GSTM1: moderate
I3C (indol-3-carbinol) Normal	CYP1A2: normal to intermediate GSTM1: probable null	induces CYP1A2	CYP1A2 inducers: accelerate caffeine and melatonin clearance (which is normal to intermediate). Null GSTM1: no known negative effect.	Pharmacology DIM/I3C <i>weak</i>	CYP1A2: high GSTM1: moderate
garlic powder (capsules) Normal	NOS3: –786CT + 298GT CYP3A5: expressor (*1/*3)	low allicin	NOS3 gives vascular rationale; AGE and black garlic have little allicin → minor effect on sirolimus. Maintain AGE; black garlic is redundant.	Genetic plan (S33) <i>weak</i>	NOS3: high CYP3A5: high
★ black garlic Normal	NOS3: –786CT + 298GT CYP3A5: expressor (*1/*3)	low allicin	Current stack (10 g) — S33 plan: optional. Redundant with AGE. NOS3 gives vascular rationale; AGE and black garlic have little allicin → minor effect on sirolimus. Maintain AGE; black garlic is redundant.	Genetic plan (S33) <i>weak</i>	NOS3: high CYP3A5: high
chrysin Normal	UGT1A1: IM (*1/*28)	no clinical effect	Inducer of UGT1A1 in vitro, but without effect on humans (very low bioavailability).	Walle 2001 <i>weak</i>	UGT1A1: high
★ extract of aged garlic (AGE) Normal	NOS3: –786CT + 298GT CYP3A5: expressor (*1/*3)	low allicin	Current stack (1.2 g) — plan S33: KEEP. The same. NOS3 gives vascular rationale; AGE and black garlic have little allicin → minor effect on sirolimus. Maintain AGE; black garlic is redundant.	Genetic plan (S33) <i>weak</i>	NOS3: high CYP3A5: high
pine bark extract (generic) ⊕ Normal	NOS3: –786CT + 298GT	useful (NO)	NOS3 –786 C/T + Glu298Asp: pine bark extract improves endothelial function. Keep in the morning, outside the rapamycin window for prudence.	Genetic plan (S31) <i>weak</i>	NOS3: high
★ olive leaf extract (oleuropein) ⊕ Normal	PON1: low expression	useful (PON1)	Current stack (oleuropein 250 mg) — plan S32: KEEP. The same, with meal. Low expression PON1: olive oil/oliveira polyphenols increase the activity of PON1.	Genetic plan (S32) <i>weak</i>	PON1: high

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pomegranate extract (punicalagines) ⊕ Normal	PON1: low expression	useful (PON1)	Low expression PON1: pomegranate increases the activity of PON1 (Aviram 2004). No relevant effect on CYP3A4 in humans.	2004 <i>weak</i>	PON1: high
flavanols from cocoa Normal	COMT: Val/Val (high)	low catechol load	Flavanols of cocoa in standard dose (500–900 mg): low catechol load; NOS3 gives vascular rationale.	COSMOS <i>weak</i>	COMT: high
glucoraphanin (without myrosinase) ⊕ Normal	GSTM1: probable null NFE2L2 (NRF2): normal	favourable response	Probable null GSTM1: isothiocyanates are more in circulation (associated with more effect); NRF2 –617 C/C: KEAP1–NRF2 intact. Maintain.	Gasper 2005; genetic plan (S25) <i>moderate</i>	GSTM1: moderate NFE2L2 (NRF2): high
★ hydroxytyrosol ⊕ Normal	PON1: low expression COMT: Val/Val (high)	useful in low dose	Current stack (25 mg) — plan S32: KEEP. The same, with meal. Low expression PON1: olive oil polyphenols rise to PON1. Low dose (25 mg) : small impact on COMT. Maintain.	Genetic plan (S32) <i>weak</i>	PON1: high COMT: high
moringin/ isothiocyanates of moringa ⊕ Normal	GSTM1: probable null NFE2L2 (NRF2): normal	favourable response	Probable null GSTM1: isothiocyanates are more in circulation (associated with more effect); NRF2 –617 C/C: KEAP1–NRF2 intact. Maintain.	Gasper 2005; genetic plan (S25) <i>moderate</i>	GSTM1: moderate NFE2L2 (NRF2): high
★ pine bark extract (marine pine hull) ⊕ Normal	NOS3: –786CT + 298GT	useful (NO)	Current stack (100 mg) — plan S31: KEEP. In the morning, outside the rapamycin window. NOS3 –786 C/T + Glu298Asp: pine bark extract improves endothelial function. Keep in the morning, outside the rapamycin window for prudence.	Genetic plan (S31) <i>weak</i>	NOS3: high
polyphenols of olive oil ⊕ Normal	PON1: low expression	useful (PON1)	Low expression PON1: olive oil/oliveira polyphenols increase the activity of PON1.	Genetic plan (S32) <i>weak</i>	PON1: high
Broccoli shoot powder ⊕ Normal	GSTM1: probable null NFE2L2 (NRF2): normal	favourable response	Probable null GSTM1: isothiocyanates are more in circulation (associated with more effect); NRF2 –617 C/C: KEAP1–NRF2 intact. Maintain.	Gasper 2005; genetic plan (S25) <i>moderate</i>	GSTM1: moderate NFE2L2 (NRF2): high
★ sulforaphane (glucoraphanine + myrosinase) ⊕ Normal	GSTM1: probable null NFE2L2 (NRF2): normal	favourable response	Current stack (glucorphanin 100 mg) — plan S25: KEEP. The same in the morning. Probable null GSTM1: isothiocyanates are more in circulation (associated with more effect); NRF2 –617 C/C: KEAP1–NRF2 intact. Maintain.	Gasper 2005; genetic plan (S25) <i>moderate</i>	GSTM1: moderate NFE2L2 (NRF2): high

No known genetic relevance (35):

anthocyanins
capsaicin (CYP2E1/1A2) — *CYP2E1: normal*
diosmin
equal (S-equil)
ericytrin (lime extract)
spermidine (wheat germ)
aronia extract
cranberry extract (PAC)
extract of maqui
blueberry extract
grape seed extract (OPC)

genistein (UGT)
gingerols
hesperetin
hesperidin
holokiol / magnol
soybean isoflavones (UGT; microbiota (equil))
isoflavones of clover (UGT)
kaempferol (UGT)
linseed lignans (SDG) (microbiota)
methylhesperidin chalcone
neohesperidin dihydrochalcone

oleanolic
oleocanthal
oligonol (lychee)
salidroside
sesamine (CYP)
tangerine/nobiletin (PMF citrus fruits)
theaflavins (black tea)
uroletin A (microbiota)
betulinic acid
ellagic acid (microbiota)
ferulic acid

ursolic acid

garlic oil (CYP2E1 (inhib)) *CYP2E1: normal*

3.6 Carotenoids and pigments — 13 supplements

dose adjustment: 1 · normal: 5 · without known genetic relevance: 7

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
★ lycopene Adjust dose	BCO1: slightly ↓ conversion	lower dose (≤ 10 mg)	Current stack (35 mg) — plan S23: ADJUST (descend). 10 mg or stop. Genetic levels of lycopene tendentially high: no genetic indication for 35 mg. 10 mg or tomato cooked with olive oil (suggestion, not applied).	Genetic plan (S23) weak	BCO1: high
alpha-carotene Normal	BCO1: slightly ↓ conversion	slightly reduced conversion, without practical relevance	BCO1 R267S het + rs6564851 G/T: conversion to retinol slightly smaller; A379V absent. It is the preferred form of vitamin A with isotretinoin (regulated conversion). Food or low dose; β-carotene at a high dose does not bring benefits (and is not advised in smokers).	BCO1 studies (Leung 2009; Lietz 2012); NIH ODS moderate (genotype) · weak (practice)	BCO1: high
beta-cryptoxanthin Normal	BCO1: slightly ↓ conversion	slightly reduced conversion, without practical relevance	BCO1 R267S het + rs6564851 G/T: conversion to retinol slightly smaller; A379V absent. It is the preferred form of vitamin A with isotretinoin (regulated conversion). Food or low dose; β-carotene at a high dose does not bring benefits (and is not advised in smokers).	BCO1 studies (Leung 2009; Lietz 2012); NIH ODS moderate (genotype) · weak (practice)	BCO1: high
★ lutein Normal	CFH/ARMS2: CFH het BCO1: slightly ↓ conversion	useful (retina)	Current stack (20 mg) — plan S24: KEEP. The same, with fat. CFH Y402H het, ARMS2 G/G; lutein/zeaxanthin do not depend on BCO1. Keeping with fat.	Genetic plan (S24); AREDS2 weak	CFH/ARMS2: high BCO1: high
meso-zeaxanthin Normal	CFH/ARMS2: CFH het BCO1: slightly ↓ conversion	useful (retina)	CFH Y402H het, ARMS2 G/G; lutein/zeaxanthin do not depend on BCO1. Keeping with fat.	Genetic plan (S24); AREDS2 weak	CFH/ARMS2: high BCO1: high
★ zeaxanthin Normal	CFH/ARMS2: CFH het BCO1: slightly ↓ conversion	useful (retina)	Current stack (4 mg) — plan S24: KEEP. The same, with fat. CFH Y402H het, ARMS2 G/G; lutein/zeaxanthin do not depend on BCO1. Keeping with fat.	Genetic plan (S24); AREDS2 weak	CFH/ARMS2: high BCO1: high

No known genetic relevance (7):

★ astaxanthin
betalains (beet)
cantaxanthin

chlorophylline
crocin/crocetin (saffron)
phycocyanin (blue spirulin)

fucoxanthin

3.7 Western medicinal plants — 140 supplements

prevention: 6 · Monitoring: 21 · indeterminate: 1 · normal: 7 · without known genetic relevance: 105

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
greater celandine Avoid	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).	LiverTox <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate
comfrey (conferred) Avoid	PNPLA3/Pi*MZ (liver): unfavourable	avoid (pyrrolizidine alkaloids)	Pyrrolizidine alkaloids cause veno-occlusive liver disease; oral use is prohibited in several countries. Genetic liver reinforces.	EMA/HMPC; LiverTox <i>strong</i>	PNPLA3/Pi*MZ (liver): moderate
germander (Teucrium) Avoid	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).	LiverTox <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate
St. John's wort Avoid	CYP3A4: normal CYP3A5: expressor (*1/*3) CYP2C19: NM TERT/melanoma: unfavourable	avoid (strong inducer)	Strong inducer of CYP3A4, P-gp and CYP2C19: low cyclosporine/tacrolimus 30–60% → predictable descent of sirolimus, which has already reduced exposure to CYP3A5 expressor; also low tadalafil. Photosensitizing hypericin (p88 + isotretinoin) (suggestion, not applied).	EMA/FDA labels (interactions); Stockley <i>strong (interaction)</i>	CYP3A4: high CYP3A5: high CYP2C19: high TERT/melanoma: high
pore (oil) Avoid	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).	LiverTox <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate
tussiling Avoid	PNPLA3/Pi*MZ (liver): unfavourable	avoid (pyrrolizidine alkaloids)	Pyrrolizidine alkaloids cause veno-occlusive liver disease; oral use is prohibited in several countries. Genetic liver reinforces.	EMA/HMPC; LiverTox <i>strong</i>	PNPLA3/Pi*MZ (liver): moderate
garlic/oil Monitoring	CYP3A5: expressor (*1/*3)	P-gp/CYP3A4 inducer	Allicin of fresh extracts induces P-gp/CYP3A4 (lowered saquinavir ~50%): it can lower sirolimus. Never in the rapamycin window (3 days before 7 days after each dose).	Piscitelli 2002 <i>moderate (interaction)</i>	CYP3A5: high
aloe vera (oral) Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver	Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.	LiverTox <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
chinese angelic (quang quai) Monitoring	CYP3A5: expressor (*1/*3) TERT/melanoma: unfavourable	furanocoumarins	Furanocoumarins: inhibition of CYP3A4 and photosensitization (p88 + isotretinoin nevi). Never in the rapamycin window (3 days before 7 days after each dose).	Pharmacology of furanocoumarins <i>weak</i>	CYP3A5: high TERT/melanoma: high
pocket Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver	Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.	LiverTox <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
willow bark (salicin) Monitoring	UGT1A6: *2 het CYP2C9: IM (*1/*3)	salicylate	Salicylate is metabolized by UGT1A6 (*2 het) and CYP2C9 (*1/*3); it adds antiplatelet effect to fish oil, garlic, ginger and pine bark extract.	Pharmacology of salicylate <i>weak</i>	UGT1A6: high CYP2C9: high
cimicifiga (black cohosh) Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver	Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.	LiverTox <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
cascara-sagrada Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	only short term	Anthraquinones: chronic use associated with rare liver injury and melanotic colon; with genetic liver, only punctual use.	LiverTox; EMA/HMPC <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
yerba (Melissa) + valerian Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver (valerian)	Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used. (valerian/scullery in mixtures).	LiverTox <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
American listener (skullcap) Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver	Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.	LiverTox <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
Cervul thorn (frangula) Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	only short term	Anthraquinones: chronic use associated with rare liver injury and melanotic colon; with genetic liver, only punctual use.	LiverTox; EMA/HMPC <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
hydrate (goldenseal) Monitoring	CYP2D6: Probable NM (without CNV) CYP3A5: expressor (*1/*3)	strong inhibitor CYP3A4/2D6	Hydrastin/berberine: clinical inhibition of CYP3A4 and CYP2D6 (Gurley 2008). Interaction with sirolimus; the probable normal CYP2D6 (without CNV) can become phenoconverted. Never in the rapamycin window (3 days before 7 days after each dose).	Gurley 2008 <i>moderate (interaction)</i>	CYP2D6: moderate CYP3A5: high
mahonia (Oregon grape) Monitoring	CYP2D6: Probable NM (without CNV) CYP2C9: IM (*1/*3) CYP3A5: expressor (*1/*3)	the genetic plan advises to add	It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012); with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures; it is added to 4 drugs that lower glucose (suggestion, not applied). Never in the rapamycin window (3 days before 7 days after each dose).	Guo 2012; genetic plan (b.3) <i>moderate (interaction)</i>	CYP2D6: moderate CYP2C9: high CYP3A5: high
pelargonium sidoides Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver	Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.	LiverTox <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
petasitis (butterbur, without PA) Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver	Even extracts without PA had cases of hepatitis (LiverTox); with genetic liver, short use.	LiverTox (butterbur) <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
angelica root (Angelica arcangelica) Monitoring	CYP3A5: expressor (*1/*3) TERT/melanoma: unfavourable	furanocoumarins	Furanocoumarins: inhibition of CYP3A4 and photosensitization (p88 + isotretinoin nevi). Never in the rapamycin window (3 days before 7 days after each dose).	Pharmacology of furanocoumarins <i>weak</i>	CYP3A5: high TERT/melanoma: high
rhubarb (root) Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	only short term	Anthraquinones: chronic use associated with rare liver injury and melanotic colon; with genetic liver, only punctual use.	LiverTox; EMA/HMPC <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
ruta (ruda) Monitoring	CYP3A5: expressor (*1/*3) TERT/melanoma: unfavourable	furanocoumarins	Furanocoumarins: inhibition of CYP3A4 and photosensitization (p88 + isotretinoin nevi). Never in the rapamycin window (3 days before 7 days after each dose).	Pharmacology of furanocoumarins <i>weak</i>	CYP3A5: high TERT/melanoma: high
sene Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	only short term	Anthraquinones: chronic use associated with rare liver injury and melanotic colon; with genetic liver, only punctual use.	LiverTox; EMA/HMPC <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
meadowsweet Monitoring	UGT1A6: *2 het CYP2C9: IM (*1/*3)	salicylate	Salicylate is metabolized by UGT1A6 (*2 het) and CYP2C9 (*1/*3); it adds antiplatelet effect to fish oil, garlic, ginger and pine bark extract.	Pharmacology of salicylate <i>weak</i>	UGT1A6: high CYP2C9: high
grape (brerberis) Monitoring	CYP2D6: Probable NM (without CNV) CYP2C9: IM (*1/*3) CYP3A5: expressor (*1/*3)	the genetic plan advises to add	It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012); with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures; it is added to 4 drugs that lower glucose (suggestion, not applied). Never in the rapamycin window (3 days before 7 days after each dose).	Guo 2012; genetic plan (b.3) <i>moderate (interaction)</i>	CYP2D6: moderate CYP2C9: high CYP3A5: high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
valerian Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver	Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.	LiverTox <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
licorice (root) Undetermined	HSD11B2: indeterminate	non-valuable individual sensitivity	Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).	EMA/HMPC; EFSA (glycyrrhizin) <i>not determinable</i>	HSD11B2: non-determinable
deglycerizin (DGL) Normal	HSD11B2: indeterminate	without glicirrizine	DGL has no glicirrizine: no mineralocorticoid effect.	EMA/HMPC <i>weak</i>	HSD11B2: non-determinable
alfalfa Normal	VKORC1: TT (sensitive)	only relevant with coumarins	Vitamin K/coumarins: irrelevant without anticoagulant (VKORC1 TT).	CPIC warfarin context <i>weak</i>	VKORC1: high
green oat (Avena sativa) Normal	HLA-DQ2/DQ8: negative	without celiac HLA	Without DQ2.5 or DQ8: very low genetic risk of celiac disease; gluten/cereals tolerated.	Genetic plan (e.6) <i>strong</i>	HLA-DQ2/DQ8: high
cardo-marian (integer seed) Normal	UGT1A1: IM (*1/*28)	food dose	Whole/milled seed: low dose of silymarin.	Pharmacokinetics <i>weak</i>	UGT1A1: high
ginkgo biloba Normal	CYP2C19: NM	Normal CYP2C19	Ginkgo induces CYP2C19 (omeprazole –40%); with CYP2C19 *1/*1 and without drug CYP2C19 in the stack, without practical effect. Light antiplatelet effect (sum of fish oil).	Yin 2004 <i>weak</i>	CYP2C19: high
mellilote (church clover) Normal	VKORC1: TT (sensitive)	only relevant with coumarins	Vitamin K/coumarins: irrelevant without anticoagulant (VKORC1 TT).	CPIC warfarin context <i>weak</i>	VKORC1: high
cinchona Normal	G6PD: normal CYP3A4: normal	Normal G6PD	Quinine: hemolysis in G6PD deficit (normal here); metabolized by CYP3A4 (normal). Arrhythmias/thrombocytopenia do not depend on these genes.	FDA quinine label <i>moderate (G6PD)</i>	G6PD: high for the tested variants CYP3A4: high

No known genetic relevance (105):

absinthe (artemisia)
 agnocasto (chasteberry)
 agrimony
 agripalma (motherwort)
 AKBA (boswelina enriched in AKBA)
 artichoke (leaf)
 rosemary (carnosic acid)
 leachate
 alquemyl
 love-perfect (heartsease)
 Anise-star
 artemisia (mugwort)
 burdock
 birch (leave/seed)
 calendula (oral)
 chamomile
 cardo-santo

oak bark
 sweet orange peel
 horse chestnut (aescin)
 horse (thioaminase)
 centaurea-menor
 Java tea (Orthosiphon)
 cimbalaria / hera (leaf)
 cranberry (concentrated juice)
 D-mannose
 dandelion (root)
 echinacea (CYP3A (module, weak)
 pale echinacea/angustifolia (CYP3A)
 eucalyptus (oral oil) (CYP)
 eufrasia
 elderberry flower
 raspberry leaf
 walnut leaf

leaf of nettle
 red grape leaf
 fumitory
 fennel
 Devil's claw (CYP2C9 (inhib, in vitro)) *CYP2C9: IM (*1/*3)*
 gerenciane
 ★ **ginger**
 ginger + equinacea (immunity) (CYP3A)
 Butcher's broom
 grass
 grindelia
 blackberries (leaves)
 hamamelis (oral)
 hibiscus
 hyssop
 peppermint (enteric oil) (CYP3A4) *CYP3A4: normal*
 wild yam

oral lavender (standard oil)
Levistic
lobelia
lima
hops
brown-white
black maroon
matrix (ferrafew)
melissa (lemon balm)
millefolium
myrrh
blueberry (leaf)
island moss
black walnut (fruit bark)
olmo-americano (slippery elm)
olibane (Boswellia serrata)
ononis (restarrow)
Calyformia poppy

passiflora (passion fruit)
pyloriteiro (pinheiro-alvar)
primule (root)
propolis (extract)
pygeum
rose (marshmallow)
root of nettle
rhodiola rosea (CYP3A4/2C9 (inhibits, in vitro)) *CYP3A4: normal*
rooibs
rosemary (hyp rose)
elderberry (berry)
parsley (leave/root) (furanocoumarins)
salsaparril
saw palmetto (Serenoa)
pumpkin seed (extract)
always-novel (knotgrass)
sawdust (wildworm)
sage

tanching
Menor tanching (ribwort)
tea tree (oral no; topic)
Thyme
tormentil
red clover
tilapia (flower)
white nettle
Ursin grape
Gold rod
fundaments
verbena
viburn (crack)
violet
visco (Viscum album)
juniper
Innula (elecampane)
Oregano oil

3.8 Ayurvedic plants — 47 supplements

prevention: 6 · Monitoring: 5 · normal: 2 · without known genetic relevance: 34

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
ashwagandha (full spectrum root extract) Avoid	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Ashwagandha: growing cholestatic cases (2017–2025).	LiverTox (ashwagandha); genetic plan (b.3) <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate
ashwagandha (root and leaf extract) Avoid	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Ashwagandha: growing cholestatic cases (2017–2025).	LiverTox (ashwagandha); genetic plan (b.3) <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate
ashwagandha (root) Avoid	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Ashwagandha: growing cholestatic cases (2017–2025).	LiverTox (ashwagandha); genetic plan (b.3) <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate
babchi (Psoralea corylifolia) Avoid	PNPLA3/Pi*MZ (liver): unfavourable TERT/melanoma: unfavourable	avoid (liver + photosensitization)	Babchi: photosensitizing psoralens (p88 + isotretinoin) and hepatitis (LiverTox) cases.	LiverTox (Psoralea) <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate TERT/melanoma: high
brahmi + ashwagandha (combined) Avoid	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Ashwagandha: growing cholestatic cases (2017–2025).	LiverTox (ashwagandha); genetic plan (b.3) <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate
giloy / guduchi (Tinospora cordifolia) Avoid	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).	LiverTox <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate
gotu kola (centella) Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver	Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.	LiverTox <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
guggul Monitoring	PNPLA3/Pi*MZ (liver): unfavourable CYP3A5: expressor (*1/*3)	CYP3A4 + liver inducer	Guggulsterone activates PXR (induces CYP3A4: it can lower sirolimus) and there are cases of hepatitis. Never in the rapamycin window (3 days before 7 days after each dose).	LiverTox; Brobst 2004 <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate CYP3A5: high
mucuna pruriens (L-dopa) Monitoring	COMT: Val/Val (high) MTHFR: 677CT/1298AC	L-dopa: COMT/ homocysteine	The L-dopa is O-methylated by COMT (Val/Val, high) spending SAM and up to homocysteine; Compound MTHFR and p92 score reinforce.	L-dopa × homocysteine literature <i>moderate</i>	COMT: high MTHFR: high
pippali (long pepper) Monitoring	CYP3A5: expressor (*1/*3) UGT1A1: IM (*1/*28)	increases drug exposure	Piperine inhibits CYP3A4, P-gp and glucuronidation: up to levels of drugs and other supplements. Never in the rapamycin window (3 days before 7 days after each dose).	Bhardwaj 2002 <i>moderate (interaction)</i>	CYP3A5: high UGT1A1: high
trikatu (peppers + ginger) Monitoring	CYP3A5: expressor (*1/*3) UGT1A1: IM (*1/*28)	increases drug exposure	Piperine inhibits CYP3A4, P-gp and glucuronidation: up to levels of drugs and other supplements. Never in the rapamycin window (3 days before 7 days after each dose).	Bhardwaj 2002 <i>moderate (interaction)</i>	CYP3A5: high UGT1A1: high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
curcuma longa (paste/ golden milk) Normal	UGT1A1 : IM (*1/*28) HLA-B*35:01 : probable negative	OK cooking dose	Culinary quantities of curcuma: not relevant for UGT1A1 *28 or liver.	LiverTox (curcuma) <i>weak</i>	UGT1A1: high HLA-B*35:01: moderate
yellow melon (karela) Normal	G6PD : normal	Normal G6PD	The melon-yellow seeds have vicin (G6PD deficit haemolysis); Normal G6PD → without genetic risk. It is added to the drugs that lower glucose.	CPIC G6PD 2022 (context) <i>moderate (G6PD)</i>	G6PD: high for the tested variants

No known genetic relevance (34):

ajwain	coleus forskohlii (forscoline)	nagarmotha
amla (brown-indian)	fenugreek	neem
andrographis (kalmegh) (CYP)	garcinia indica (kokum)	punarnava
arjuna (shell)	gymnema sylvestre	safe musli
ashoka (Saraca)	haritaki	Shankhpushpi
assa-fetida (hing)	jatamansi	shatavari
bacopa (brahmi)	kalonji (nigel seed)	shilajit
bhringraj	kutki (Picrorhiza kurroa)	tribulus terrestris
bibhitaki	lodhra	tryphal
cardamom (extract)	manjistha	tulsi (sacred basil)
chyawanprash	moringa (leaf)	vivari kand
	musli-negro	

3.9 Traditional Chinese medicine plants and Kampo — 76 supplements

prevention: 8 · Monitoring: 10 · indeterminate: 5 · without known genetic relevance: 53

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
He Shou Wu (Polygonum multiflorum, fo-ti) Avoid	PNPLA3/Pi*MZ (liver): unfavourable HLA-B*35:01: probable negative	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Polygonum multiflorum: HLA-B*35:01-linked lesion (probable negative), but intrinsic hepatotoxicity remains.	LiverTox; Li 2019 (HLA-B*35:01) <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate HLA-B*35:01: moderate
Jin Bu Huan (tetra-hydropalmatin) Avoid	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).	LiverTox <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate
astragalosid IV Avoid	TERT/melanoma: unfavourable	avoid (telomerase activator)	Used as telomerase activators. With 5 alleles TERT of oncologic risk and nevi at the 88th percentile (melanoma by chip is p6.5/p25), precaution is asymmetric (hypothetical benefit vs theoretical risk). Avoid purified extracts of astragalus, cycloastragenol and astragalus (suggestion, not applied).	Genetic plan (d); GWAS TERT <i>weak (precaution)</i>	TERT/melanoma: high
astralis (root) Avoid	TERT/melanoma: unfavourable	avoid (telomerase activator)	Used as telomerase activators. With 5 alleles TERT of oncologic risk and nevi at the 88th percentile (melanoma by chip is p6.5/p25), precaution is asymmetric (hypothetical benefit vs theoretical risk). Avoid purified extracts of astragalus, cycloastragenol and astragalus (suggestion, not applied).	Genetic plan (d); GWAS TERT <i>weak (precaution)</i>	TERT/melanoma: high
cycloastragenol Avoid	TERT/melanoma: unfavourable	avoid (telomerase activator)	Used as telomerase activators. With 5 alleles TERT of oncologic risk and nevi at the 88th percentile (melanoma by chip is p6.5/p25), precaution is asymmetric (hypothetical benefit vs theoretical risk). Avoid purified extracts of astragalus, cycloastragenol and astragalus (suggestion, not applied).	Genetic plan (d); GWAS TERT <i>weak (precaution)</i>	TERT/melanoma: high
purified extract of astrogalum (telomerase activator) Avoid	TERT/melanoma: unfavourable	avoid (telomerase activator)	Used as telomerase activators. With 5 alleles TERT of oncologic risk and nevi at the 88th percentile (melanoma by chip is p6.5/p25), precaution is asymmetric (hypothetical benefit vs theoretical risk). Avoid purified extracts of astragalus, cycloastragenol and astragalus (suggestion, not applied).	Genetic plan (d); GWAS TERT <i>weak (precaution)</i>	TERT/melanoma: high
ma huang (ephedra) Avoid	CETP/9p21/APOB: genetic vascular risk NOS3: −786CT + 298GT	avoid	Prohibited sympathomimetics (UE/EUA): Stroke/farce; 9p21 risk homozygous and NOS3 reinforce.	FDA 2004; NCCIH <i>strong</i>	CETP/9p21/APOB: high NOS3: high
tripterygium (thunder Godvine) Avoid	PNPLA3/Pi*MZ (liver): unfavourable	avoid	Tripterygium: hepatic, renal and reproductive toxicity; immunosuppressant.	NCCIH; LiverTox <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate
Sho-saiko-to / Xiao Chai Hu Tang Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver	Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.	LiverTox <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
angelica dahurica (bai zhi) Monitoring	CYP3A5: expressor (*1/*3) TERT/melanoma: unfavourable	furanocoumarins	Furanocoumarins: inhibition of CYP3A4 and photosensitization (p88 + isotretinoin nevi). Never in the rapamycin window (3 days before 7 days after each dose).	Pharmacology of furanocoumarins <i>weak</i>	CYP3A5: high TERT/melanoma: high
bupleur Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver	Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.	LiverTox <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
coptis (huang lian) Monitoring	CYP2D6: Probable NM (without CNV) CYP2C9: IM (*1/*3) CYP3A5: expressor (*1/*3)	the genetic plan advises to add	It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures; it is added to 4 drugs that lower glucose (suggestion, not applied). Never in the rapamycin window (3 days before 7 days after each dose).	Guo 2012; genetic plan (b.3) <i>moderate (interaction)</i>	CYP2D6: moderate CYP2C9: high CYP3A5: high
corydalis (yan hu suo) Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver	Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.	LiverTox <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
Chinese mantle (Scutellaria baicalensis) Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	cases of liver injury	Scutellaria baicalensis was associated with liver injury (LiverTox); with genetic liver, avoiding prolonged use.	LiverTox (hearing) <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
schisandra (Schisandra chinensis) Monitoring	CYP3A5: expressor (*1/*3) PNPLA3/Pi*MZ (liver): unfavourable	inhibits CYP3A4/P-gp	Schisandra rises the levels of tacrolimus (used for this in China); by analogy, it rises the sirolimus. Never in the rapamycin window (3 days before 7 days after each dose).	Wuzhi × tacrolimus studies <i>moderate (interaction)</i>	CYP3A5: high PNPLA3/Pi*MZ (liver): moderate
hu zhang (Polygonum cuspidatum, knotweed) Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	only short term	Anthraquinones: chronic use associated with rare liver injury and melanotic colon; with genetic liver, only punctual use.	LiverTox; EMA/HMPC <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
phellodendron Monitoring	CYP2D6: Probable NM (without CNV) CYP2C9: IM (*1/*3) CYP3A5: expressor (*1/*3)	the genetic plan advises to add	It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures; it is added to 4 drugs that lower glucose (suggestion, not applied). Never in the rapamycin window (3 days before 7 days after each dose).	Guo 2012; genetic plan (b.3) <i>moderate (interaction)</i>	CYP2D6: moderate CYP2C9: high CYP3A5: high
cassava seed (jue ming zi) Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	only short term	Anthraquinones: chronic use associated with rare liver injury and melanotic colon; with genetic liver, only punctual use.	LiverTox; EMA/HMPC <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
Hochuekkito (Kampo) Undetermined	HSD11B2: indeterminate	non-valuable individual sensitivity	Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).	EMA/HMPC; EFSA (glycyrrhizin) <i>not determinable</i>	HSD11B2: non-determinable
Juzentaiho (Kampo) Undetermined	HSD11B2: indeterminate	non-valuable individual sensitivity	Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).	EMA/HMPC; EFSA (glycyrrhizin) <i>not determinable</i>	HSD11B2: non-determinable
Kakkonto (Kampo) Undetermined	HSD11B2: indeterminate	non-valuable individual sensitivity	Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).	EMA/HMPC; EFSA (glycyrrhizin) <i>not determinable</i>	HSD11B2: non-determinable
Shakuyakukanzoto (Kampo) Undetermined	HSD11B2: indeterminate	non-valuable individual sensitivity	Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).	EMA/HMPC; EFSA (glycyrrhizin) <i>not determinable</i>	HSD11B2: non-determinable

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
Yokukansan (Kampo) Undetermined	HSD11B2: indeterminate	non-valuable individual sensitivity	Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).	EMA/HMPC; EFSA (glycyrrhizin) <i>not determinable</i>	HSD11B2: non-determinable

No known genetic relevance (53):

achyranthes	evodia (evodiamine) (CYP1A2/3A4) — <i>CYP1A2: normal to intermediate</i>	ligusticum (chuanxiong)
blackberry (fruit, sang shen)	lotus leaf	ligustrum
ashitaba (chalcones)	forsythia	madresilva (lonicera)
atractylodes	gardenia (zhi zi) (genipina)	morinda officinalis
magnolia peel	gastrodia (tian ma)	notoginseng (Panax notoginseng)
tangerine peel (chen pi)	american ginseng	ophiopogon (male dong)
chrysanthemum tea	ginseng (Panax ginseng) (CYP3A4 (module); ginsenosides) <i>CYP3A4: normal</i>	white peony (bai shao)
cystanche	black ginseng	Chinese pestiler (shan zha)
codonopsis (dang shen)	siberian ginseng (eleuter)	pinellia
coix (yi yi ren)	Korean red ginseng (CYP3A4) <i>CYP3A4: normal</i>	polygonatum
cuscuta	goji (Lycium barbarum) (CYP2C9 (inhib, cases)) <i>CYP2C9: IM (*1/*3)</i>	polygala (yuan zhi)
Daikenchuto (Kampo)	houltuynia	Pueraria mirifica (phytoestrogens)
danshen (Salvia miltiorrhiza) (CYP; antiplatelet)	huperzine A (CYP1A2) — <i>CYP1A2: normal to intermediate</i>	rehmannia
dendrobium	isatis (ban lan gen)	scrophilia (xuan shen)
dioscorea (shan yao)	jiaogulan (Gynostemma)	lotus seed
epimedium (horny drop weed, icariin) (PDE5 (inhibits, weak))	jujube (fruit)	black sesame
Seabream (sea buckthorn)	kudzu (Pueraria lobata)	yin chen (Artemisia capillaris)
eucommia		ziphus (suan zao ren)

3.10 Plants of the Americas and Africa; superfoods — 61 supplements

prevention: 3 · dose adjustment: 6 · Monitoring: 5 · normal: 9 · without known genetic relevance: 38

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
pharmacaria (Larrea) Avoid	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).	LiverTox <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate
kava (Piper methylicum) Avoid	PNPLA3/Pi*MZ (liver): unfavourable CYP2D6: Probable NM (without CNV)	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Kava: severe hepatitis and transplants; removal in several countries.	LiverTox (kava) <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate CYP2D6: moderate
kratom (Mitragyna speciosa) Avoid	PNPLA3/Pi*MZ (liver): unfavourable CYP2D6: Probable NM (without CNV) OPRM1: A/G	avoid	Atypical opioid (dependence), cholestatic liver injury (LiverTox); CYP3A4/2D6 (CNV not evaluated) and OPRM1 A/G.	LiverTox (kratom); FDA <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate CYP2D6: moderate OPRM1: high
green coffee (extract) Adjust dose	CYP1A2: normal to intermediate ADORA2A: C/C COMT: Val/Val (high) MTHFR: 677CT/1298AC	caffeine + chlorogenic acid	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied). The concentrated chlorogenic acid can rise to homocysteine (COMT/MTHFR).	Cornelis 2006; Olthof 2001 <i>moderate</i>	CYP1A2: high ADORA2A: high COMT: high MTHFR: high
yerba mate Adjust dose	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).	Cornelis 2006; genetic plan (e.4) <i>moderate</i>	CYP1A2: high ADORA2A: high
guarana Adjust dose	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).	Cornelis 2006; genetic plan (e.4) <i>moderate</i>	CYP1A2: high ADORA2A: high
guayusa Adjust dose	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).	Cornelis 2006; genetic plan (e.4) <i>moderate</i>	CYP1A2: high ADORA2A: high
walnut Adjust dose	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).	Cornelis 2006; genetic plan (e.4) <i>moderate</i>	CYP1A2: high ADORA2A: high
yaupon Adjust dose	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).	Cornelis 2006; genetic plan (e.4) <i>moderate</i>	CYP1A2: high ADORA2A: high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
cinnamon-cassava (coumarin) Monitoring	CYP2A6 : probable normal PNPLA3/Pi*MZ (liver) : unfavourable	hepatotoxic coumarin	Cassia cinnamon has coumarin (EFSA fixed a low tolerable daily intake); 7-hydroxylation depends on CYP2A6 (*2/*9 absent; non-visible deletions). To prefer cinnamon-do-ceil.	EFSA 2008 (coumarin) <i>weak-moderate</i>	CYP2A6: low PNPLA3/Pi*MZ (liver): moderate
yohimbine HCl Monitoring	CYP2D6 : Probable NM (without CNV)	CYP2D6 (confirm CNV)	Yohimbine depends on CYP2D6: slow metabolizers have often higher exposure. Probable *1/*2 (normal), but not evaluated CNV; is also cardiovascular stimulant.	Le Corre 2004 <i>moderate</i>	CYP2D6: moderate
noni Monitoring	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring liver	Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.	LiverTox <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
concentrated beet juice (nitrate) Monitoring	NOS3 : –786CT + 298GT	avoid concentrated nitrate (tadalafil)	NOS3 gives rational, but concentrated inorganic nitrate is added to the tadalafil (hypotension). Only vegetal nitrates (suggestion, not applied).	Genetic plan (b.3, f.13) <i>weak</i>	NOS3: high
yohimbe (shell) Monitoring	CYP2D6 : Probable NM (without CNV)	CYP2D6 (confirm CNV)	Yohimbine depends on CYP2D6: slow metabolizers have often higher exposure. Probable *1/*2 (normal), but not evaluated CNV; is also cardiovascular stimulant.	Le Corre 2004 <i>moderate</i>	CYP2D6: moderate
beet powder Normal	NOS3 : –786CT + 298GT	OK food dose	Nitrate quantity of a food powder is moderate; with tadalafil, avoid "shot" doses.	Genetic plan (f.13) <i>weak</i>	NOS3: high
cocoa (theobromine extract) Normal	CYP1A2 : normal to intermediate	theobromine (CYP1A2)	Theobromine is metabolized by CYP1A2 (normal to intermediate) but it is a weak stimulant; counting in large amounts.	Pharmacology <i>weak</i>	CYP1A2: high
raw cocoa powder Normal	CYP1A2 : normal to intermediate	theobromine (CYP1A2)	Theobromine is metabolized by CYP1A2 (normal to intermediate) but it is a weak stimulant; counting in large amounts.	Pharmacology <i>weak</i>	CYP1A2: high
Cinnamon Normal	CYP2A6 : probable normal	low coumarin	Ceylon cinnamon almost without coumarin.	EFSA 2008 <i>weak</i>	CYP2A6: low
chlorella Normal	VKORC1 : TT (sensitive)	vitamin K (only relevant with coumarins)	Rich in vitamin K; it only weighs if you take coumarins (VKORC1 TT).	Genetic plan (e.6) <i>weak</i>	VKORC1: high
Spirulina Normal	PAH : no variants	normal for the tested variants	No pathogenic variants of PAH among those tested (improbable phenylketonuria).	ClinVar (chip) <i>moderate</i>	PAH: partial
sacha inchi (oil) Normal	FADS1/FADS2 : high activity	good conversion ALA→EPA	With high activity FADS, the conversion of ALA/SDA to EPA is one of the best; however, it does not replace the preformed DHA.	FADS × ALA studies <i>moderate</i>	FADS1/FADS2: high
chia seeds Normal	FADS1/FADS2 : high activity	good conversion ALA→EPA	With high activity FADS, the conversion of ALA/SDA to EPA is one of the best; however, it does not replace the preformed DHA.	FADS × ALA studies <i>moderate</i>	FADS1/FADS2: high
ground linseed seeds Normal	FADS1/FADS2 : high activity	good conversion ALA→EPA	With high activity FADS, the conversion of ALA/SDA to EPA is one of the best; however, it does not replace the preformed DHA.	FADS × ALA studies <i>moderate</i>	FADS1/FADS2: high

No known genetic relevance (38):

açaí
baobab (post)
hypoxis (Hypoxys) (CYP3A4 (inhib, in vitro)) *CYP3A4: normal*
Buchu

camu camu
catuaba
tart cherry
chanca piedra (Phyllanthus)

chuchuhuasi
clove
damiana
jackfruit (barley grass)

wheatgrass (wheatgrass)
stevia
Yellow leaf (Vernonia)
Cat claw (CYP3A4 (inhibits, weak)) *CYP3A4: normal*
soursop
hoodia gorgonii
jatoba
kanna (Sceletium tortuosum)
lucuma

maca (Lepidium meyenii)
black maca
irvingia (Irvingia gabonensis)
mangostion
manuka honey
Wild blueberry powder
muira puama
pau d'archo (lapacho)
kayenne pepper (CYP1A2) — *CYP1A2: normal to intermediate*

quinoa / amaranth powder
hemp seeds
abstract (Brazilian ginseng)
supergreen powder (greens) (various)
sutherlandia
Cider vinegar (capsules)
yacon (syrup/FOS)
castor oil (oral)

3.11 Medicinal mushrooms — 21 supplements

Monitoring: 1 · normal: 3 · without known genetic relevance: 17

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
reishi (Ganoderma lucidum) Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver	Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.	LiverTox weak	PNPLA3/Pi*MZ (liver): moderate
mushroom king trumpet (ergothionein) Normal	SLC22A4/5 (OCTN1/2): typical	typical capture	OCTN1 (SLC22A4) L503 C/C: typical ergothioneine uptake. Maintain.	Genetic plan (S20) weak	SLC22A4/5 (OCTN1/2): high
★ ergothioneine Normal	SLC22A4/5 (OCTN1/2): typical	typical capture	Current stack (10 mg) — plan S20: KEEP. The same. OCTN1 (SLC22A4) L503 C/C: typical ergothioneine uptake. Maintain.	Genetic plan (S20) weak	SLC22A4/5 (OCTN1/2): high
pleurotes (mushroom-ostral) Normal	SLCO1B1: increased function	natural lovastatin (low)	SLCO1B1 *1/*14 (augmented function): without genetic risk with low natural lovastatin.	CPIC statins 2022 (context) weak	SLCO1B1: high

No known genetic relevance (17):

agaricus blazei (sun mushroom)	cordyceps militaris	mesima (Phellinus linteus)
AHCC (cultured shiitake extract)	cordyceps sinensis (CS-4)	7-10 mushroom mixture
yeast beta-glucan (1.3/1.6)	enoki	ear (Auricularia)
peru tail (Trametes, PSK/PSP)	lion's mantle (Hericium erinaceus)	poria coconuts
chaga (Inonatus obliquus) (oxalates)	lentin	shiitake (Lentinula edodes)
	mayake (Grifola frondosa)	trenella

3.12 Probiotics, Prebiotics and Fibers — 69 Supplements

dose adjustment: 2 · normal: 20 · without known genetic relevance: 47

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
★ PHGG (partially hydrolyzed guar gum) ⊕ Adjust dose	TCF7L2/FTO/GCKR: unfavourable CETP/9p21/APOB: genetic vascular risk FUT2: secretor	adjust dose (plan)	Current stack (6 g) — plan S35: ADJUST (subir). 10 g/day. TCF7L2/FTO/MTNR1B/GCKR (postprandial glycosis) and CETP/9p21/APOB (low to LDL). More fiber is always better.	Genetic plan (S35) strong	TCF7L2/FTO/GCKR: high CETP/9p21/APOB: high FUT2: high
★ psyllium (ispaghula bark) ⊕ Adjust dose	TCF7L2/FTO/GCKR: unfavourable CETP/9p21/APOB: genetic vascular risk FUT2: secretor	adjust dose (plan)	Current stack (10 g) — plan S35: ADJUST (subir). 15–20 g/day in 2–3 takes with water; drugs ≥ 1 h before or 2–4 h after. TCF7L2/FTO/MTNR1B/GCKR (postprandial glycosis) and CETP/9p21/APOB (low to LDL). More fiber is always better.	Genetic plan (S35) strong	TCF7L2/FTO/GCKR: high CETP/9p21/APOB: high FUT2: high
2'-fucosyl-lactose (HMO) Normal	FUT2: secretor	typical colonization (secretor)	FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.	Wacklin 2011; genetic plan (S36) weak	FUT2: high
Akkermansia muciniphila (pasteurized) Normal	FUT2: secretor	typical colonization (secretor)	FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.	Wacklin 2011; genetic plan (S36) weak	FUT2: high
Bifidobacterium animalis lactis BB-12 Normal	FUT2: secretor	typical colonization (secretor)	FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.	Wacklin 2011; genetic plan (S36) weak	FUT2: high
Bifidobacterium bifidum Normal	FUT2: secretor	typical colonization (secretor)	FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.	Wacklin 2011; genetic plan (S36) weak	FUT2: high
Bifidobacterium breve Normal	FUT2: secretor	typical colonization (secretor)	FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.	Wacklin 2011; genetic plan (S36) weak	FUT2: high
Bifidobacterium lactis HN019 Normal	FUT2: secretor	typical colonization (secretor)	FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.	Wacklin 2011; genetic plan (S36) weak	FUT2: high
Bifidobacterium longum Normal	FUT2: secretor	typical colonization (secretor)	FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.	Wacklin 2011; genetic plan (S36) weak	FUT2: high
Bifidobacterium longum 1714 (psychobiotic) Normal	FUT2: secretor	typical colonization (secretor)	FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.	Wacklin 2011; genetic plan (S36) weak	FUT2: high
Bifidobacterium longum subsp. Normal	FUT2: secretor	typical colonization (secretor)	FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.	Wacklin 2011; genetic plan (S36) weak	FUT2: high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
L. helveticus R0052 + B. longum R0175 Normal	FUT2 : secretor	typical colonization (secretor)	FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.	Wacklin 2011; genetic plan (S36) <i>weak</i>	FUT2: high
Lactacaseibacillus casei (fermented milk strain) Normal	AOC1 (DAO) : slightly ↓ HNMT : Thr105Ile het	histamine: only if there are symptoms	Some strains produce histamine; AOC1 and HNMT heterozygotes (light reduction). No restriction, except symptoms.	Genetic plan (e.6) <i>weak</i>	AOC1 (DAO): high HNMT: high
Lactobacillus delbrueckii bulgaricus Normal	AOC1 (DAO) : slightly ↓ HNMT : Thr105Ile het	histamine: only if there are symptoms	Some strains produce histamine; AOC1 and HNMT heterozygotes (light reduction). No restriction, except symptoms.	Genetic plan (e.6) <i>weak</i>	AOC1 (DAO): high HNMT: high
Limosylactobacillus reuteri (DSM 17938) Normal	AOC1 (DAO) : slightly ↓ HNMT : Thr105Ile het	histamine: only if there are symptoms	Some strains produce histamine; AOC1 and HNMT heterozygotes (light reduction). No restriction, except symptoms.	Genetic plan (e.6) <i>weak</i>	AOC1 (DAO): high HNMT: high
oat beta-glucan ☺ Normal	CETP/9p21/APOB : genetic vascular risk TCF7L2/FTO/GCKR : unfavourable	useful for LDL and glucose	β-glucan 3 g/day low to LDL (genetic priority) and postprandial glucose.	EFSA (β-glucan) <i>strong</i>	CETP/9p21/APOB: high TCF7L2/FTO/GCKR: high
barley beta-glucan ☺ Normal	CETP/9p21/APOB : genetic vascular risk TCF7L2/FTO/GCKR : unfavourable	useful for LDL and glucose	β-glucan 3 g/day low to LDL (genetic priority) and postprandial glucose.	EFSA (β-glucan) <i>strong</i>	CETP/9p21/APOB: high TCF7L2/FTO/GCKR: high
bovine colostrum Normal	LCT : persistent	Lactose tolerated	LCT –13910 C/T: persistence of lactase, without genetic lactose intolerance.	Genetic plan (e.6) <i>strong</i>	LCT: high
Wheat bran/wheat fiber Normal	HLA-DQ2/DQ8 : negative	without celiac HLA	Without DQ2.5 or DQ8: very low genetic risk of celiac disease; gluten/cereals tolerated.	Genetic plan (e.6) <i>strong</i>	HLA-DQ2/DQ8: high
kefir (grains/liophilized) Normal	LCT : persistent	Lactose tolerated	LCT –13910 C/T: persistence of lactase, without genetic lactose intolerance.	Genetic plan (e.6) <i>strong</i>	LCT: high
high potency probiotic (8 strains formula) Normal	FUT2 : secretor	typical colonization (secretor)	FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.	Wacklin 2011; genetic plan (S36) <i>weak</i>	FUT2: high
★ multistrain (generic) Normal	FUT2 : secretor	typical colonization (secretor)	Current stack (multistrain) — S36 plan: KEEP. The same. FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.	Wacklin 2011; genetic plan (S36) <i>weak</i>	FUT2: high

No known genetic relevance (47):

resistant starch (raw potato fecule) (microbiota)
larch arabinogalactan (microbiota)
Bacillus clausii (microbiota)
Bacillus coagulans (microbiota)
Bacillus subtilis (spores) (microbiota)
sodium butyrate/calcium
Clostridium butyricum (MIYAIRI 588) (microbiota)

E. coli Nissle 1917 (microbiota)
Enterococcus faecium SF68 (microbiota)
acacia fiber (arabic gum) (microbiota)
FOS (fruit-oligosaccharides) (microbiota)
glucomannan (konjac)
GOS (galacto-oligosaccharide) (microbiota)
bovine serum immunoglobulins (SBI)

inulin (microbiota)
L. rhamnosus GR-1 + L. reuteri RC-14 (microbiota)
Lactacaseibacillus paracasei (microbiota)
Lactacaseibacillus rhamnosus HN001 (microbiota)
Lactiplantibacillus plantarum (other strains) (microbiota)
Lactobacillus acidophilus (NCFM) (microbiota)
Lactobacillus crispatus (microbiota)

Lactobacillus gasseri (microbiota)
Lactobacillus helveticus (microbiota)
Lactobacillus johnsonii (microbiota)
Lactobacillus plantarum 299v (microbiota)
Lactobacillus rhamnosus GG (microbiota)
Lactococcus lactis (microbiota)
lactoferrin (iron)
lactulose (microbiota)
Levilactobacillus brevis (microbiota)

Ligilactobacillus salivarius (microbiota)
Limosylactobacillus fermentum (microbiota)
methylcellulose
pectin (incl. modified citrus pectin) (microbiota)
Pediococcus acidilactici (microbiota)
polydextrose (microbiota)
probiotic of spores (multistrain) (microbiota)
Propionibacterium freudenreichii (microbiota)
potato protein (PI2 satiety)

postbiotic (L. paracasei inactivated)
chitosan
Saccharomyces boulardii (microbiota)
symbiotic (probiotic + prebiotic) (microbiota)
Streptococcus salivarius K12 (oral) (oral microbiota)
Streptococcus thermophilus (microbiota)
★ **tributyrene**
XOS (xylo-oligosaccharides) (microbiota)

3.13 Enzymes — 23 supplements

indeterminate: 1 · normal: 5 · without known genetic relevance: 17

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
amylase Undetermined	AMY1: indeterminate	non-valuable AMY1 copies	The number of copies of AMY1 (salivary amylase) is not visible in the chip.	AMY1 Literature not determinable	AMY1: non-determinable
AN-PEP (prolyl endopeptidase) Normal	HLA-DQ2/DQ8: negative	without genetic indication	Without celiac HLA; these enzymes do not treat celiac disease in any way.	Genetic plan (e.6) strong	HLA-DQ2/DQ8: high
DAO (diamine oxidase) in capsules Normal	AOC1 (DAO): slightly ↓ HNMT: Thr105Ile het	optional if symptoms	AOC1 Thr16Met/His645Asp and HNMT Thr105Ile heterozygotes: slight reduction; DAO only if there are histamine symptoms.	Genetic plan (e.6); §12 profile weak	AOC1 (DAO): high HNMT: high
DPP-IV (gluten enzyme) Normal	HLA-DQ2/DQ8: negative	without genetic indication	Without celiac HLA; these enzymes do not treat celiac disease in any way.	Genetic plan (e.6) strong	HLA-DQ2/DQ8: high
lactase Normal	LCT: persistent	not necessary	Lactase persistence (LCT –13910 C/T): no genetic requirement.	Genetic plan (e.6) strong	LCT: high
melon superoxide dismutase Normal	SOD2: Ala/Ala	without genetic indication	SOD2 Ala/Ala (efficient mitochondrial import): without deficit to compensate.	Profile §16 weak	SOD2: high

No known genetic relevance (17):

alpha-galactosidase	digestive enzymes (mixture)	pancreatine
bromelain	systemic enzymes (proteolytic formula)	papain
bile of cattle (biliary salts)	invertase/sucrose	pepsin
catalase (CAT) — CAT: normal	lipase	fungal protease
cellulase/hemicelulase	lumbrokinase	serrapeptase
	nattokinase	xylose isomerase (fructose)

3.14 Nootropics, stimulants, sleep and mood — 32 supplements

dose adjustment: 10 · Monitoring: 3 · indeterminate: 1 · normal: 4 · without known genetic relevance: 14

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
caffeine + L-theanine Adjust dose	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).	Cornelis 2006; genetic plan (e.4) <i>moderate</i>	CYP1A2: high ADORA2A: high
anhydrous caffeine Adjust dose	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).	Cornelis 2006; genetic plan (e.4) <i>moderate</i>	CYP1A2: high ADORA2A: high
prolonged release caffeine Adjust dose	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).	Cornelis 2006; genetic plan (e.4) <i>moderate</i>	CYP1A2: high ADORA2A: high
magnesium + B6 (cognition/stress) Adjust dose	PDXK/ALPL/NBPF3 (B6): typical/high MTHFR: 677CT/1298AC DHFR: slow reduction (CC)	choose formula (low B6, without folic acid)	Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).	Genetic plan (S3, S6); EFSA UL B6 <i>moderate</i>	PDXK/ALPL/NBPF3 (B6): high MTHFR: high DHFR: moderate
melatonin (immediate release) Adjust dose	MTNR1B: C/G (carrier) CYP1A2: normal to intermediate	low dose (0.3–1 mg)	MTNR1B rs10830963 C/G: in carrier G, melatonin reduces insulin secretion (5 mg worsened glucose tolerance in carriers); Garaulet 2015). Normal CYP1A2 extends it (apigenin, with suggested pause, would add up). 0.3–1 mg, 30–60 min before bedtime and ≥ 3 h after the last meal (suggestion, not applied).	Garaulet 2015; genetic plan (S30) <i>moderate</i>	MTNR1B: high CYP1A2: high
prolonged release melatonin Adjust dose	MTNR1B: C/G (carrier) CYP1A2: normal to intermediate	low dose (0.3–1 mg)	MTNR1B rs10830963 C/G: in carrier G, melatonin reduces insulin secretion (5 mg worsened glucose tolerance in carriers); Garaulet 2015). Normal CYP1A2 extends it (apigenin, with suggested pause, would add up). 0.3–1 mg, 30–60 min before bedtime and ≥ 3 h after the last meal (suggestion, not applied).	Garaulet 2015; genetic plan (S30) <i>moderate</i>	MTNR1B: high CYP1A2: high
★ melatonin high dose (≥ 10 mg) Adjust dose	MTNR1B: C/G (carrier) CYP1A2: normal to intermediate	low (0.3–1 mg)	Current stack (up to 100 mg) — plan S30: ADJUST (desert much). 0.3–1 mg, 30–60 min before bedtime. MTNR1B rs10830963 C/G: in carrier G, melatonin reduces insulin secretion (5 mg worsened glucose tolerance in carriers); Garaulet 2015). Normal CYP1A2 extends it (apigenin, with suggested pause, would add up). 0.3–1 mg, 30–60 min before bedtime and ≥ 3 h after the last meal (suggestion, not applied). Doses ≥ 10 mg (up to 100 mg) have no genetic basis and prolong the effect on insulin.	Garaulet 2015; genetic plan (S30) <i>moderate</i>	MTNR1B: high CYP1A2: high
melatonin in spray/sublingual Adjust dose	MTNR1B: C/G (carrier) CYP1A2: normal to intermediate	low dose (0.3–1 mg)	MTNR1B rs10830963 C/G: in carrier G, melatonin reduces insulin secretion (5 mg worsened glucose tolerance in carriers); Garaulet 2015). Normal CYP1A2 extends it (apigenin, with suggested pause, would add up). 0.3–1 mg, 30–60 min before bedtime and ≥ 3 h after the last meal (suggestion, not applied).	Garaulet 2015; genetic plan (S30) <i>moderate</i>	MTNR1B: high CYP1A2: high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
paraxanthine Adjust dose	CYP1A2 : normal to intermediate ADORA2A : C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).	Cornelis 2006; genetic plan (e.4) <i>moderate</i>	CYP1A2: high ADORA2A: high
tryptophan + B6 (combined) Adjust dose	PDXK/ALPL/NBPF3 (B6) : typical/high MTHFR : 677CT/1298AC DHFR : slow reduction (CC)	choose formula (low B6, without folic acid)	Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).	Genetic plan (S3, S6); EFSA UL B6 <i>moderate</i>	PDXK/ALPL/NBPF3 (B6): high MTHFR: high DHFR: moderate
CBD (cannabidiol) Monitoring	CYP2C19 : NM CYP2C9 : IM (*1/*3) PNPLA3/Pi*MZ (liver) : unfavourable CYP3A5 :expressor (*1/*3)	inhibits CYP; transaminase	CBD is metabolized by CYP2C19 (*1/*1, normal) and inhibits CYP2C19/2C9/3A4; increased levels of tacrolimus/everolimus in cases. At a high dose, transaminases (Epidiolex trials) are relevant with genetic liver. Never in the rapamycin window (3 days before 7 days after each dose).	Epidiolex label; clinical cases <i>moderate</i>	CYP2C19: high CYP2C9: high PNPLA3/Pi*MZ (liver): moderate CYP3A5: high
galantamine (winter bell) Monitoring	CYP2D6 : Probable NM (without CNV)	CYP2D6 (confirm CNV)	Galantamine is a substrate of CYP2D6/3A4; probable normal without CNV.	FDA label <i>moderate</i>	CYP2D6: moderate
Hemp oil (total spectrum) Monitoring	CYP2C19 : NM CYP2C9 : IM (*1/*3) PNPLA3/Pi*MZ (liver) : unfavourable CYP3A5 :expressor (*1/*3)	inhibits CYP; transaminase	CBD is metabolized by CYP2C19 (*1/*1, normal) and inhibits CYP2C19/2C9/3A4; increased levels of tacrolimus/everolimus in cases. At a high dose, transaminases (Epidiolex trials) are relevant with genetic liver. Never in the rapamycin window (3 days before 7 days after each dose).	Epidiolex label; clinical cases <i>moderate</i>	CYP2C19: high CYP2C9: high PNPLA3/Pi*MZ (liver): moderate CYP3A5: high
nicotine (chipelets/gums) Undetermined	CYP2A6 : probable normal	debugging not fully evaluated	Nicotine is eliminated by CYP2A6 (*2/*9 absent; non-visible *4 deletions): non-determinable clearance rate. Dependence; no indication.	CYP2A6 Literature <i>not determinable</i>	CYP2A6: low
L-theanine + magnesium (sleep) Normal	TRPM6 : typical	typical absorption	TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.	TRPM6 (Song 2009) studies; genetic plan (S13) <i>weak</i>	TRPM6: high
methylene blue (low dose) Normal	G6PD : normal	Normal G6PD	Normal G6PD: no genetic risk of hemolysis. MAO-A inhibitor: not to combine with serotonergic drugs.	CPIC G6PD 2022 <i>strong (G6PD)</i>	G6PD: high for the tested variants
bacopa + ginkgo (combined) Normal	CYP2C19 : NM	Normal CYP2C19	Ginkgo induces CYP2C19 (omeprazole –40%); with CYP2C19 *1/*1 and without drug CYP2C19 in the stack, without practical effect. Light antiplatelet effect (sum of fish oil).	Yin 2004 <i>weak</i>	CYP2C19: high
citicoline (patented form) ☺ Normal	PEMT : V175M het FMO3 : typical PNPLA3/Pi*MZ (liver) : unfavourable	Choline preferred form	PEMT V175M het + genetic liver give rational to choline; this form generates little TMAO. 250–500 mg of choline/day in total (suggestion, not applied).	Genetic plan (S38) <i>weak-moderate</i>	PEMT: high FMO3: high PNPLA3/Pi*MZ (liver): moderate

No known genetic relevance (14):

saffron (extract, affron)

bacopa + rhodiola (combined)

CBG (cannabigerol) (CYP (inhib, in vitro)

CBN (cannabinol) (CYP2C9/3A4) — *CYP2C9: IM (*1/*3)*
centrofenoxin
DMAE (dimethylaminoethanol)
Catnip

GABA + theanine (combined)
methylliberine (ADORA2A) — *ADORA2A: C/C*
nootropic mixture (commercial stack) (various)

PQQ (pyrroloquinoline quinone)
teacrine (ADORA2A) — *ADORA2A: C/C*
uridine monophosphate
vinpocetin

3.15 Sport and performance — 36 supplements

prevention: 2 · dose adjustment: 4 · Monitoring: 7 · normal: 7 · without known genetic relevance: 16

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
DMHA / DMAA (prohibited stimuli) Avoid	CETP/9p21/APOB: genetic vascular risk NOS3: −786CT + 298GT	avoid	Prohibited sympathomimetics (UE/EUA): Stroke/farce; 9p21 risk homozygous and NOS3 reinforce.	FDA 2004; NCCIH <i>strong</i>	CETP/9p21/APOB: high NOS3: high
prohormones/SARMs (sold as supplement) Avoid	PNPLA3/Pi*MZ (liver): unfavourable	avoid	Synthesis steroids and SARMs: severe cholestatic lesion (LiverTox) and hormonal axis; genetic liver reinforces. They are not legal supplements in UE.	LiverTox <i>strong</i>	PNPLA3/Pi*MZ (liver): moderate
electrolytes with caffeine Adjust dose	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).	Cornelis 2006; genetic plan (e.4) <i>moderate</i>	CYP1A2: high ADORA2A: high
energy gel with caffeine Adjust dose	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).	Cornelis 2006; genetic plan (e.4) <i>moderate</i>	CYP1A2: high ADORA2A: high
pre-training with caffeine Adjust dose	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).	Cornelis 2006; genetic plan (e.4) <i>moderate</i>	CYP1A2: high ADORA2A: high
tauro-caffeine (energy drink powder) Adjust dose	CYP1A2: normal to intermediate ADORA2A: C/C	limit dose and time	CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p89). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).	Cornelis 2006; genetic plan (e.4) <i>moderate</i>	CYP1A2: high ADORA2A: high
BCAA + electrolytes Monitoring	TCF7L2/FTO/GCKR: unfavourable	the genetic plan advises to add	High BCAA are associated with insulin resistance; with risk TCF7L2/FTO and genetic levels of leucine/valine tendentially high, prefer complete protein (suggestion, not applied).	Genetic plan (b.3) <i>weak</i>	TCF7L2/FTO/GCKR: high
waxy corn starch Monitoring	TCF7L2/FTO/GCKR: unfavourable	prefer intact hydrates	TCF7L2/FTO/GCKR risk + 4 drugs that move on glucose: rapid absorption hydrates only during long training.	Genetic plan (e.1) <i>weak</i>	TCF7L2/FTO/GCKR: high
gainer (hydrates + protein) Monitoring	TCF7L2/FTO/GCKR: unfavourable	prefer intact hydrates	TCF7L2/FTO/GCKR risk + 4 drugs that move on glucose: rapid absorption hydrates only during long training.	Genetic plan (e.1) <i>weak</i>	TCF7L2/FTO/GCKR: high
leucine + HMB (combined) Monitoring	TCF7L2/FTO/GCKR: unfavourable	the genetic plan advises to add	High BCAA are associated with insulin resistance; with risk TCF7L2/FTO and genetic levels of leucine/valine tendentially high, prefer complete protein (suggestion, not applied).	Genetic plan (b.3) <i>weak</i>	TCF7L2/FTO/GCKR: high
maltodextrin/cyclic dextrin Monitoring	TCF7L2/FTO/GCKR: unfavourable	prefer intact hydrates	TCF7L2/FTO/GCKR risk + 4 drugs that move on glucose: rapid absorption hydrates only during long training.	Genetic plan (e.1) <i>weak</i>	TCF7L2/FTO/GCKR: high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
synephrine (Orange-yellow) Monitoring	CYP3A5 :expressor (*1/*3) CETP/9p21/APOB : genetic vascular risk	stimulant + CYP3A4	Yellow orange furanocoumarins inhibit CYP3A4 (grapefruit effect); synephrine is a cardiovascular stimulant, especially with caffeine; 9p21 homozygous. Never in the rapamycin window (3 days before 7 days after each dose).	NCCIH (yellow orange) <i>weak</i>	CYP3A5: high CETP/9p21/APOB: high
deer horn velvet (IGF-1) Monitoring	TERT/melanoma : unfavourable	IGF-1 axis	Sold by IGF-1: the plan avoids the GH/IGF-1 axis (risk TERT alleles).	Genetic plan (d) <i>weak</i>	TERT/melanoma: high
D-ribose Normal	AMPD1 : normal	without genetic indication	AMPD1 Q12X absent: no myoadenylate deaminase deficit (where ribose was studied).	AMPD1 Literature <i>weak</i>	AMPD1: high
L-arginine + L-ornithine Normal	NOS3 : –786CT + 298GT	rational NO (small effect)	NOS3 –786 T/C + Glu298Asp G/T: a slightly smaller production of NO. Oral arginine has low bioavailability; with tadalafil the vasodilator effect is added (mild).	NOS3 studies; NIH ODS (arginine) <i>weak</i>	NOS3: high
Nitric oxide booster (mix) ☺ Normal	NOS3 : –786CT + 298GT	rational NO	NOS3 –786 T/C + Glu298Asp: citrulline rises arginine better than arginine itself. With tadalafil, slight vasodilator sum.	NOS3 studies; citrulline trials <i>weak</i>	NOS3: high
pre-training without stimulants ☺ Normal	NOS3 : –786CT + 298GT	rational NO	NOS3 –786 T/C + Glu298Asp: citrulline rises arginine better than arginine itself. With tadalafil, slight vasodilator sum.	NOS3 studies; citrulline trials <i>weak</i>	NOS3: high
ketone salts (BHB) Normal	APOE : ε3/ε3	typical response	APOE ε3/ε3: typical ketone metabolism (the ketogenic advantage described in ε4 does not apply).	APOE × ketones studies <i>weak</i>	APOE: high
whey + creatine (combined) ☺ Normal	AMPD1 : normal ACTN3 : R/X GNMT : TT	useful (muscle + save SAM)	Normal AMPD1; ACTN3 R/X; high fragility genetic score. Exogenous creatine saves SAM spent on endogenous synthesis (GAMT), useful with compound MTHFR/GNMT TT. Monohydrate 3–5 g is the form with evidence.	Genetic plan (S10); ISSN (<i>muscle</i>) · <i>moderate</i> (<i>methylation</i>)	AMPD1: high ACTN3: high GNMT: high
ketone esters (D-BHB) Normal	APOE : ε3/ε3	typical response	APOE ε3/ε3: typical ketone metabolism (the ketogenic advantage described in ε4 does not apply).	APOE × ketones studies <i>weak</i>	APOE: high

No known genetic relevance (16):

Oral ATP (PEAK ATP)
 sodium bicarbonate (buffer)
 capsimax (capsaicinoids)
 sodium citrate
 ecdysterone (beta-ecdysterone)
 fadogia agreste

glycerol (hyperhydration)
 hordenin (MAO-B)
 inosine (uric acid)
 isomaltulose
 laxogenin

octopamine (adrenergic)
 plant protein (mixture)
 tongkat there (Eurycoma longifolia)
 turksterone
 phosphatidic acid

3.16 Hormones, precursors and glandulars — 15 supplements

prevention: 2 · without known genetic relevance: 13

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
desiccated liver (capsules) Avoid	HFE: H63D het BCO1: slightly ↓ conversion PNPLA3/Pi*MZ (liver): unfavourable	avoid (retinol + iron)	Desiccated liver concentrates retinol (isotretinoin) and iron heme (HFE H63D).	Genetic plan (b.3, e.3) moderate	HFE: high BCO1: high PNPLA3/Pi*MZ (liver): moderate
organ glandular (mixture) Avoid	HFE: H63D het BCO1: slightly ↓ conversion PNPLA3/Pi*MZ (liver): unfavourable	avoid (retinol + iron)	Desiccated liver concentrates retinol (isotretinoin) and iron heme (HFE H63D).	Genetic plan (b.3, e.3) moderate	HFE: high BCO1: high PNPLA3/Pi*MZ (liver): moderate

No known genetic relevance (13):

7-keto-DHEA	glandular thymus	ortho + saw palmetto (prostate)
calcium-D-glucarate (β-glucuronidase)	adrenal glandular	pregnenolone (CYP17)
DHEA (CYP17/CYP19; SULT2A1)	yam-savage (diogenin)	progesterone in cream (OTC) (CYP3A4) — CYP3A4: normal
glandular prostate/testis	menopause isoflavones (UGT)	desiccated thyroid (glandular) (deiodination)
	hops (8-prenilnaringenina)	

3.17 Longevity, mitochondria and antioxidants — 21 supplements

prevention: 2 · indeterminate: 1 · normal: 8 · without known genetic relevance: 10

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
apigenin + magnesium (sleep) Avoid	UGT1A1 : IM (*1/*28) CYP3A5 :expressor (*1/*3) TRPM6 : typical	avoid / pause (by apigenin)	Magnesium does not raise a problem (typical TRPM6), but for apigenin a test break (UGT1A1 *28) is suggested. CYP3A4/P-gp with sirolimus: pause while ALT is high (current protocol). Prefer magnesium without apigenin (suggestion, not applied).	Genetic plan (S13, S28) <i>weak</i>	UGT1A1: high CYP3A5: high TRPM6: high
synolytic combined (fisetin + quercetin) Avoid	UGT1A1 : IM (*1/*28) COMT : Val/Val (high) MTHFR : 677CT/1298AC CYP3A5 :expressor (*1/*3)	avoid/pause	UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.	Genetic plan (S29 + coordination note) <i>weak</i>	UGT1A1: high COMT: high MTHFR: high CYP3A5: high
carnosine + ALCAR (combined) Undetermined	CNDP1 : indeterminate	indeterminate (carnosinase)	Serum carnosinase (CNDP1) is regulated by a repetition (CTG) _n that the chip does not read: the fraction of oral carnosine that arrives intact is not predictable. The plan maintains 2 g (1 g × 2).	CNDP1 Literature (Janssen 2005) <i>not determinable</i>	CNDP1: non-determinable
L-ergathione + PQQ (combined) Normal	SLC22A4/5 (OCTN1/2): typical	typical capture	OCTN1 (SLC22A4) L503 C/C: typical ergothioneine uptake. Maintain.	Genetic plan (S20) <i>weak</i>	SLC22A4/5 (OCTN1/2): high
astaxanthin + DHA ☺ Normal	FADS1/FADS2 : high activity APOE : ε3/ε3	EPA/DHA useful preformed	FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied).	Genetic plan (S1 + coordination note); FADS studies <i>moderate</i>	FADS1/FADS2: high APOE: high
coenzyme Q10 (ubiquinone) Normal	NQO1 : *1/*1	normal; no genetic indication	NQO1 *1/*1 (reduces ubiquinone to ubiquinol and activates idebenone normally); COQ2 without V393A. No genetic indication (suggestion, not applied).	Genetic plan (b.3) <i>weak</i>	NQO1: high
creatine + glycine (longevity) ☺ Normal	CPS1 : low glycine GSTM1 : probable null GNMT : TT	genetically useful glycine	CPS1 rs715 T/T + rs1047891 C/C: genetically lower plasma glycine; Probable null GSTM1 (glycine is a glutathione substrate); GNMT TT. 5 g/day in total (suggestion, not applied).	GWAS glycine (CPS1); genetic plan (S8) <i>moderate</i>	CPS1: high GSTM1: moderate GNMT: high
glycine + NAC + taurine (combined) ☺ Normal	CPS1 : low glycine GSTM1 : probable null GNMT : TT	genetically useful glycine	CPS1 rs715 T/T + rs1047891 C/C: genetically lower plasma glycine; Probable null GSTM1 (glycine is a glutathione substrate); GNMT TT. 5 g/day in total (suggestion, not applied).	GWAS glycine (CPS1); genetic plan (S8) <i>moderate</i>	CPS1: high GSTM1: moderate GNMT: high
idebenone Normal	NQO1 : *1/*1	normal; no genetic indication	NQO1 *1/*1 (reduces ubiquinone to ubiquinol and activates idebenone normally); COQ2 without V393A. No genetic indication (suggestion, not applied).	Genetic plan (b.3) <i>weak</i>	NQO1: high
ubiquinol Normal	NQO1 : *1/*1	normal; no genetic indication	NQO1 *1/*1 (reduces ubiquinone to ubiquinol and activates idebenone normally); COQ2 without V393A. No genetic indication (suggestion, not applied).	Genetic plan (b.3) <i>weak</i>	NQO1: high
vitamin E + selenium Normal	GPX1 : Pro198Leu het	Nutritional dose	GPX1 Pro198Leu het: activity slightly lower, but without benefit of high doses; selenium > 200 µg/day is associated with 2 type diabetes. The stack already gives 55 µg; without Brazil nuts.	SELECT; genetic plan (S12) <i>weak</i>	GPX1: high

No known genetic relevance (10):

★ calcium alpha-ketoglutarate (Ca-AKG)

sinolitic herbs (piperlongumin)
fullerene C60 in olive oil
molecular hydrogen (compressed)

mitoquinone (CoQ10 directed to mitochondria)
oxaloacetate
powder of antioxidant berries (ORAC)

★ R-ALA (R-lipoic acid)

trehalosis
alpha-lipoic acid (racemic)

3.18 Articulations, skin, hair and connective tissue — 22 supplements

prevention: 1 · normal: 5 · without known genetic relevance: 16

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
turmeric + boswellia (articular) Avoid	UGT1A1 : IM (*1/*28) PNPLA3/Pi*MZ (liver) : unfavourable HLA-B*35:01 : probable negative CYP3A5 :expressor (*1/*3)	avoid/pause	UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).	Genetic plan (S26); LiverTox (curcuma, HLA-B*35:01) <i>weak-moderate</i>	UGT1A1: high PNPLA3/Pi*MZ (liver): moderate HLA-B*35:01: moderate CYP3A5: high
Polypodium leucotomos (oral photoprotection) ☯ Normal	TERT/melanoma : unfavourable	Useful oral photoprotection	P88 + isotretinoin (photosensitivity): Polypodium reduces erythema UV in trials; it complements (not) the sunscreen.	Polypodium (Nestor 2015) trials <i>weak-moderate</i>	TERT/melanoma: high
wheat ceramides Normal	HLA-DQ2/DQ8 : negative	without celiac HLA	Without DQ2.5 or DQ8: very low genetic risk of celiac disease; gluten/cereals tolerated.	Genetic plan (e.6) <i>strong</i>	HLA-DQ2/DQ8: high
collagen + vitamin C Normal	COL1A1/COL5A1 : normal	without risk alleles	COL1A1 Sp1 C/C and COL5A1 T/T: without risk alleles; useful source of glycine (CPS1).	Genetic plan (S9) <i>weak</i>	COL1A1/COL5A1: high
collagen type I+III (skin) Normal	COL1A1/COL5A1 : normal	without risk alleles	COL1A1 Sp1 C/C and COL5A1 T/T: without risk alleles; useful source of glycine (CPS1).	Genetic plan (S9) <i>weak</i>	COL1A1/COL5A1: high
calcium + D3 + K2 (bone) Normal	VDR : typical GC (DBP) : favourable HFE : H63D het	typical absorption	VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.	NIH ODS (calcium); genetic plan (S14) <i>weak</i>	VDR: high GC (DBP): high HFE: high

No known genetic relevance (16):

ASU (unaponifiable of avocado-soy)
high dose biotin (hair)
shark cartilage
chondroitin
hair / skin / nails formula (biotin; several)

glucosamine (chloride)
glucosamine (sulfate)
glucosamine + chondroitin + MSM
L-cysteine + B (hair)
eggshell membrane (NEM)
mussel of green lips (post)

keratin
silicon (MMST)
bamboo silica
hyaluronic acid (oral)
Marine pine oil (skin)

3.19 Metabolism, appetite and cardiovascular — 25 supplements

prevention: 5 · Monitoring: 7 · normal: 1 · without known genetic relevance: 12

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
caffeine + green tea + garcinia (combined) Avoid	PNPLA3/Pi*MZ (liver): unfavourable COMT: Val/Val (high) HLA-B*35:01: probable negative UGT1A1: IM (*1/*28)	avoid (liver)	Concentrated green tea extract is a known cause of liver injury (LiverTox, probability B; risk in fasting and > 800 mg EGCG/day). Genetic liver (PNPLA3, MBOAT7, Pi*MZ); EGCG is catechol (COMT Val/Val). In favor: HLA-B*35:01 probable negative. Tea in drink is OK (suggestion, not applied).	LiverTox; EFSA 2018 (catechins); genetic plan (b.3) <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate COMT: high HLA-B*35:01: moderate UGT1A1: high
thermogenic green tea (extract) Avoid	PNPLA3/Pi*MZ (liver): unfavourable COMT: Val/Val (high) HLA-B*35:01: probable negative UGT1A1: IM (*1/*28)	avoid (liver)	Concentrated green tea extract is a known cause of liver injury (LiverTox, probability B; risk in fasting and > 800 mg EGCG/day). Genetic liver (PNPLA3, MBOAT7, Pi*MZ); EGCG is catechol (COMT Val/Val). In favor: HLA-B*35:01 probable negative. Tea in drink is OK (suggestion, not applied).	LiverTox; EFSA 2018 (catechins); genetic plan (b.3) <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate COMT: high HLA-B*35:01: moderate UGT1A1: high
red rice yeast (monacoline K) Avoid	SLCO1B1: increased function PNPLA3/Pi*MZ (liver): unfavourable CYP3A5: expressor (*1/*3)	avoid (unregulated statin)	Monacholine K = lovastatin with non-standardized dose and possible citrinine. SLCO1B1 *1/*14 is favourable, but it already takes pitavastatin (statin duplication) and there are genetic livers and cases of hepatitis (suggestion, not applied).	LiverTox; EFSA 2018; genetic plan (b.3) <i>moderate</i>	SLCO1B1: high PNPLA3/Pi*MZ (liver): moderate CYP3A5: high
multi-ingredient thermogenic Avoid	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Multi-ingredient thermogenic formulas were associated with acute hepatitis (LiverTox); genetic liver.	LiverTox <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate
Usnic acid (fat burner) Avoid	PNPLA3/Pi*MZ (liver): unfavourable	avoid (liver)	Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).	LiverTox <i>moderate</i>	PNPLA3/Pi*MZ (liver): moderate
Garcinia + green coffee (combined) Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver	Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/ Pi*MZ), use only for short periods, if used.	LiverTox <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
sine detox tea Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver	Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/ Pi*MZ), use only for short periods, if used.	LiverTox <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
chromium + cinnamon (glucose) Monitoring	CYP2A6: probable normal PNPLA3/Pi*MZ (liver): unfavourable	hepatotoxic coumarin	Cassia cinnamon has coumarin (EFSA fixed a low tolerable daily intake); 7-hydroxylation depends on CYP2A6 (*2/*9 absent; non-visible deletions). To prefer cinnamon-do-ceil.	EFSA 2008 (coumarin) <i>weak-moderate</i>	CYP2A6: low PNPLA3/Pi*MZ (liver): moderate
garcinia cambogia (HCA) Monitoring	PNPLA3/Pi*MZ (liver): unfavourable	monitoring liver	Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/ Pi*MZ), use only for short periods, if used.	LiverTox <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
sodium nitrate/ potassium Monitoring	NOS3: –786CT + 298GT	avoid concentrated nitrate (tadalafil)	NOS3 gives rational, but concentrated inorganic nitrate is added to the tadalafil (hypotension). Only vegetal nitrates (suggestion, not applied).	Genetic plan (b.3, f.13) <i>weak</i>	NOS3: high

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
black pepper (extract) Monitoring	CYP3A5 :expressor (*1/*3) UGT1A1 : IM (*1/*28)	increases drug exposure	Piperine inhibits CYP3A4, P-gp and glucuronidation: up to levels of drugs and other supplements. Never in the rapamycin window (3 days before 7 days after each dose).	Bhardwaj 2002 <i>moderate (interaction)</i>	CYP3A5: high UGT1A1: high
thermogenic (caffeine + synephrine) Monitoring	CYP3A5 :expressor (*1/*3) CETP/9p21/APOB : genetic vascular risk	stimulant + CYP3A4	Yellow orange furanocoumarins inhibit CYP3A4 (grapefruit effect); synephrine is a cardiovascular stimulant, especially with caffeine; 9p21 homozygous. Never in the rapamycin window (3 days before 7 days after each dose).	NCCIH (yellow orange) <i>weak</i>	CYP3A5: high CETP/9p21/APOB: high
Liquid L-carnitine (thermogenic) Normal	SLC22A4/5 (OCTN1/2) : typical FMO3 : typical	Typical transport; TMAO form	Typical OCTN2. Typical FMO3: oral carnitine generates TMAO by the microbiota; with genetic vascular priority, moderate and non-chronic dose at a high dose.	Genetic plan (S19); Koeth 2013 <i>weak</i>	SLC22A4/5 (OCTN1/2): high FMO3: high

No known genetic relevance (12):

lipoic + biotin (glucose)	raspberry ketones	white bean (phaseolamine)
banaba (corosolic acid)	artichoke extract + bergamot (LDL)	glucomannan + chromium (society)
bergamot (BPF)	apple peel extract	Morosil (Orange-sanguinea)
Caralluma fimbriata	whiteberry leaf extract (DNJ)	grifonia seed (5-HTP) (AADC)

3.20 Others (bee products, algae, formulas, detoxification, several) — 38 supplements

dose adjustment: 4 · Monitoring: 7 · normal: 5 · without known genetic relevance: 22

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
energy formula (B + caffeine) Adjust dose	PDXK/ALPL/NBPF3 (B6) : typical/high MTHFR : 677CT/1298AC DHFR : slow reduction (CC)	choose formula (low B6, without folic acid)	Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).	Genetic plan (S3, S6); EFSA UL B6 <i>moderate</i>	PDXK/ALPL/NBPF3 (B6): high MTHFR: high DHFR: moderate
homocysteine formula with folic acid Adjust dose	MTHFR : 677CT/1298AC DHFR : slow reduction (CC)	prefer 5-MTHF (form)	MTHFR 677CT + 1298AC (activity ~50–60%) and DHFR rs408626 C/C (slow reduction of folic acid; may leave folic acid unmetabolized in circulation). Exchange by 5-MTHF or food folate; no product with folic acid (suggestion, not applied).	Genetic plan (S3, b.3); MTHFR/DHFR literature <i>moderate</i>	MTHFR: high DHFR: moderate
methylation formula (B12 + folate + B6 + TMG) Adjust dose	PDXK/ALPL/NBPF3 (B6) : typical/high MTHFR : 677CT/1298AC DHFR : slow reduction (CC)	choose formula (low B6, without folic acid)	Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).	Genetic plan (S3, S6); EFSA UL B6 <i>moderate</i>	PDXK/ALPL/NBPF3 (B6): high MTHFR: high DHFR: moderate
lysine + vitamin C (Pauling protocol) Adjust dose	HFE : H63D het G6PD : normal SLC23A1/2 (vit. C) : typical	drinking between meals (hour)	HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).	Genetic plan (S16); CPIC G6PD 2022 (analogy) <i>moderate (iron interaction)</i>	HFE: high G6PD: high for the tested variants SLC23A1/2 (vit. C): high
extract of fava (Vicia faba, L-dopa natural) Monitoring	G6PD : normal COMT : Val/Val (high) MTHFR : 677CT/1298AC	Normal G6PD; L-dopa (COMT)	Normal G6PD (probable favism); the L-dopa of the fava is O-methylated by COMT Val/Val and rises homocysteine (compound MTHFR).	CPIC G6PD 2022; L-dopa literature <i>moderate</i>	G6PD: high for the tested variants COMT: high MTHFR: high
anti-stress formula (adaptogen) Monitoring	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring liver	Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.	LiverTox <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
hepatic support formula (detox) Monitoring	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring liver	Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.	LiverTox <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate
hangover formula (DHM, dihydromyricetin) Monitoring	COMT : Val/Val (high) MTHFR : 677CT/1298AC	catechol: prefer low doses	Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). Food doses without a problem.	Genetic plan (f.2) <i>weak</i>	COMT: high MTHFR: high
hanging formula with NAC + cardo Monitoring	UGT1A1 : IM (*1/*28) CYP2C9 : IM (*1/*3) PNPLA3/Pi*MZ (liver) : unfavourable	without proven gain; inhibits UGT1A1/CYP2C9	Silibinin is glucuronidated by UGT1A1 and inhibits it (bilirubin can rise in *28) and inhibits CYP2C9 (effect on losartan greater in *1/*1 than in *3; Han 2009). It does not protect the liver in a proven way in PNPLA3/Pi*MZ.	Han 2009; silymarin NASH trials <i>weak</i>	UGT1A1: high CYP2C9: high PNPLA3/Pi*MZ (liver): moderate
Sleep formula (mixture of herbs) Monitoring	PNPLA3/Pi*MZ (liver) : unfavourable	monitoring liver (valerian)	Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used. (valerian/scullery in mixtures).	LiverTox <i>weak</i>	PNPLA3/Pi*MZ (liver): moderate

Supplement	Genes (state)	Predicted response	Recommendation (summary) and implication for Sérgio F	Source · evidence	Trust
nutritional yeast Monitoring	MTHFR : 677CT/1298AC DHFR : slow reduction (CC)	check fortification	Many brands are fortified with synthetic folic acid (DHFR CC): choosing not fortified or with 5-MTHF.	Genetic plan (b.3) <i>weak</i>	MTHFR: high DHFR: moderate
eggshell calcium Normal	VDR : typical GC (DBP) : favourable HFE : H63D het	typical absorption	VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.	NIH ODS (calcium); genetic plan (S14) <i>weak</i>	VDR: high GC (DBP): high HFE: high
immunity formula (zinc + C + D) ☺ Normal	SLC30A8 : C/C (usual zinc)	genetically useful zinc	SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20-25 mg/day with 1.5-2 mg of copper (Zn:Cu ≈ 10-15:1; DAO depends on copper, AOC1 het) (suggestion, not applied).	Kanoni 2011 (CHARGE); genetic plan (S12) <i>moderate</i>	SLC30A8: high
formula for the eye (lutein + bileberry) Normal	CFH/ARMS2 : CFH het BCO1 : slightly ↓ conversion	useful (retina)	CFH Y402H het, ARMS2 G/G; lutein/zeaxanthin do not depend on BCO1. Keeping with fat.	Genetic plan (S24); AREDS2 <i>weak</i>	CFH/ARMS2: high BCO1: high
wheat germ Normal	HLA-DQ2/DQ8 : negative	without celiac HLA	Without DQ2.5 or DQ8: very low genetic risk of celiac disease; gluten/cereals tolerated.	Genetic plan (e.6) <i>strong</i>	HLA-DQ2/DQ8: high
Wheat germ oil Normal	CYP4F2 : normal	normal metabolism	CYP4F2 V433M C/C (normal): vitamin E ω-hydroxylation is typical. Nutritional dose; no genetic benefit for high doses.	CYP4F2 × tocopherol (Bardowell 2012) studies <i>weak</i>	CYP4F2: high

No known genetic relevance (22):

kombu/wakame algae powder (iodine)
 sodium alginate
 bentonite clay
 activated carbon
 kombucha tea powder (microbiota)
 Oral EDTA (quelation)
 fucus/goat (iodine)

Prostate formula
 royal jelly
 beer yeast
 monolaurine
 Sea moss (sea moss, Chondrus) (iodine)
 bee pollen
 spray of propolis + zinc (throat)
 diatomaceous earth

TUDCA (Tauroursodeoxycholic acid) (biliary conjugation)
 zeolite (clinoptilolite)
 fulvic acid
 malic acid
 alkaline water/alkaline salts
 isotonic seawater
 essential oils in capsules (mixture)

4. Limitations

- **Chip + imputation, no sequencing.** Only the variants that the chip measures or that the imputation reconstructs with $DR2 \geq 0.8$ are seen. Rare and new variants are not seen; "normal" means "no variants tested".
- **Evidence of weaker nutrition than pharmacogenetics.** There are no CPIC guidelines for supplements. Most of the gene-nutrient interactions come from small, observational or mechanistic studies; the column "evidence" indicates the degree. Colors also reflect the context of stack (sirolimus, isotretinoin, telmisartan, tadalafil, glucose drugs), when genetics changes the importance of an interaction.
- **CNVs and non-visible repetitions:** GSTM1 (inferred) and GSTT1, CYP2D6 (*5, duplications), CYP2A6 (deletions), AMY1, HP (haptoglobin), CNDP1 (CTG)_n HSD11B2 not tested.
- **Haplotype phase:** MTHFR 677/1298 assumes compound heterozygosity (cis/trans not determined).
- **Only genetics, on purpose.** Doses of vitamin D, omega-3, zinc, iron, B12 and homocysteine should be tuned with measurements; therefore, where there is a decision based on measurements (D3 10 000 IU; omega-3 3.2–4 g; curcumin pause, luteolin, quercetin and apigenin while ALT is high; monthly pulse fisetin + quercetin only with normal ALT), this decision prevails (Note of coordination of the genetic plan; current protocol).
- **Product quality:** adulteration (stimulants, undeclared drugs), contamination (metals, pyrrolizidine alkaloids) and dose variation between brands are not predictable by genetics and explain much of the hepatotoxicity described.
- **List of supplements:** constructed from NIH ODS and NCCIH chips and curation of the usual commercial categories; it does not derive from sales data. It includes chemical forms, strains and combined formulas as distinct items.
- **It is not a diagnosis.** It is support to the decision to discuss with the doctor. All indications are suggestion (not applied).

5. Sources and methods

- NIH Office of Dietary Supplements — list A-Z of supplements (read to 02-10-2026); NCCIH — Herbs at a Glance (67 inputs); LiverTox (NIH/NIDDK) — plant chapters and supplements (probability of hepatotoxicity).
- EFSA — upper limits of ingestion (B6 2023, selenium, zinc), opinions on green tea catechins (2018), monacolins (2018) and coumarin; EMA/HMPC — plant monographs (pyrrolizidine alkaloids, glycirizine, anthraquinones).
- Pharmacogenetics: CPIC/DPWG profile and guidelines of the 632 drug report (same genotype), used by analogy (e.g. tacrolimus–CYP3A5 for sirolimus, G6PD 2022, 2022 statins, warfarin 2017).
- Gene-nutrient interactions cited by line (e.g. Kanoni 2011 zinc–SLC30A8; Garaulet 2015 melatonin–MTNR1B; Cornelis 2006 caffeine–CYP1A2; Olthof 2001 chlorogenic acid–homocysteine; Chow 2010 resveratrol–CYP; Gurley 2008 hydrate; Guo 2012 berberina; Han 2009 silimarina–CYP2C9; Riche 2014 pterostilbene–LDL).
- Genotypes: chip (GRCh38) + Beagle 5.5 imputation (public reference panel of complete genomes to 30x), direct queries to the 02-10-2026 chip (AMPD1, CYP2E1, UGT1A6, EPHX1, MTR) and the genetic plan of 02-10-2026 (including the coordination note).